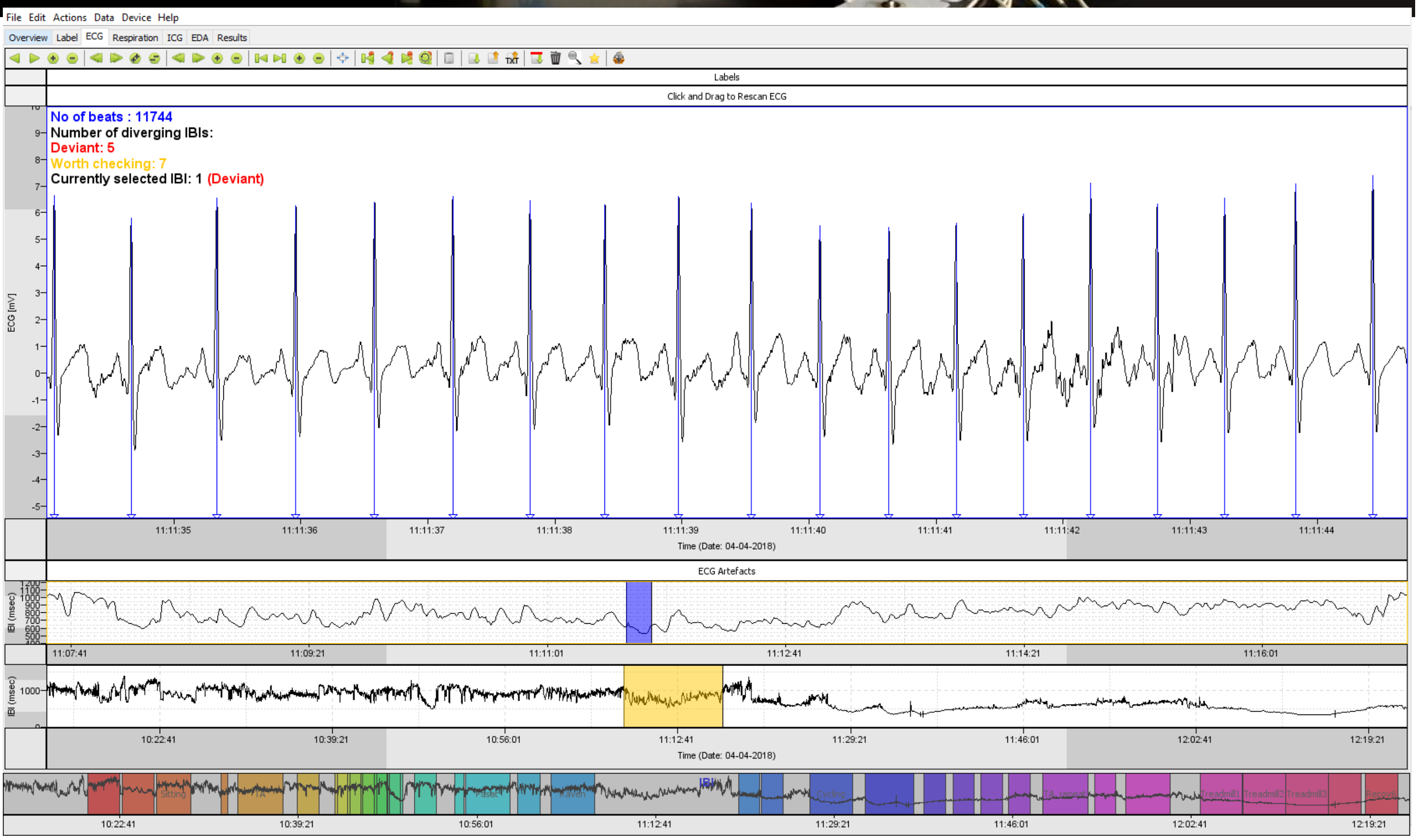
DATA ANALYSIS AND MANAGEMENT SOFTWARE (DAMS) *FOR THE* VRIJE UNIVERSITEIT AMBULATORY MONITORING SYSTEM (VU-AMS)

Manual version 1 10 januari 2022 (written for VU-DAMS v5.0)



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1. SAFETY INFORMATION

To prevent injury to yourself and others, or damage to your device, read all of the following information before using your device.

Intended purpose

The product is an Ambulatory Monitoring System, which records ECG, ICG, Electrodermal Activity and Tri-Axial Accelerometry for stress and emotion measurements strictly for non-medical, psychological research of people in their daily habitat. The device is not intended to be used for medical purpose.
The electronic device connects to the wearer with 7 ECG electrodes and 2 EDA electrodes and is powered by 2 AA batteries. It requires to be connected to a computer for configuration, by infrared communication. After that there is no communication. The data is stored on compact flash memory card. At the end the memory card can be removed, and the data copied with a standard card reader.

Store and handle your device with proper care

Keep your device dry

Humidity and liquids may damage the parts or electronic circuits in your device.
Do not turn on your device if it is wet. If your device is already on, turn it off and remove the battery immediately (if the device will not turn off or you cannot remove the battery, leave it as-is). Then, dry the device with a towel and contact an authorized VU-AMS service centre.

Store your device only on flat surfaces. Do not drop your device or cause impacts to your device.

If your device falls, it may be damaged.
If bent or deformed, your device may be damaged or parts may malfunction.

Do not store your device near or in heaters, microwaves, hot cooking equipment, or high-pressure containers

Never place batteries or devices on or in heating devices, such as microwave ovens, stoves, or radiators.
Do not store or use your device in very hot or very cold areas.

It is recommended to use your device at temperatures from 2 °C to 40 °C. Extreme temperatures can damage the device.

Proper use and cleaning

Do not use the device for anything other than its intended use

Your device is not a toy. Do not allow children to play with it as they could hurt themselves and others, or damage the device. If children use the device, make sure that they use the device properly.

This device is for external use only

Do not insert the device or supplied accessories into the eyes, ears, mouth or other bodily orifices. Doing so may cause suffocation or serious injuries.

Apply the device according to the guidelines in the manual

Make sure the cables do not become entangled in your arms or on nearby objects. Do not loop cables around the neck or other body parts. Secure loose cables. Make sure the carrying belt has a snug fit.

Do not re-use disposable accessories such as electrodes

These accessories were designed with a single use in mind, and re-using them will negatively impact data quality and may be unhygienic.

Properly clean re-usable accessories

Desinfect accessories which may come in contact with the body of the wearer.

Clean your device with a damp cleaning cloth

Wipe your device with a cleaning cloth. Do not use chemicals or detergents on the device, the cables, and other accessories.

Only use manufacturer-approved batteries, accessories, and supplies.

Unapproved third-party accessories may cause the device to malfunction or may result in damage to the device.

Only use undamaged batteries, accessories, and supplies.

Never use a damaged battery.
Never use damaged lead wires.
Never attempt connecting lead wires to any other device or power source.

Do not disassemble, modify, or repair your device

Any changes or modifications to your device can void your manufacturer's warranty. If your device needs servicing, take your device to an authorized VU-AMS Service Centre.

Allow only qualified personnel to service your device

Allowing unqualified personnel to service your device may result in damage to your device and will void your manufacturer's warranty.

Data handling and data security

Do not remove batteries while recording

Do not remove the battery while the device is recording information, as this could result in loss of data and/or damage to the card or device.

Handle storage cards with care

Protect cards from strong shocks, static electricity, and electrical noise from other devices.
Do not touch gold-colored contacts or terminals with your fingers or metal objects. If dirty, wipe the card with a soft cloth.

Back-up your data

While using your device, be sure to back up important data. VU-AMS is not responsible for the loss of any data.

Keep your data secure

Data recorded with the VU-AMS potentially contains sensitive and personal information, follow the appropriate privacy and data protection guidelines when handling and storing the recorded data.
When disposing of your device, back up all data and then reset your device to prevent misuse of your personal information.

Caution: Follow all safety warnings and regulations when using your device in restricted areas.

Do not use your device near other electronic devices

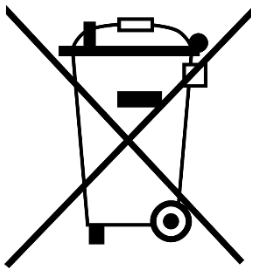
Most electronic devices use radio frequency signals. Your device may interfere with other electronic devices.

Do not use your device near a pacemaker

Avoid using your device within a 15 cm range of a pacemaker, if possible, as your device can interfere with the pacemaker.

Correct disposal of this product

Never dispose of batteries or devices in a fire. Follow all local regulations when disposing of used batteries or devices.



(Waste Electrical & Electronic Equipment)

(Applicable in countries with separate collection systems)
This marking on the product, accessories or literature indicates
that the product and its electronic accessories (e.g. lead
wires, USB/IR cable) should not be disposed of with other household waste.
To prevent possible harm to the environment or human health from uncontrolled waste disposal, please separate these items from other types of waste and recycle them responsibly to promote the sustainable reuse of material resources.

This product and its electronic accessories should not be mixed with other commercial wastes for disposal.

Regulatory compliance

This device complies with the relevant CE technical legislation as defined in horizontal New Legislative Framework package on free movement of goods from 2008 (Regulation 764/2008/EC and 765/2008/EC plus Decision 768/2008/EC) and sectoral technical regulations (known as New Approach Directives).

The device meets the following standards:
EN 55024:2010 - Information technology equipment - Immunity characteristics - Limits and methods of measurement - CISPR 24:2010 CISPR 24:2010/A1:2015
EN 55032:2012 - Electromagnetic compatibility of multimedia equipment - Emission requirements

Each VU-AMS 5fs device carries the CE marking and a unique serial number which identifies each individual device.

2. COMPONENTS

	Core system components	
1	VU-AMS*5fs*	The main unit of the VU-Ambulatory Monitoring System, version 5fs
2	Combined ECG/ICG leadset	Seven-wire lead set with blue connector for the recording of the ECG and thorax impedance. A three-wire variety for ECG-only recording is available.
3	EDA leadset	Two-wire lead set with yellow connector for electrodermal activity.
4	Compact Flash card	Removable memory card. The VU-AMS5fs has been extensively tested with the following Compact Flash cards. 1GB 80x from Transcend (TS1GCF80) 2GB Ultra from SanDisk (SDCFH-002G-U46) 4GB Ultra from SanDisk (SDCFHS-004G-G46).
5	Compact Flash card reader	Standard card reader unit to extract the VU-AMS data from the Compact Flash card after recording and to erase the card for a next recording.
7	VU-AMS*i* cable	An infrared interface cable that either connects to the RS232 serial port of a PC or to an USB port.
8	Data Analysis and Management Software (DAMS)	The VU-AMS device is configured using this program (referred to as 'DAMS') and measurements can be started and stopped with the program. The DAMS program is also used for primary data extraction and for data reduction. It can be downloaded from the VU-AMS website (www.vu-ams.nl).

	Consumables	
9	ECG electrodes	We use the 'Kendall ARBO H98SG' single use ECG electrode with wet gel. Three are required to record ECG, 4 for ICG and 1 for EDA.
10	EDA electrodes	A single Biopac EL507 electrode is used for each EDA recording

11	Alcohol (pads)	To prepare the skin for optimal conductance and adhesion of electrodes, use pre-packaged alcohol pads or bottled alcohol.
7	Skin preparation paper	3M™ Red Dot™ Trace Prep, 2236 18 mm x 5 mm - to prepare the ground electrode site on the lower arm for EDA measurement.
8	Micropore tape	We use 2,5 cm micropore tape to affix the electrode wires to the body. This will improve signal quality and participant comfort.
7	Two AA batteries	We recommend non-rechargeable 1.5V alkaline batteries.

Optional accessories

9	Belt	For ambulatory data collection the device is placed in a holder on a belt that participants wear around their hip.
10	Elastic band	To increase comfort we advise that during ambulatory data collection the blue electrode cable set is worn with the elastic band.

3. SETTING UP A MEASUREMENT

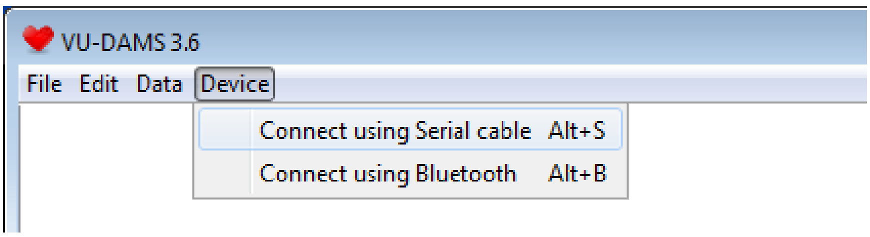
3.1 PREPARING THE VU-AMS DEVICE

Always use an empty Compact Flash card with all previous files removed from the card before each new measurement. Put the flash card with the blue VU-AMS sticker facing up in the memory card slot of the VU-AMS. Then place two completely charged AA batteries in the battery holder. Make sure you use batteries where the bottom contact sticks out (in some batteries it is covered by an outer plastic ring; these won't work properly with the VU-AMS). Battery clips are vulnerable, so place batteries carefully, with the blue ribbon under the batteries. The ribbon should be used to remove the batteries after the measurement. Successful battery placement is signaled by a triple beep tone.

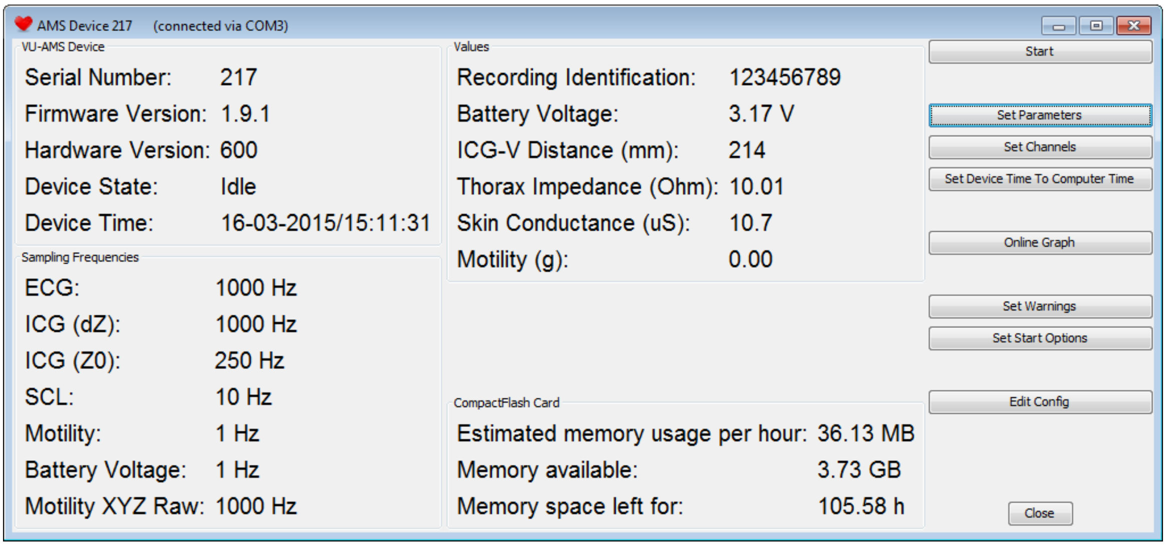
The VU-AMS is now on standby and the green light will flash twice every ten seconds. This indicates the VU-AMS is ready, but not recording. When the VU-AMS is recording the green light will flash once every three seconds.

3.2 CONFIGURING THE VU-AMS DEVICE FOR RECORDING

Connect the VU-AMS to the PC with the interface cable. Connect the infrared end of the interface cable to the VU-AMS; the electronic end of the interface cable goes to the serial port or the USB port of the PC. Start the DAMS program and select the *Device* tab in the main menu. Choose to *Connect using Serial cable*.



This will open up the configuration screen:



3.2.1 CLOCK SYNCHRONIZING

Before starting, make sure to set date and time of the PC correctly. All dates and times in the VU-AMS data files will be based on the time and date read from the PC at start-up, so it is important to make sure your PC has the correct time and date. Do this by clicking on *Set Device Time To Computer Time*.

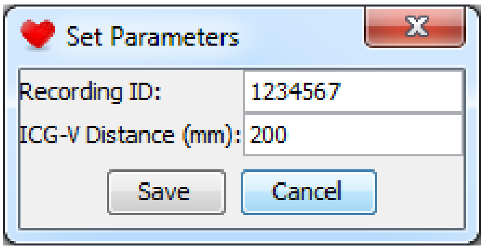
> ***TIP1:*** *Synchronize the watch of the subject/observer to the exact time of the PC used to start up the VU-AMS for optimal time-locked self-report diaries and physiological data. When electronic diaries are used make sure that their clocks are synchronized with the configuration PC too.*

3.2.2 BATTERY

Check battery voltage indication (should be about 3.1 Volt for alkaline and about 2.7 Volt for rechargeable NiMH batteries) and **re-check** time and date. Though battery voltage gives an indication of battery capacity left, for 24-hour recordings it is safest to start with new alkaline batteries or batteries that were in the charger up till the very last moment. When re-using rechargeable batteries or alkaline batteries (for short recordings), keep them together in sets with the same usage histories. Never use batteries which are not part of the same set.

3.2.3 SET PARAMETERS

Click *Set Parameters* and fill in the recording identification field. Also fill in the distance measured in millimeters between the two front ICG electrodes for later stroke volume estimations (see section 4 electrode hook up and 5.5 ICG scoring).

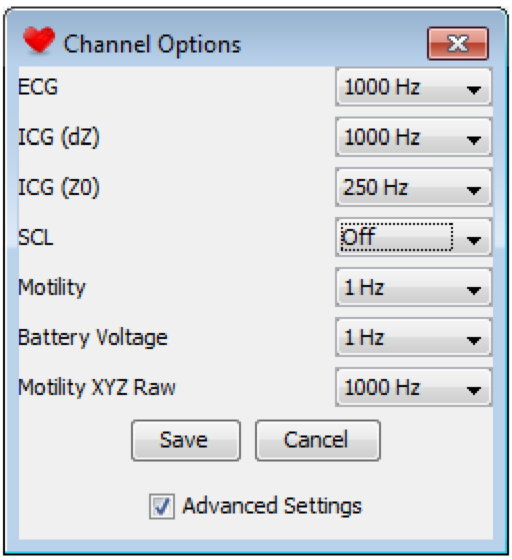


3.2.4 SET CHANNELS

The recommended sampling frequencies are as follows.

ECG	1000 Hz
ICG (dZ)	1000 Hz
ICG (Z0)	250 Hz
SCL	10 Hz
Motility	1 Hz
Battery voltage	1 Hz
Motility XYZ raw	1000 Hz

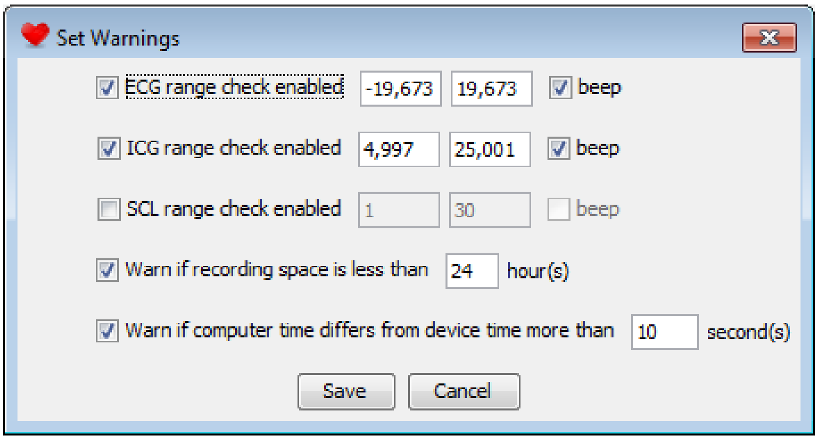
By selecting on *Set Channels* you are able to set sampling frequencies for the various signals. When you are not measuring a particular signal (e.g. no SCL cable with yellow plug connected) you should disable it by setting it to *'Off'* (as in the example



below). When changing any setting, make sure to save the settings to the device before closing the DAMS program!

3.2.5 SET WARNINGS

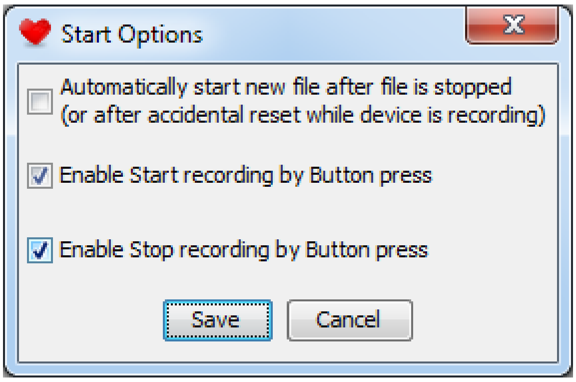
Here you can activate or deactivate audio warning signals when ICG, ECG or SCL signals exceed their boundary values. You can also activate warning for low recording space and time-sync problems. These beeps are useful in field recordings when subjects are able and sufficiently instructed to reconnect electrodes themselves. Otherwise, it is advisable to turn the warning signals off because the beeps will persist as long as the electrodes are not properly reattached.



TIP2: *With the use of the new electrodermal activity electrodes the maximum skin conductance levels (SCL) have become higher. We therefore advise to edit the upper value to 90 when using the Biopac EL507 electrode with ECG electrode as ground configuration.*

3.2.6 SET START AND STOP OPTIONS

Pressing the button on the VU-AMS device shortly always results in a time marker in the data file. Hence the button can be used as an event marker by the subject during field recordings. You can further program the button on the VU-AMS device to act as a start and/or stop button. If the event marker function is used it is advised to either disable the stop button or to explicitly instruct your subjects to only *shortly* press the black button to mark specific events. Accidentally pressing the button for more than 3 seconds may otherwise stop the recording.

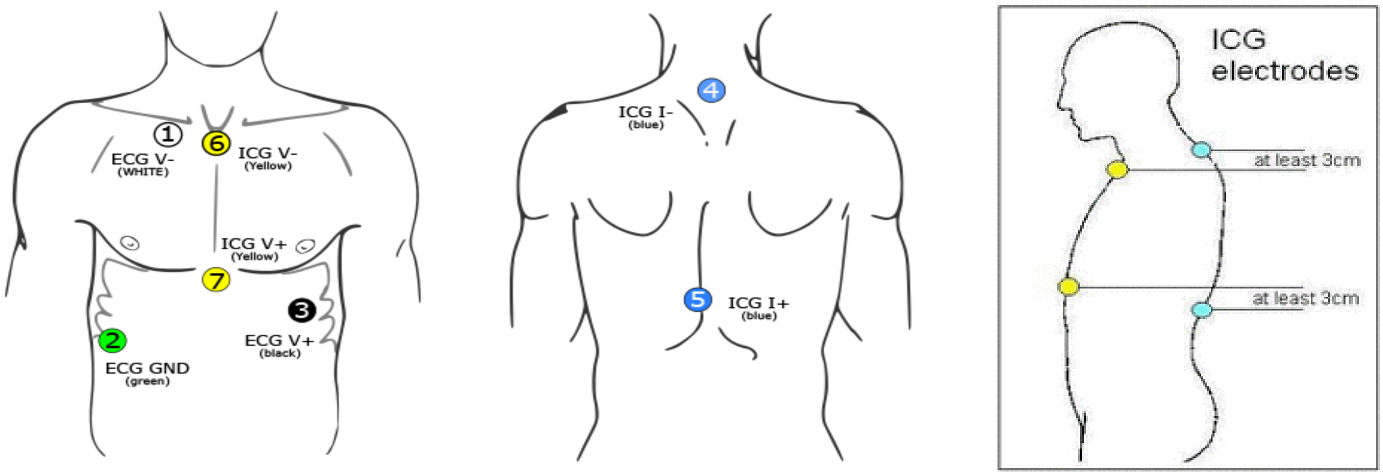


When the top option is selected a new recording will immediately start when the battery is replaced (which may be needed during recordings lasting longer than 48 hours) or when a subject tries to stop the recording using the black event button. When this option is checked the only way to definitively stop the device from recording is from the configuration screen in the DAMS program.

4. ELECTRODE HOOK UP

4.1 ATTACHMENT OF THE ECG/ICG ELECTRODES

Clean the skin at the 7 positions indicated in the figure. Rub the skin firmly with an alcohol soaked tissue or, if alcohol is not available, with a clean dry tissue. Attach an electrode by pressing the sticky plastic brim of the electrode on the skin and subsequently pushing the metal stud at the center of the electrode firmly, to properly spread the contact gel.



Chest Back **Side**

ECG:

1 Slightly below the right collar bone 4 cm to the right of the sternum

2 (GND) On the right side, between the lower two ribs

3 At the apex of the heart on the left lateral margin of the chest approximately at the level of the processus xiphodius

ICG: ***Electric current generating electrodes***

4 At the back, on the spine, at least 3 cm (1") above electrode 6

5 At the back, on the spine, at least 3 cm (1") below electrode 7

Impedance measuring electrodes

6 At the suprasternal notch above the top of the sternum

7 At the processus xiphodius at the bottom of the sternum

ICG-V distance: with a tape measure, determine the distance between yellow front electrodes 6 and 7 in millimeters.

> ***TIP3:*** *See "Recording with VU-AMS" tutorial video for a demonstration: www.vu-ams.nl/support/tutorials/hardware/vu-ams-recording*

4.1 ATTACHMENT OF THE ECG/ICG ELECTRODES

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4.2 ATTACHMENT OF SCL ELECTRODES (OPTIONAL)

To measure electrodermal activity we recommend placing a dedicated EDA electrode (for example Biopac EL507) on the thenar eminence of the non-dominant hand and a regular ECG electrode on the lower arm as a ground (Figure 1). The thenar site should not be pre-treated with alcohol, to retain normal fluid balance in the skin. After applying the electrode sticker, secure the edges with micropore tape to help keep it in place. The site of the ground electrode should be lightly abraded with skin preparation paper and cleaned with alcohol before applying the standard ECG electrode sticker.

Figure 1. Placement of the EDA electrodes

The VUAMS also allows for other electrode configurations, such as Velcro straps with an electrode holder filled with gel on the medial phalanges of the index and middle or ring finger, or SCL electrodes at both the thenar and hypothenar eminences of the hand palms. However, we have found the configuration above advantageous in terms of both signal quality and feasibility and tolerability in extended ambulatory recording.

As the sweat ducts acts as parallel conductors it is essential to keep the surface area of measurement constant across subjects by using the same electrodes and electrode positioning within a single experiment.

4.3 ATTACHMENT OF THE LEAD WIRES AND LEAD WIRE CONNECTOR

The blue ECG/ICG lead wire connector has to be plugged in the blue socket. The yellow SCL lead wire connector is plugged into the yellow socket. Both the plug and socket have arrows which should be aligned with each other for the plug to fit, without having to apply force.

4.4 WEARING THE DEVICE

Put the VU-AMS device in its carrier bag with the lead wire connector facing up. Fasten the device with the strap in the bag and gird it on with the VU-AMS belt (you can also use your own belts or the participants' belts). Make sure the device is attached in a vertical position on the participants left hip. The placement on the left hip is required for correct interpretation of the accelerometer data.

4.5 SIGNAL QUALITY CONTROL

After connecting the lead wire plug(s) to the VU-AMS device, and while still connected to the computer using the AMSi IR cable, always check signal quality using the the *Online Graph* option. In this screen ECG, Z0, dZ (≈ change in impedance due to respiration and heartbeat) and dZ/dt (= Impedance CardioGram), as well as the skin conductance signal can be streamed live to check for proper quality of the recorded signals.

TIP4: *For ambulatory data collection we advise to use an elastic waist band to further secure the ECG/ICG wires, reducing pull on the electrodes and the hazards of dangling wires. This improves both signal quality and participant comfort.*

4.5.1 ECG

The ECG registers the electrical activity of the heart. Before contracting, cardiac muscle cells depolarize. Three different stages of depolarization are usually visible on the ECG.

1. P wave: depolarization of the atria
2. QRS complex: depolarization of the ventricles
3. T wave: repolarization of the ventricles

The R peak in a normal ECG has a much higher amplitude than the P wave, because there is more muscle involved. This makes the R peak easy to detect for an algorithm. A clear QRST-complex should be detectable in the ECG. The R-wave should be upward and it should be the peak with the largest (absolute) amplitude. If either S-wave or T-wave are of comparable magnitude re-attach the black(+) ECG electrode first more laterally then more medially until a satisfactory QRS complex is seen (Figure 2).

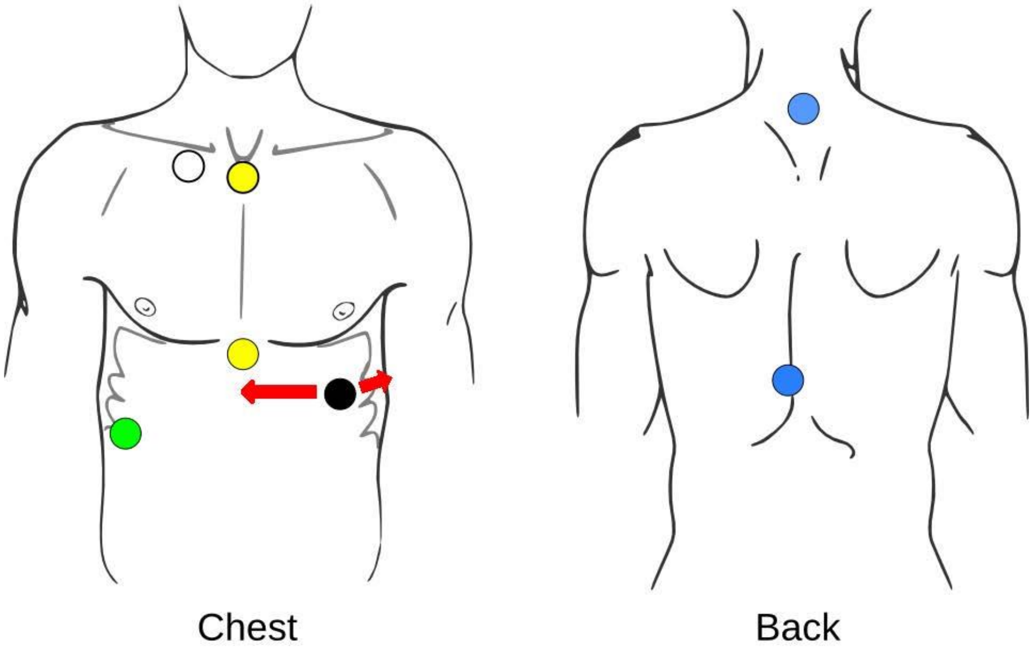


Figure 2. How to relocate the black(+) electrode to improve S / T-wave magnitude

4.5.2 ICG

Z0

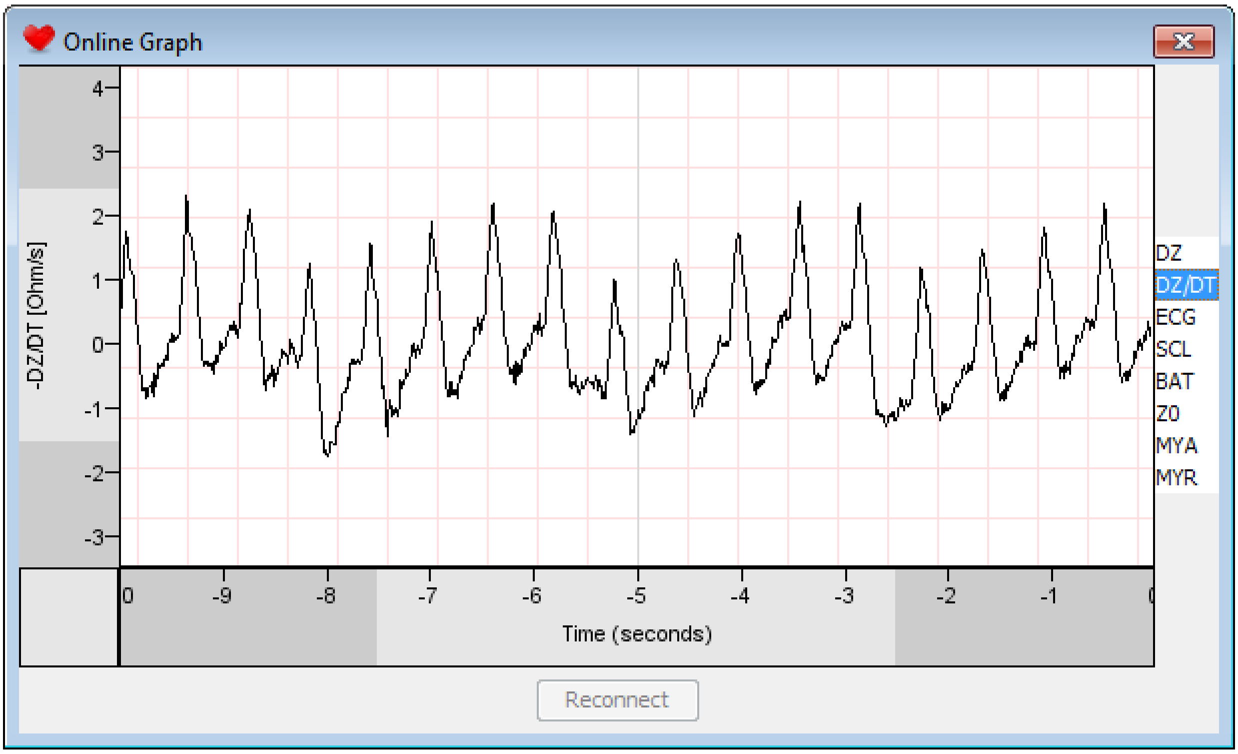
Z0 should always stay within an 8 to 20 Ohm range.

DZ

The dZ should be within -0.5 and +0.5 Ohm most of the time. The dZ signal should reflect deep breathing clearly and stay in range (between -1 and +1 Ohm).

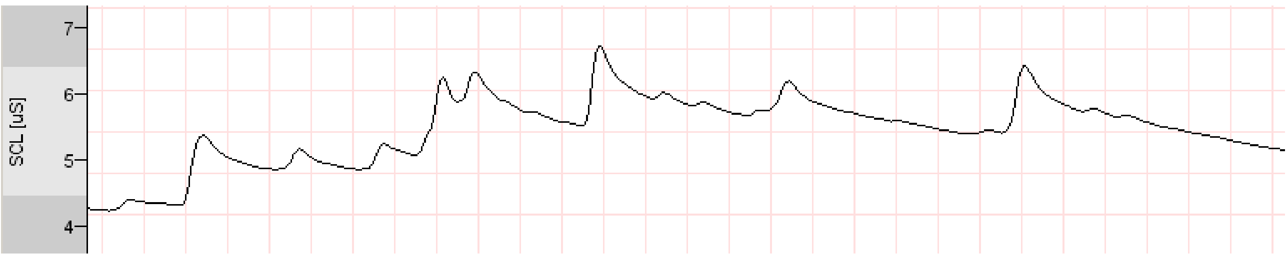
DZ/DT

In the ICG (dZ/dt) the typical upward waveform of the cardiac ejection phase should be clearly detectable. Light movement of the subject should not overly distort it. If these criteria are not met, re-attach the electrodes in the order 7,6,1,3,4,5,2 (see illustration at 1.4 Electrode hook up) until satisfactory signals are obtained. The scrollbar on the Y-axis of the *online graph* can be used to scale the signals (or hit F5 to auto scale).



4.5.3 EDA

Levels of skin conductance depend heavily on the type and configuration of electrodes used. When using the recommended electrode configuration, the SCL signal should be above 2 microSiemens but, depending on electrode type and positioning, it is not uncommon to see signal levels of 40 to 50 microSiemens or higher. Spontaneous phasic responses should be discernible in most subjects, recognizable by a steep rise and a slower downward slope. An orienting response to a sudden unexpected stimulus should also give a phasic increase (e.g. clapping hands behind the back of the subject). Since EDA is a relatively slow signal, it is best checked on a 1-2 minute time scale, when close to vertical lines should be visible in your data.



NOTE: In ambulatory paradigms, online signal inspection is <u>your only opportunity</u> to re-attach faulty electrodes.

4.6 STARTING A MEASUREMENT

When satisfied, start data recording by pressing *Start* in the configuration screen. A beep will be heard to acknowledge the start of the recording and the green light will start flashing once every three seconds as long as the device is recording. Close the configuration screen of the DAMS program and disconnect the VU-AMS device from the interface.

4.7 MARKING SPECIAL EVENTS

A small black button is placed on top of the VU-AMS device next to the two lead wire plug connectors. To mark a special event during the recording, push this button shortly. Pushing it will be confirmed by a short beep.

4.7 STOPPING A MEASUREMENT

The measurement by the VU-AMS device can be stopped by:

1) Reconnecting the device to the PC and connecting again by serial cable by choosing the appropriate action in the DAMS menu under *Device*. You can now press *Stop*. This is the preferred method. Close the DAMS program and then disconnect the device from the interface cable.
2) Pressing the event button for more than three seconds. This requires that the option of stopping with the event button was **enabled** in the *Set Start Options* menu. After this action the light will flash every 10 seconds to indicate standby mode. The method is preferred when subjects have to stop the recordings themselves at home at a designated time. N.B.: When the device is returned to you after a day of ambulatory research, make sure to check whether (1) the measurement has been stopped already with the button (the light flashes twice

every ten seconds), (2) is still recording (the light flashes every three seconds), or (3) has stopped because of empty batteries (the light does not flash at all).

3) Removing the batteries. This is strongly discouraged as it will lead to corrupted data files (0KB). In designs where participants are required to stop the recording themselves, always ask them to long-press the marker button to stop the recording first, before they remove batteries.

Once the VU-AMS has stopped recording, the yellow and/or blue lead wire plug(s) may be disconnected from the VU-AMS device and the lead wires from the electrodes. Now remove the batteries and place the Compact Flash Card in the card reader. Move the *.5FS* file containing the raw data to a designated directory. After confirming your data is securely stored, delete all files from the Compact Flash card. To prevent data loss, never use the compact flash card for long term storage, and never add any data to it in any other way than recording with the VU-AMS device.

4.8 SAVING AN .AMSDATA FILE

Double click the *.5FS* data file to open the recording. When you close DAMS it will automatically save the data in a new file with the extension *.amsdata.* This file can be opened by using the *Open data* option in the main menu or just double-clicking it. Using *.amsdata* files (once these are created) will make DAMS load the data much faster. Also all manual scoring will be saved in the .amsdata file. Please archive the .5fs raw data file.

N.B.: you can batch convert .5fs files to the .amsdata format. Files recorded with the older AMS 4.6 system can be converted too, but no visible ECG signal will be present as only the inter beat intervals were stored with this device and not the ECG signal itself. If label information was present there are two ways to keep the labels intact while the batch conversion from .5fs to .amsdata is done. One option is to place the .lbl and the .cfg files in the same folder as the .5fs files. In the second option, a text file which holds information about the start time and end time of the labels together with the label codes is provided (see further Explanation in *2.3.3).* *Click on* ***Batch convert data files*** *in the menu and select the folder with all .5fs or .ams raw data files . After conversion the folder will contain an .amsdata file for each of the raw data input files.*

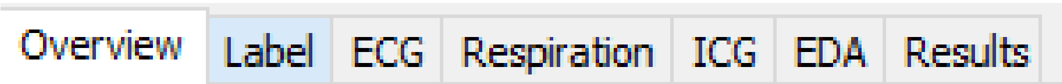
4.9 MERGING MULTIPLE .5FS FILES

If the recording has been interrupted by the experimenter or by the subject (because the participant took a shower or batteries have been replaced), multiple *.5FS* data files with different start times will be generated. There is a possibility to use a standalone tool, *AmsMerge,* that concatenates the *.5FS* files into a single *.5FS* file that spans the entire recording. The AmsMerge program can be found in Start menu >> Programs >> VU-DAMS folder from version 2.2 and up.

5. THE DAMS PROGRAM

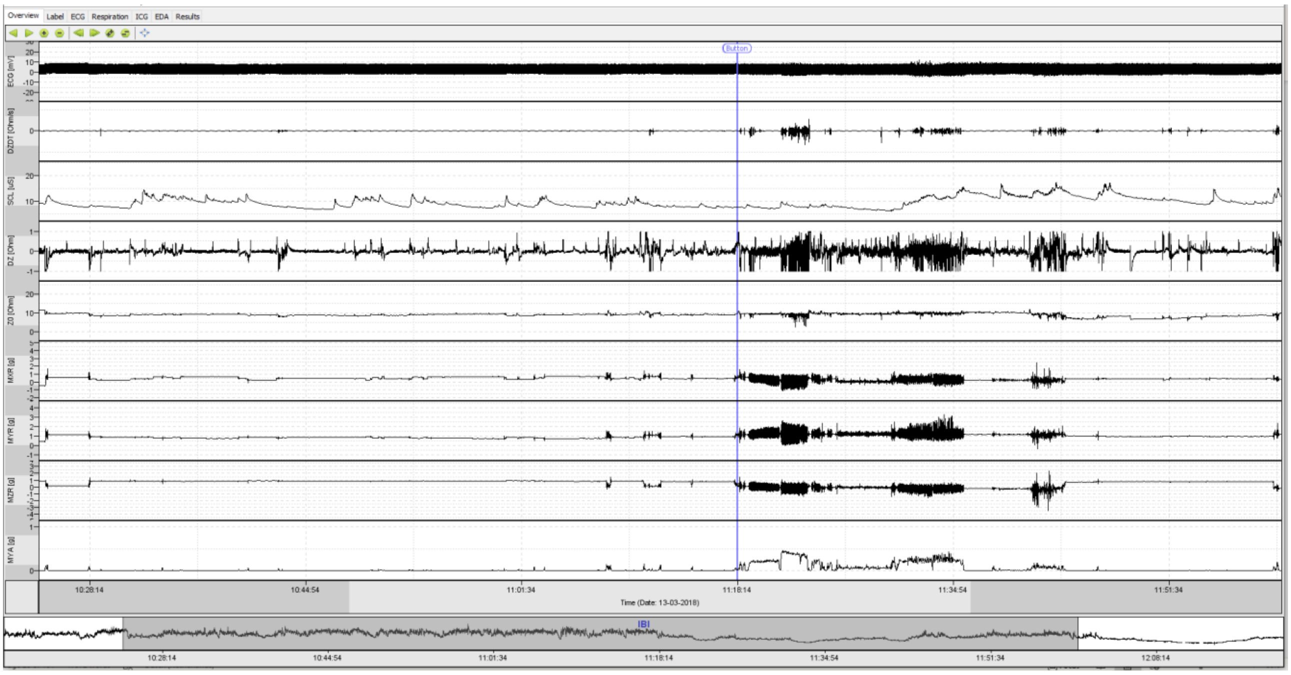
Use the DAMS program to process the VU-AMS data. Double clicking on a *.5FS* file will open the file once the DAMS program is set to be the default program to open it with.

The typical flow of VU-AMS data analysis and management is represented by a series of tabs in the main screen:



1. Overview - Inspecting the raw data
2. Label - Labeling your data
3. ECG - R-peak detection
4. Respiration – Respiration & RSA scoring
5. ICG – Impedance complex scoring
6. EDA - Electrodermal activity scoring
7. Results – Average values of variables per label

5.1 OVERVIEW

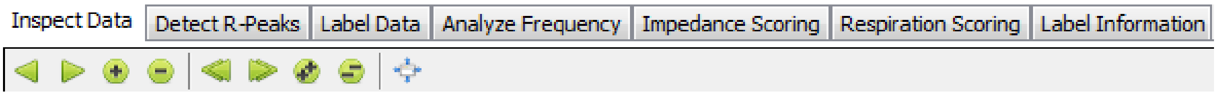


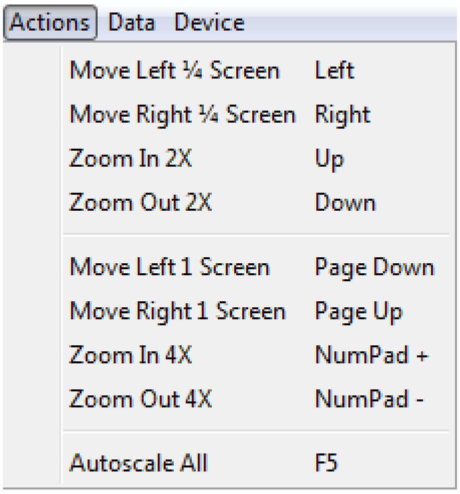
The *Overview* tab simply gives you an overview of your data. All recorded signals are shown as continuous time series. Each of the signals can be auto scaled separately by right clicking on the Y-axis of the signal. At the bottom of the screen the recording time is given. The data presented on the screen is a selected subsection of this recording time. This time of this subsection is presented on a second timeline, depicted above it. Between the two timelines, the Inter Beat Interval (IBI) time series is given as extracted from the ECG. The bottom timeline and IBI signal of the entire recording will be presented in all data analysis tabs and will function as orientation point.

By clicking and dragging either the dark grey bar in the IBI time series you can “scroll through time”. Zooming can be done by changing the size of the bar. The mouse can also be used to zoom in and out; by using the mouse wheel a shorter or a longer period of the recorded data can be shown.

Vertical blue lines with a circle containing the word “Button” above it represent the times at which the event button was pressed.

All actions for the *Overview* tab are presented in the form of buttons. The function of these buttons should be self-explanatory. Hovering the mouse cursor over a button will display a short description. These actions and their keyboard shortcuts are also available in the dropdown menu *Actions*. This setup goes for all tabs in the DAMS program. Only the type of actions will differ per tab.





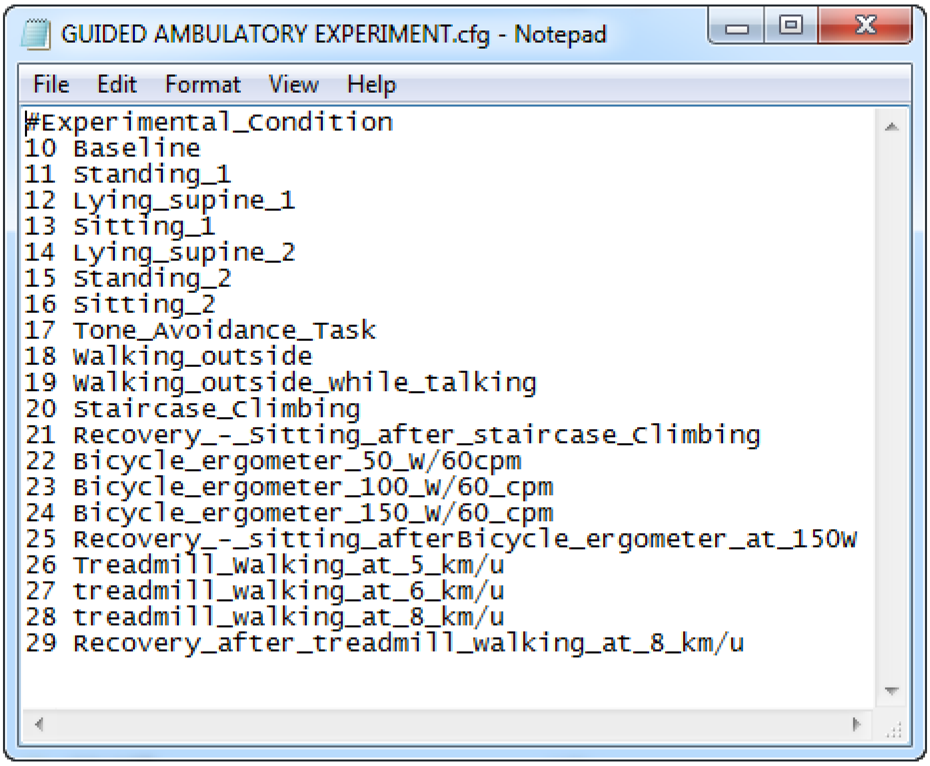
5.2 LABELING YOUR DATA

In the second tab you can divide the continuous data collected with the VU-AMS device into logical periods for further analysis. We call this labeling. The aim of labeling is to get an average value for each available parameter, such as HR, PEP and RSA, for each experimental condition or ambulatory activity. Each label will be represented by a row in the *Results tab (see section 5.7).*

> ***TIP5:*** *See the Data Labeling tutorial videos for a demonstration: www.vu-ams.nl/support/tutorials/software/data-labeling*

5.2.1 LABEL CONFIGURATION FILE

Before we can actually begin labeling the data, we need to create the blueprint for all possible labels first. This blueprint lists all the laboratory experimental conditions or ambulatory activities in our experiment. This is done in the so called label configuration file (.cfg file). The easiest way to create a label configuration file that suits your purpose is by manually creating or editing a text file.




```
#Experimental_Condition
10 Baseline
11 Standing_1
12 Lying_supine_1
13 Sitting_1
14 Lying_supine_2
15 Standing_2
16 Sitting_2
17 Tone_Avoidance_Task
18 Walking_outside
19 Walking_outside_while_talking
20 Staircase_Climbing
21 Recovery_-_Sitting_after_staircase_Climbing
22 Bicycle_ergometer_50_W/60cpm
23 Bicycle_ergometer_100_W/60_cpm
24 Bicycle_ergometer_150_W/60_cpm
25 Recovery_-_sitting_afterBicycle_ergometer_at_150W
26 Treadmill_Walking_at_5_km/u
27 treadmill_walking_at_6_km/u
28 treadmill_walking_at_8_km/u
29 Recovery_after_treadmill_walking_at_8_km/u
```

A main category is always indicated by a hash key (#) followed by the name you want for that main category, as the first line of the file. Then each level of that category, in the example below these were all the experimental conditions, needs to have a unique numerical value followed by a space and the name for each condition. Save it under a logical name with the extension *.cfg*. (Choose "All Files (*.*)" for "Save as type:")
The label.cfg file needs to be present in the same folder as the .5fs files for DAMS to read in the labels automatically upon opening the .5fs file.

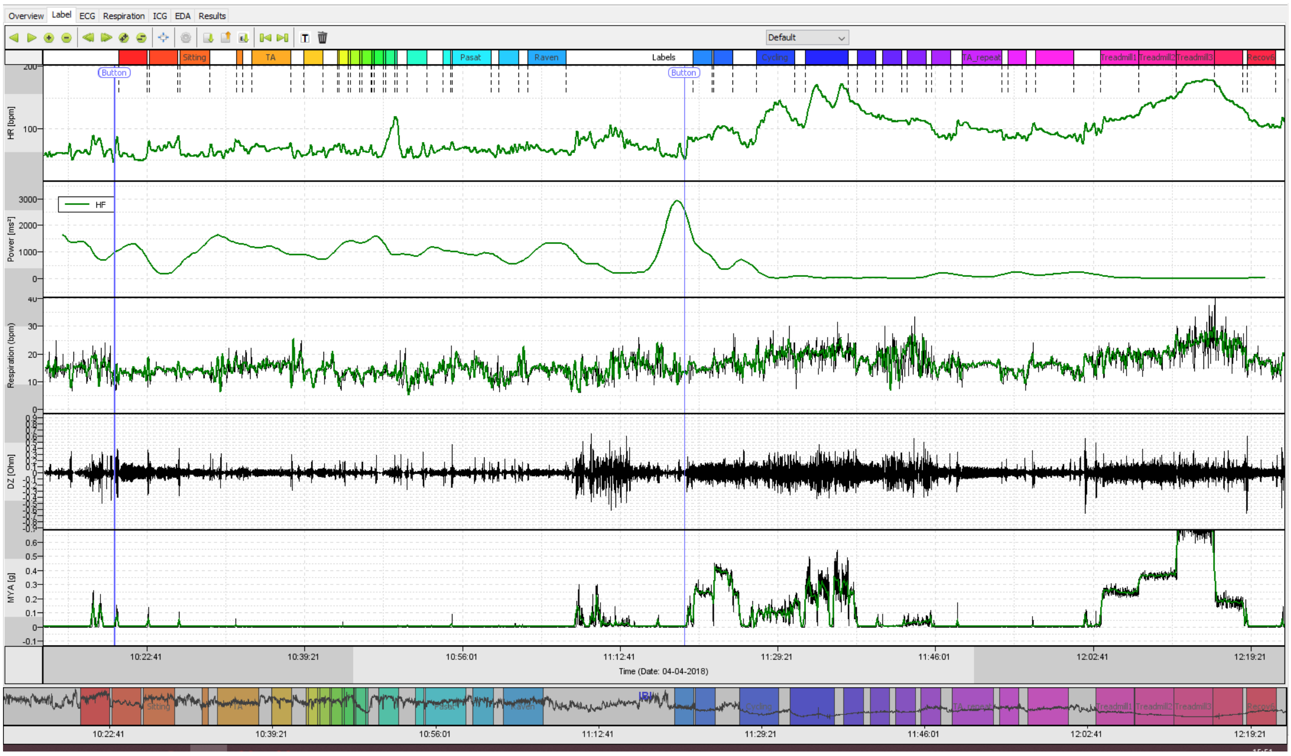
TIP6: *Make sure that the DAMS program recognizes the .cfg label file. To do this click the Label Data tab → Actions → Edit Label Configuration → Import Label Config From File. Now select the label configuration file you manually created and click open.*

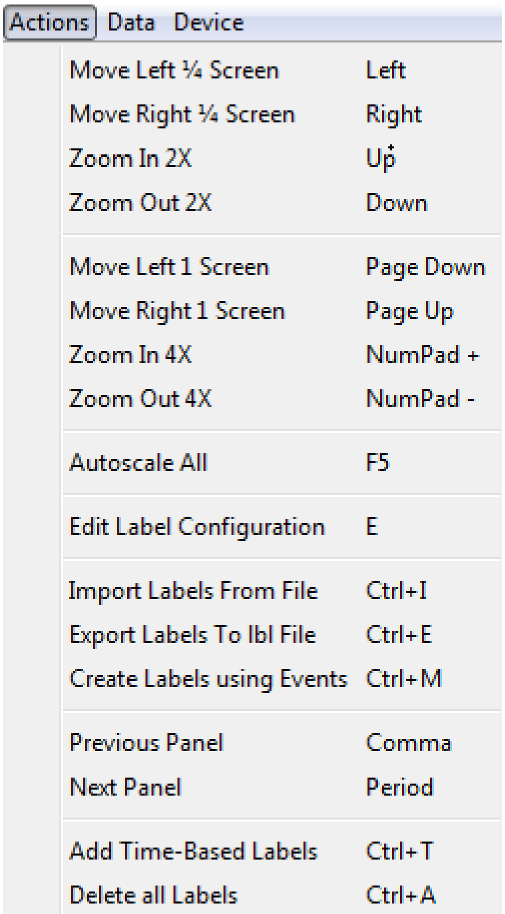
TIP7: *When the same label configuration file is used for a longer time, like for an entire research project, you could click on the option Set Label Configuration As Default.*

5.2.2 LABELING YOUR DATA

You are now ready to label your data in the *Label* tab. In the upper window of the tab we see the smoothed (and hence lightly time lagged) heart rate signal. In the second window we can see the HF-HRV (*see section 5.3.1.2 for an explanation of this signal*). In the third window you see the respiration signal in breaths per minute (*for an explanation see section 5.4.1.1 Respiration Rate*). The fourth window shows the impedance signal (*see section 5.5 Impedance Scoring*). The fifth window shows the 'motility' signal. This signal is based on the Y axis of the tri-axial accelerometer. The clock time and IBI time series are presented in the two bottom windows, which will help us to quickly locate our current position in the total recording. The signals displayed can aid in determining the start and ending times of a condition when you are uncertain. For example, when a participant goes from sitting down to standing up a peak can be observed in the motility signal.

All actions for the *Label* tab are presented in the form of buttons. The function of these buttons should be self-explanatory.



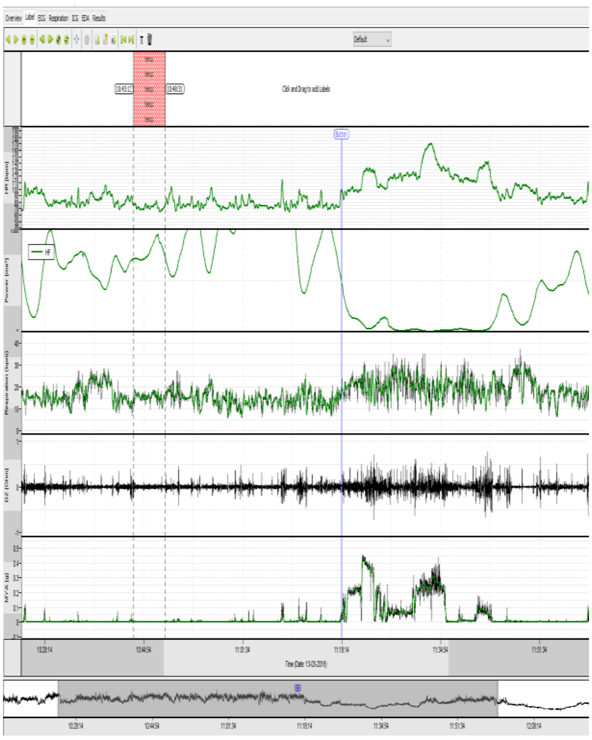


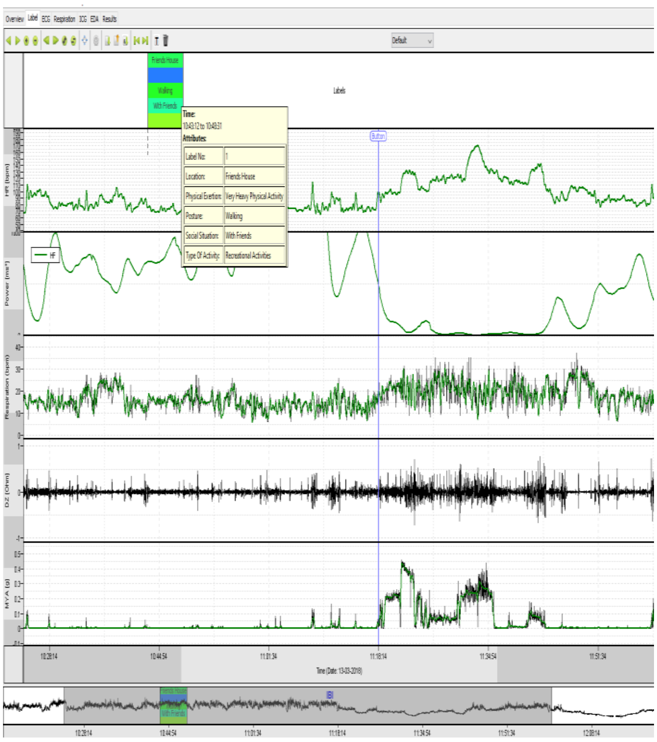
5.2.2.1 LABELING IN FIXED SETTING

There are multiple ways in which you can label your data during an experimental/fixed setting. You can have notes, markers or you made use of an external digital device to keep track of your experimental conditions. The process of labeling for each of these situations is discussed below in separate sections.

NOTES

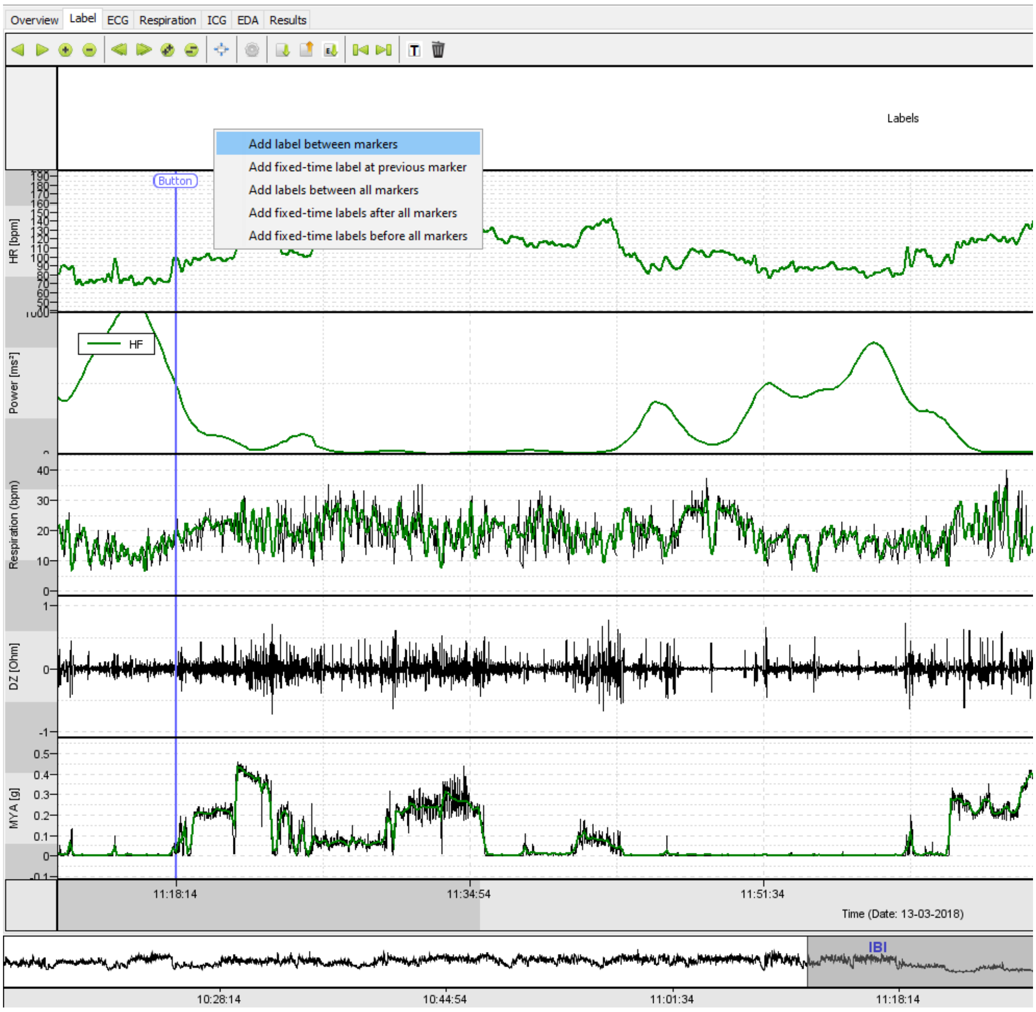
If you have notes containing the start and stop times of each condition, you can use the “Label” bar. When hovering the mouse on the bar you will see an indicator of the time window appear. This will help you place the starting point of your label accurately. Click with the left mouse button in the top bar at your starting point and drag the mouse to the right until the desired end time of the label is reached.
When releasing the mouse, a window will pop-up in which you can select the condition as indicated in the label.cfg file. When a condition is selected the pop-up window will close and a colored bar appears representing the new label. You can verify the information in the label by hovering with the mouse on top of it.





MARKERS

In case of an experiment, you have the option to use the event button of the VU-AMS device to mark the beginning and ending of each condition. Shortly pressing the event button on the VU-AMS device creates a marker in the data at the exact time the button was pressed. These markers are displayed as green lines with a circle contained a here and help us to more easily detect the start and stop times of each experimental condition. If markers are present you can click on the on the "Label" bar in-between the two markers you want to add your label.



A menu will pop-up from which various options appear. To label the region between two markers choose the first option "add label between markers". When selected, a window will pop-up in which you can select the condition as indicated in the label.cfg file.

Another way to automatically place labels using marker information is to simply create a formatted text file and import this file by clicking the *Label Data* tab →

Actions → Create Labels using Events. One can also use this in batch mode when choosing *File → Batch convert data files*

The file always starts with the header line "*SM, EM, D1, D2, LC*". The first column corresponds to the Start Marker (SM), an integer specifying which marker to use as a reference for the label. The second column corresponds to the End Marker (EM), and is only used to specify a label between the SM and the EM, in which case it is an integer specifying at which marker the label ends, otherwise it is set to missing (-9999). The third column is Duration1 (D1), meaning the interval from desired label start to the SM in seconds (where negative values indicate the label should start *after* the marker, and positive values have the label start *before* the marker). The fourth column is Duration2 (D2), meaning the interval from the SM to the desired label end in seconds. The fifth column is the Label Code (LC), or the integer describing which experimental condition the label corresponds to. These variables can be used to generate labels in several ways. An example text file used for automated labeling is shown below:

CreateLabelsUsingMarkers.txt

```
1  SM, EM, D1, D2, LC
2  1,-9999,-1,3,10
3  1,-9999,-4,5,11
4  2,3,-9999,-9999,12
5  4,-9999,1,1,13
6  5,-9999,2,-1,14
7  5,-9999,-1,2,15
8
```

In the above example:

- Lines 2 and 3 create labels 10 and 11 after marker 1.

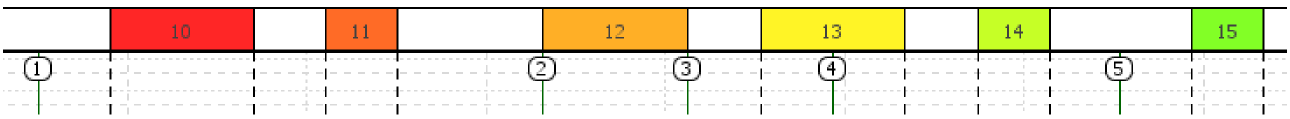
- Line 4 creates label 12 in between marker 2 and 3.

- Line 5 creates in label 13 around marker 4, one second before (D1) and one second after (D2).

- Line 6 creates label 14 before marker 5.

- Line 7 creates label 15 after marker 5.

This would result in the labels 10 to 15 below.

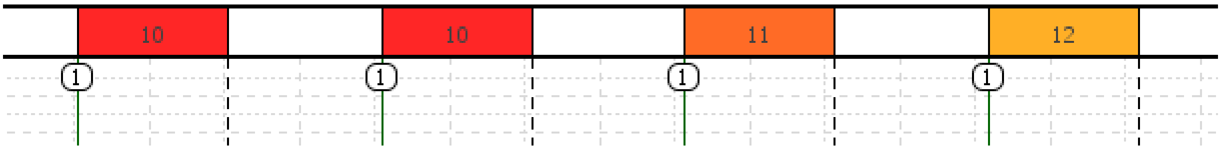


In recordings with non-unique marker codes, e.g. when the markers originate from a button press on the VU-AMS device and all have marker code 0, one can also repeat text lines to place the same (or different) label after a later instance of that marker like in the example below. Note a to-be-placed label will only be placed if the middle of the to-be-placed label is not on an existing label.

CreateLabelsUsingMarkers2.txt

```
SM, EM, D1, D2, LC
1,-9999,0,1,10
1,-9999,0,1,10
1,-9999,0,1,11
1,-9999,0,1,12

```

This would result in the labels 10 to 12 below.



NOTE: *Text files in any of the three formats which were supported in previous versions of VU-DAMS are still accepted for backwards compatibility.*

NOTE: *In laboratory studies that use a stimulus computer, this computer can be used to send markers through the VU-AMSi infrared interface cable. Please refer to VU-AMS website: www.vu-ams.nl/support/downloads/extras*

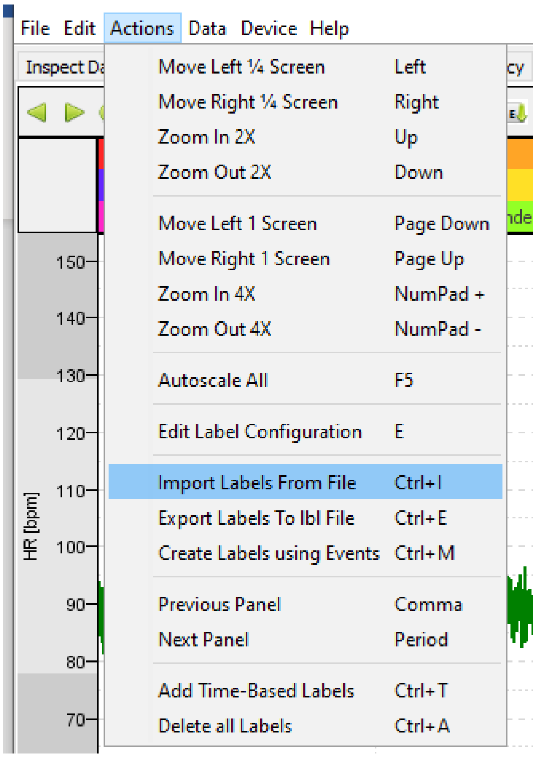
_ICG.LBL FILES

It is also possible to read in a predefined file that contains all the label information. Such a file is particularly useful when the timing of the conditions is kept track of in an external digital device (for example a time stamp application on a smartphone).

```
Label
 14-12-17/10:26:47  14-12-17/10:42:39   1
 14-12-17/10:42:40  14-12-17/10:47:40   2
 14-12-17/10:47:41  14-12-17/10:49:41   3
 14-12-17/10:49:42  14-12-17/10:54:42   4
 14-12-17/10:54:43  14-12-17/11:42:25   5
 14-12-17/11:42:26  14-12-17/11:47:26   6
 14-12-17/11:47:27  14-12-17/11:49:27   7
 14-12-17/11:49:28  14-12-17/11:54:28   8
 14-12-17/11:54:29  14-12-17/12:46:54   9
```

To file needs to adhere to a specific format to be read into DAMS and needs to end fit the suffix _icg.lbl (see example above). The file needs to start with word “Label”. On the next rows the information on the start and end time of the labels and the numbers corresponding to the conditions as specified in the label.cfg file are given in the following format: The start time is preceded by a space. The start date and end date are separated by two spaces and need to adhere to the format: dd-mm-yy/hh:mm:ss. Condition number are preceded by 3 spaces. The file needs to end with 4 empty rows.

When the _icg.lbl file is saved under the same name and is present in the same folder as .amsdata file it will be read in automatically when opening the file. A file can also be manually red in using the *Actions > Import Labels From File option* in the menu.



5.2.2.2 Labeling Naturalistic Recordings

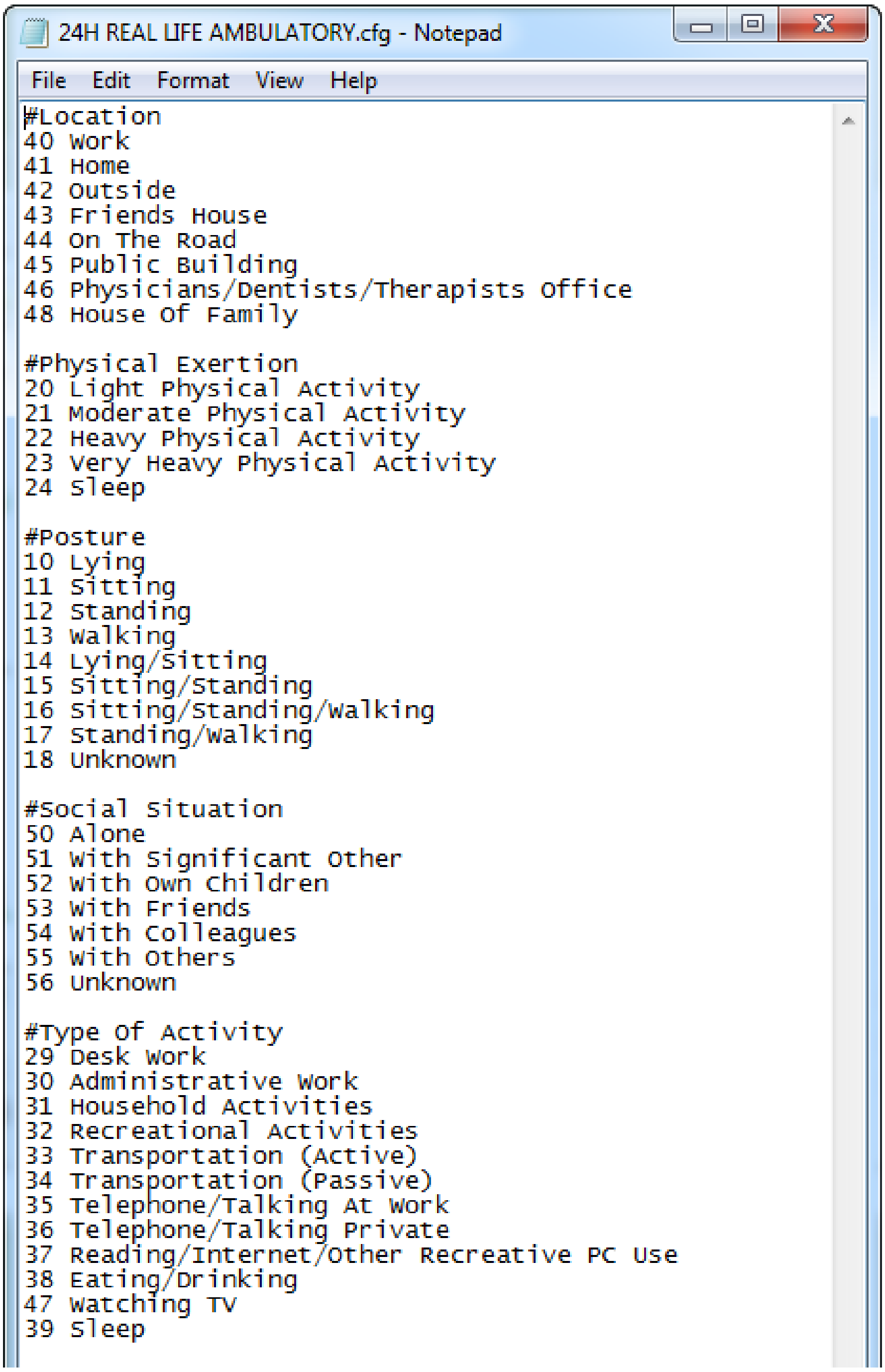
The previous example assumed a supervised setting with fixed experimental conditions of which start and end times were under the experimenter’s control. This is of course not the case in ambulatory recordings in naturalistic settings. Still, many of the principles apply, as “real life” can also be divided into periods of ‘fixed’ and frequent occurring activities.

To create a label file that will capture most daily activities of the subjects in your target population is a crucial but feasible step in ambulatory data analysis. As the autonomic nervous system is highly sensitive to changes in posture and physical activity it is first of all very important to obtain information about these aspects of a subject’s daily routine. This is usually done by asking all subjects to keep a detailed diary (paper-and-pencil or electronic hand held devices) during the ambulatory measurement day. This diary information about (changes in) posture and physical activity is then used during the labeling procedure. The VU-AMS device also contains a tri-axial accelerometer to support this self-report with objective data. Further categories to be used in the ambulatory labels will entirely depend on the research question and the target population. Here are some examples of categories that could be considered:

Category	Levels of Category
Type of activity	Reading, attending a meeting, eating, dancing, PC work, conversing, watching TV, exercising, attending a musical, driving a car, ironing, etc.
Posture	Sitting, standing, lying, walking, etc.
Physical activity	Heavy, moderate, light, none
Social situation	Alone, with significant other, with friends, with colleagues, etc.
Location	At home, at the office, travelling, at a restaurant, etc.
Time of day	Work, leisure time, sleep
Mood state	Angry, friendly/happy, sad, anxious, tired

When you have decided on the type of categories you want to use during labeling you again need to summarize these in a *label configuration file*, just like you would do for the fixed labels (*see section 5.2.1 Label Configuration file*). However, for ambulatory recordings the label configuration file is extended by adding multiple categories. To name a new category you just add a new section that starts with a #

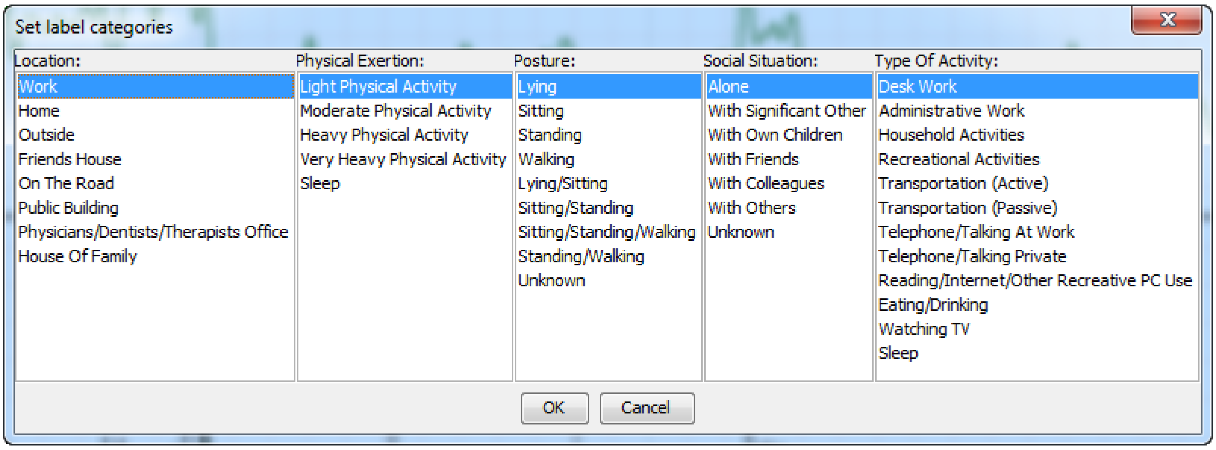
followed by the name of the category and below give the numbers and names for the different “conditions”. An example for a 24-hour recording project is given in the figure below.




```
#Location
40 Work
41 Home
42 Outside
43 Friends House
44 On The Road
45 Public Building
46 Physicians/Dentists/Therapists Office
48 House Of Family

#Physical Exertion
20 Light Physical Activity
21 Moderate Physical Activity
22 Heavy Physical Activity
23 Very Heavy Physical Activity
24 Sleep

#Posture
10 Lying
11 Sitting
12 Standing
13 Walking
14 Lying/Sitting
15 Sitting/Standing
16 Sitting/Standing/Walking
17 Standing/Walking
18 Unknown

#Social Situation
50 Alone
51 With Significant Other
52 With Own Children
53 With Friends
54 With Colleagues
55 With Others
56 Unknown

#Type Of Activity
29 Desk Work
30 Administrative Work
31 Household Activities
32 Recreational Activities
33 Transportation (Active)
34 Transportation (Passive)
35 Telephone/Talking At Work
36 Telephone/Talking Private
37 Reading/Internet/Other Recreative PC Use
38 Eating/Drinking
47 Watching TV
39 Sleep
```

In the label configuration file, all ambulatory activities you might want to use as separate conditions in future analyses should be present as all further processing in the DAMS program is tailored to the labels.

Similar to labeling fixed conditions there are several options to label your data (see 5.2.2.1 Labeling in fixed setting). However, when labeling you can now select a condition from each of the categories you created in the pop-up window.



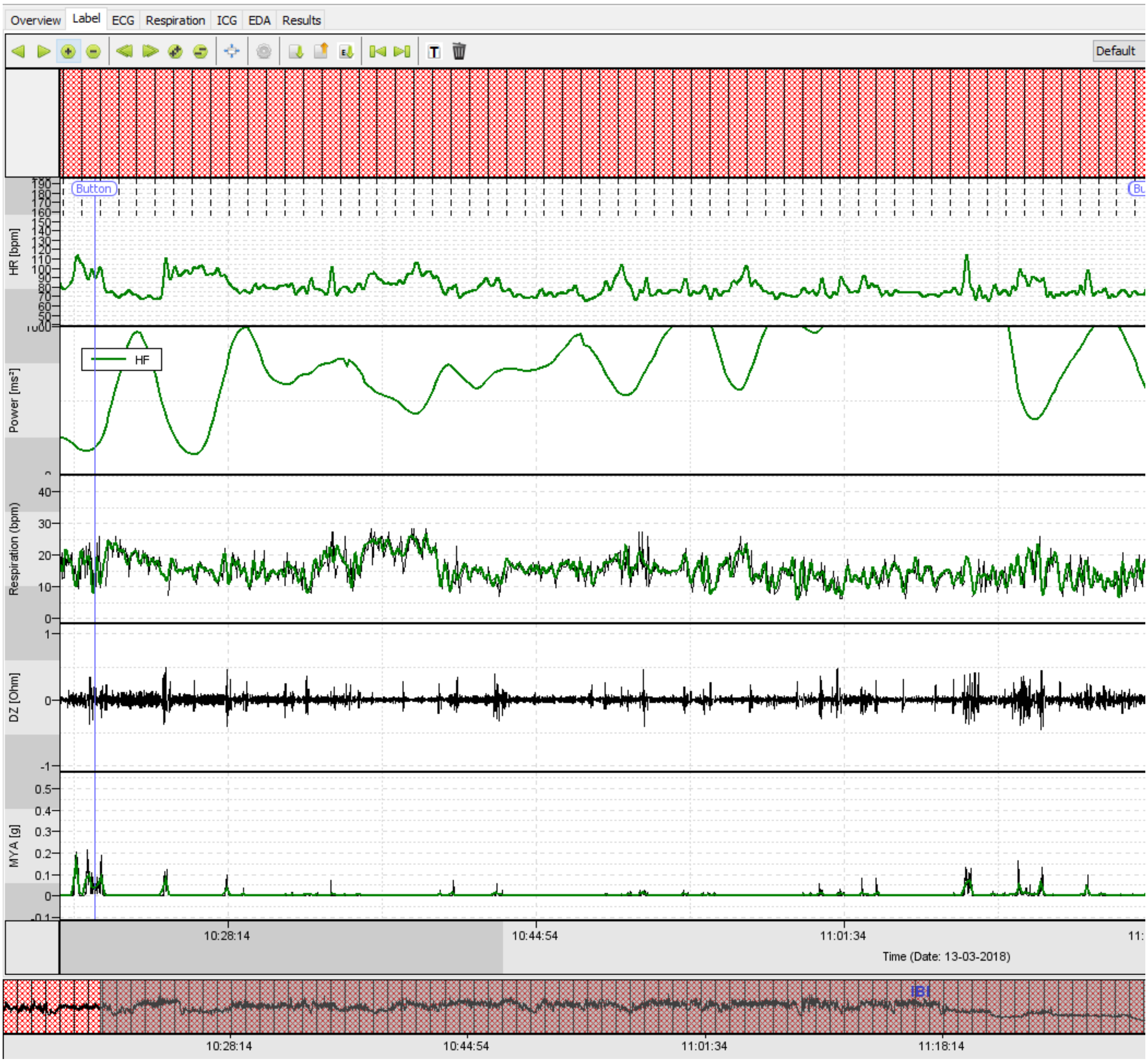
When creating a _icg.lbl file the numbers of the number of additional categories can be added after the first number preceded by three spaces.

```
Label
 19-01-18/10:34:25  19-01-18/12:37:25   40   21   17   54   33
 19-01-18/10:37:26  19-01-18/13:15:46   40   20   11   54   38
 19-01-18/13:15:47  19-01-18/15:58:11   40   20   11   54   29
 19-01-18/17:15:43  19-01-18/18:21:58   42   21   16   50   33
 19-01-18/18:21:59  19-01-18/18:50:15   41   20   11   51   38
 19-01-18/20:15:34  19-01-18/22:48:37   41   20   11   51   47
```

TIME-BASED LABELS

Instead of predefined category labels representing a specific condition, you can also choose to create time-based labels, where all labels represent a fixed amount of time that can last in duration between 10 seconds and 1 hour. This is especially useful if you want to generate continuous 60 sec ensemble averages of the impedance cardiogram for PEP scoring instead of a grand ensemble average that spans an entire condition.

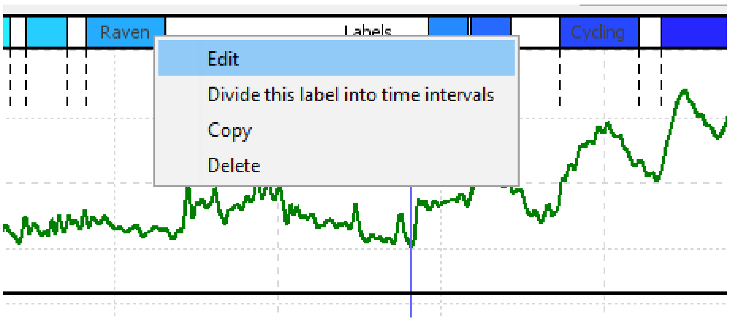
To do this click the *Label* tab → *Actions* → *Add Time-Based Labels* and type in the number of seconds you want each label to be. The entire data is now cut up into labels with your pre-defined length. The experimental label information, if present, can optionally be included in these time-based labels.

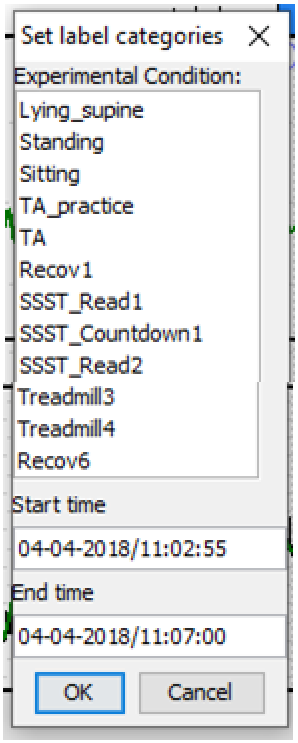


TIP8: *Make sure you have labeled all periods that might be considered of interest. There seems to be no urgent need to also label periods that you consider to be "irrelevant" or that are expected to occur in only a few subjects. However, we advise to always label the entire 24 hour recording as completely as possible. With an average length of labeled periods of 20 minutes this would result in about 72 labels per subject.*

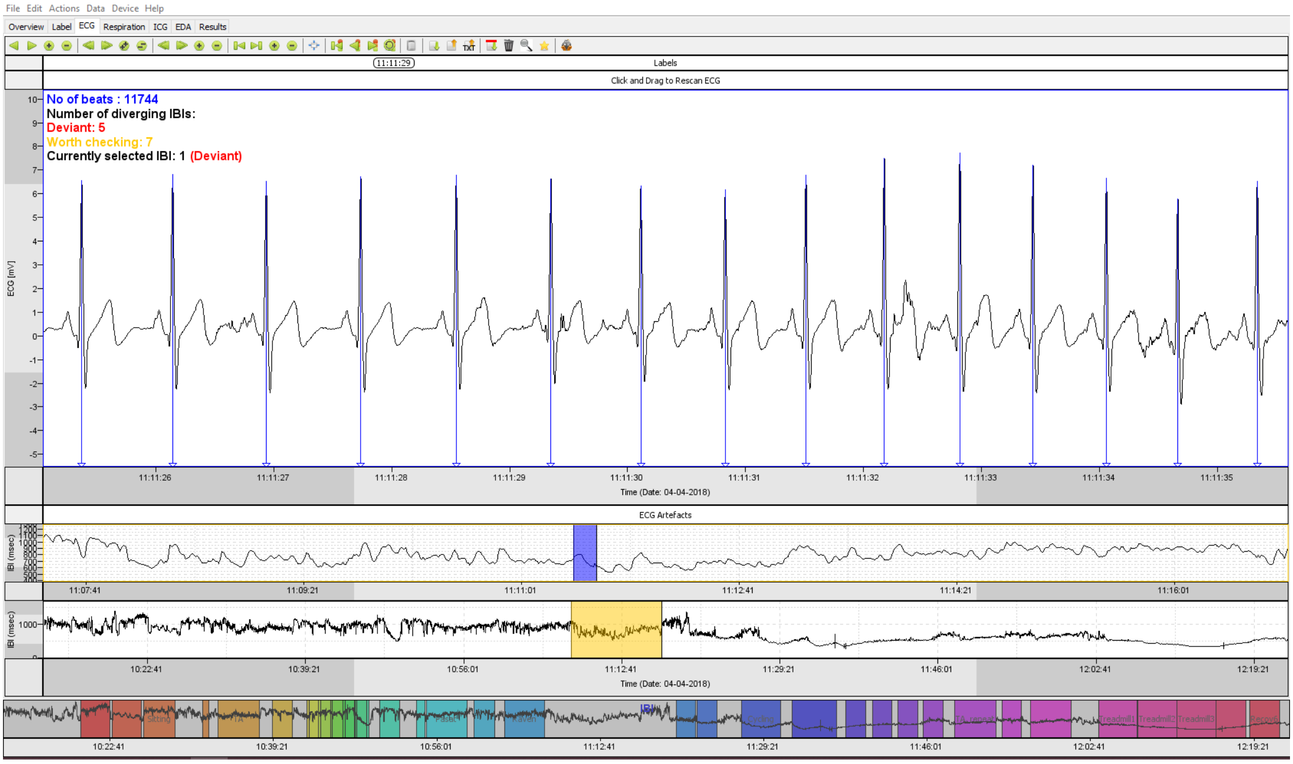
5.2.3 EDITING LABELS

If you want to edit or delete a label, just right click on it and choose the required action from the popup menu. If you want to change the length of the label click and drag with the mouse on the edges of the label. To reposition a label, click and hold the left mouse button on the label to grab and move it around. However, if you want to change the start/end time with millisecond precision, we advise to adjust the times at the bottom of the label popup menu.





5.3 ECG



As with all other tabs, the clock time and IBI time series are presented in the two bottom windows, to quickly locate our current position in the total recording. To further ease navigation, the labels (as labelled in the *Label* tab, *section 5.2*) are printed on top of the IBI time series.

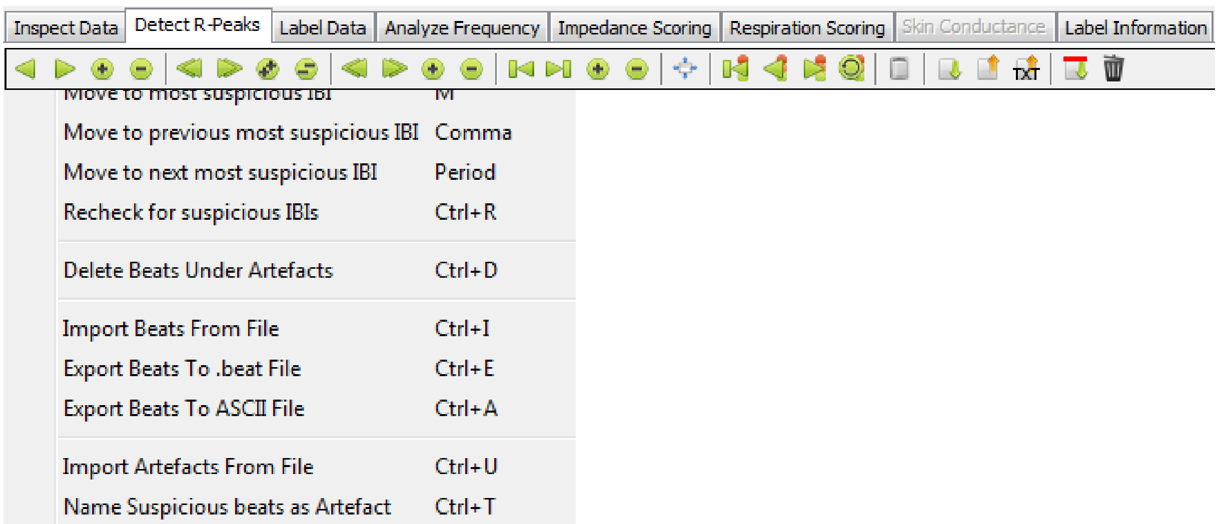
Because the time frame of a single heartbeat is in the order of milliseconds, and complete recordings can be 24 hours or longer, two additional navigation bars are present, containing colored rectangles. Each colored rectangle corresponds to the signal segment drawn in the frame above it. By clicking, dragging, and resizing these rectangles you can navigate to the desired segment and resolution.

The *ECG* tab will assist you to create an artefact free IBI signal as quickly as possible by applying automated artefact and peak detection. Visual inspection and correction of the resulting IBI signal is facilitated by using automated detector for diverging beats, the QRS detector.

The QRS detector runs three separate automated analyses on the ECG signal. The first automated analysis detects and marks periods with missing data or clipping of the electrocardiographic signal. These periods are called artefacts. A second automated analysis of the QRST waveform detects the occurrence of all R-peaks. Each R-peak detected by the algorithm is marked in the top frame with a line (either blue, yellow or red).
The R-peaks are converted to the inter beat intervals time series which is simply the time in milliseconds between two consecutive R-peaks, and plotted as a continuous line in all navigation bars. The third automated analysis checks the plausibility of the duration of each IBI in the context of its surrounding inter-beat intervals.

> ***TIP9:*** *When your data is noisy or your participant has a non-standard ECG the default setting of the peak detection algorithm might need to be adjusted. This is addressed in Appendix A.*

It is important to note any unexpected deviation in time between two consecutive beats is marked divergent by the algorithm. However, when a beat is marked red, this does not necessarily mean that the beat is invalid and something that needs to be corrected. This will be addressed further in *section 5.3.3* Suspicious beat correction.
All actions for the *ECG* tab are presented in the form of buttons. The function of these buttons should be self-explanatory. Like in all other tabs hovering the mouse cursor over a button will display a short description. The last set of options is useful for assessing suspicious beats and to export beat-to-beat data.



5.3.1 VARIABLES

Several variables are derived from the R-peaks signal: heart rate (HR), inter-beat-interval (IBI) and heart rate variability (HRV). For each variable an average per label is calculated (*see section 5.2.2 Labeling your data*). The values are displayed in the *Results tab (section 5.7).*

HR/IBI

HR and IBI are two measures of the same construct. The HR is the number of R-peaks which occurs in one minute (beats per minute), while the IBI is the time between two consecutive beats in milliseconds. HR can be derived from IBI by the formula: HR = 60000 / IBI. HR is the oldest known index of ANS activity. Each heart beat is initiated by the sinoatrial (SA) node, the “pacemaker” of the heart. The SA node has an intrinsic rate between 60 and 100 beats per minute, varying depending on an individual’s age and gender (Kashou, Basit, & Chhabra, 2020). However, resting state HR is usually lower than this intrinsic rate due to tonic parasympathetic influences on the SA node. When PNS activity is high it acts as a “brake” on the intrinsic rate by lowering it, while this brake almost disappears when PNS activity is low (Porges, 2011). However, the rate of the heart at any given moment in time is determined by the interplay between the sympathetic and parasympathetic nervous system on the SA node. Increased SNS activity leads to an increase in HR while increased PNS activity leads to a decrease in HR (Kashou, Basit, & Chhabra, 2020; Smith, Thayer, Khalsa, & Lane, 2017). Due to the mixed effects of both SNS and PNS activity on the resulting HR, it can be used as an index of arousal but the relative contribution of these separate ANS branches cannot be detangled.

HRV

A frequently used measure of PNS activity is HRV. As its name implies it represents the variability in the time interval between successive heartbeats. Due to the fast temporal kinetics of the parasympathetic signaling (<1s) at the SA node, changes in PNS activity are detectable on a beat-to-beat scale. Sympathetic signaling on the other hand has much slower effect (>5s) and cannot change HR so quickly (Chapleau & Sabharwal, 2011). Therefore, fast changes in HR can be attributed to changes in

PNS activity. Because PNS signaling to the heart is delivered by the 11th cranial nerve, called the vagus nerve, PNS activity is often referred to as vagal tone. PNS activity can be indexed by changes in HRV in both the time domain and the frequency domain.

TIME BASED

Time domain measures quantify the amount of variability in the IBI over time (Shaffer & Ginsberg, 2017). DAMS calculates two time-based measures of HRV: the root-mean-square of successive differences (RMSSD) and the standard deviation of the N-N interval (SDNN).

RMSSD is calculated by taking the difference between successive R-R intervals in milliseconds, squaring these differences to make negative values positive and then taking the root. The average value for all successive IBIs in a label is the final RMSSD. SDNN is calculated by taking the standard deviation of a set of IBIs in milliseconds. The term normal-to-normal (N-N) interval is just another name for the R-R interval.

FREQUENCY BASED

Spectral analysis is performed on the IBI time series according to obtain high frequency (HF) and low frequency (LF) HRV by the following method:

The IBI time series within each label is interpolated with a cubic spline and the resulting function is resampled at 4 Hz. The resampled signal is split into overlapping periods of 256 seconds, each with 1024 data points. The overlap between two consecutive periods is 128 seconds. Periods that have intervals longer than 5 seconds without IBIs are discarded. Missing data from the final period are padded with zeros. Each period of 1024 data points is convoluted with a smoothness prior matrix (see: An advanced detrending method with application to HRV analysis, Mika P. Tarvainen, Perttu O. Ranta-aho, and Pasi A. Karjalainen) to yield a stationary signal on which a discrete Fourier analysis is performed after additional convolution with a quadratic window. Power values for each of the 1024 data points are then averaged across all available periods in the condition (Welch method). Next the total power in the 0.0001 Hz to 0.4 Hz range is computed (TP) as well as the power in the 0.04-0.15 Hz band (LF) and the 0.15-0.40 Hz band (HF).

LF power is caused by blood pressure oscillations affecting both SNS and PNS cardiac control. HF power is caused by the effects of respiration on PNS. It is important to note that the IBI time series is first detrended and ‘corrected’ by interpolation to deal with too short and too long IBIs (e.g. in case of an extrasystolic beat) because slow trends as well as strongly deviant beats can distort the spectrum. Because at least 3 minutes are required to obtain a reliable estimate of the LF power, these values are not supplied for labels with a duration shorter than 3 minutes.

TIP10: *In case you want to change the default settings of the power spectral analysis select Edit → Settings → Frequency Analysis. Settings for preprocessing affect detrending and deviant beat removal ('artefact'). The frequency bands reflect the typical bands now commonly used in literature, but can be changed if desired (also see appendix A).*

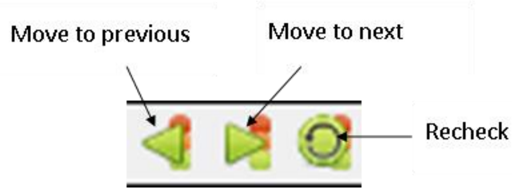
5.3.2 VISUAL INSPECTION AND MANUAL CORRECTION

Automated artefact labeling reliably detects clipping and signal loss, but detection of noisy ECG is not perfect. Manual selection of bad ECG signal parts may be additionally needed. The three main windows will help you select the parts of the IBI time series that need to be manually labeled as artefacts. The *bottom window* is our overview of the IBI signal of the entire recording that is also present in the other tabs of DAMS.

TIP11: *view the "R-Peak Detection" tutorial video for a demonstration: www.vu-ams.nl/support/tutorials/software/r-peak-detection*

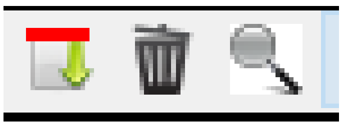
In the main window with the ECG signal the detected R-peaks are marked by vertical lines, mostly blue. A blue line means that the beat was considered to be correct according to the automatic beat detector. Potential mistakes in automated beat detection are termed 'diverging IBIs' and are flagged by a red (deviant) or yellow (worth checking) color. The red marker beats are strongly divergent from the algorithm, while the yellow ones are only slightly divergent.
A useful strategy is to navigate through the most divergent beats to check and possibly adjust them until you reach several beats in a row which have been

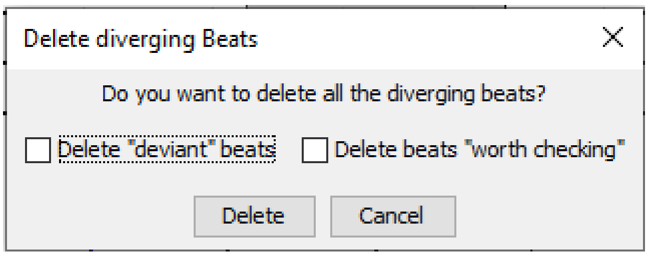
detected correctly. You can navigate from most diverging to least diverging beat by pressing the period or > key on your keyboard. You can go backwards by using the comma or > key. You can also use the following buttons:



After your corrections (or in fact at any time) you can select the menu button *re-check deviant IBI`s* by clicking the circle with traffic light (or press ctrl + R) to see how many beats are still considered deviant. ***Just remember that deviant is not wrong per se!***

TIP12: *In case of a very noisy long recording (24 hours for example), that would take too much time for the researcher to delete manually, VU-DAMS offers the option to mark all deviant beats as artefacts. You can find this feature by clicking the garbage can icon.*





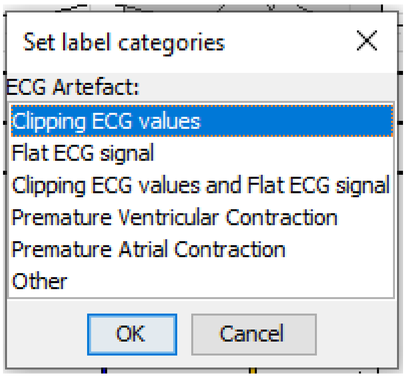
A deviant beat can have several origins which will be discussed in the following order:

1) Noisy data
2) Misplaced R peaks
3) Physiological artefacts

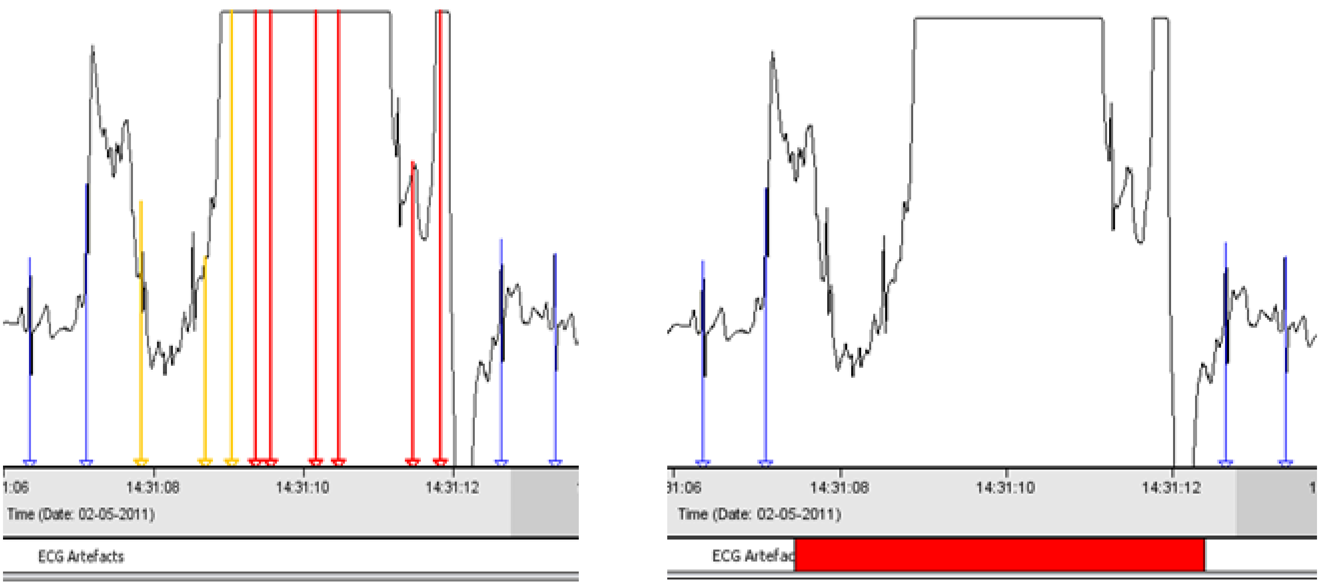
5.3.2.1 NOISY DATA

Below the ECG signal time axis is the bar named *"ECG Artefacts".* All automatically detected artefacts by QRS detector software are labeled by a red bar. However, some parts of the data might not be bad enough to be detected by the automated artefact detector, while already too noisy for the R-peak detector to function properly. Hence the R-peak detector will try to make something out of noise and may still score occasional beats as being correct (blue) where they are not. These periods of noisy data will also contain a lot of red and yellow lines which makes them easy to detect.

The origin of the noise can be caused by either rigorous movements, detached electrodes or other technical difficulties. However, we handle this data in a similar manner: manually labeling them as artefacts. To mark the bad ECG as an artefact, click with the left mouse button in the artefact bar and drag it from left to right until it covers the entire period that you want to be marked as an artefact. When you let go of the mouse the following window will pop-up from which you can select the origin of the artefact (when unsure choose the option “Other”). All beats that fall within an artefact are deleted from all further data analyses. However, when <5 successive beats are deleted, the VU-DAMS software interpolates the IBI times using a cubic spline.

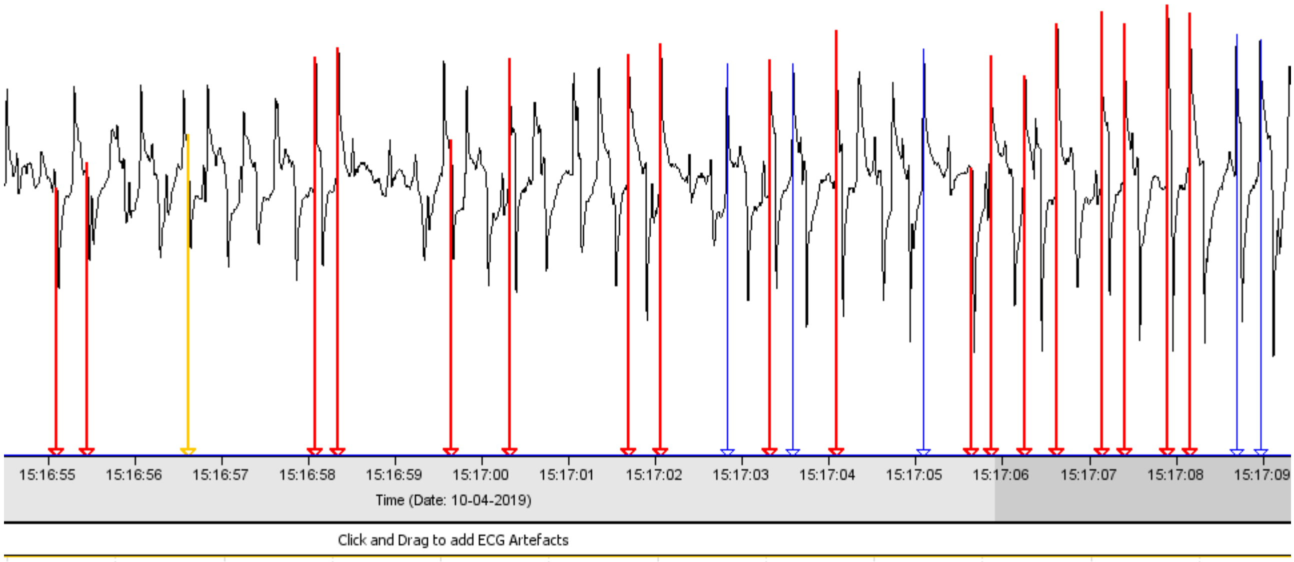


Example of a clipping signal caused by an electrode temporarily coming loose.



5.3.2.2 MISPLACED R PEAKS

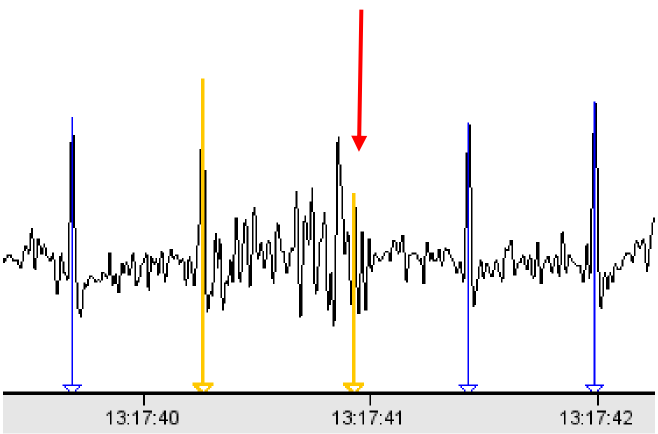
Example of noisy data due to movement of the torso:



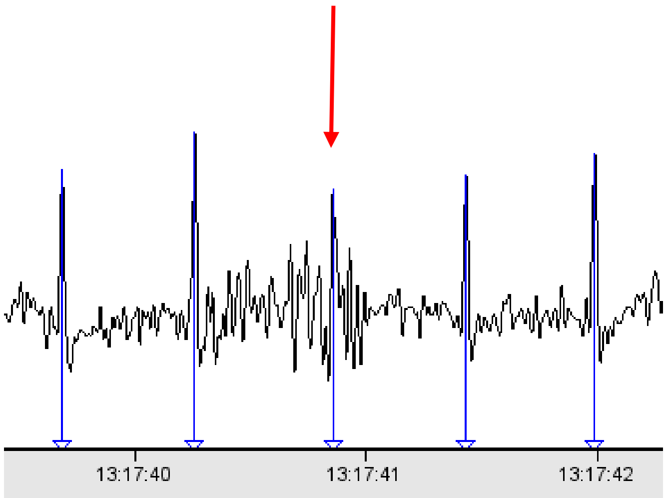
> ***TIP13:*** *when removing large stretches of data containing artefacts, zoom out so the entire artefact period is visible in your screen.*

5.3.2.2 MISPLACED BEATS

When encountering a misplaced divergent beat several actions can be performed. You can delete a beat by right clicking on it, e.g. when a beat was placed in an obvious wrong location in between beats. Or you can add a beat by left clicking on the correct location of the R-peak, e.g. when a beat was completely missed. You can also move a beat by placing the left mouse button on the vertical line and then move it left or right. Releasing the mouse button will lock the vertical line to its new location. Notice that the surrounding beats might also change color when adding, deleting or moving a beat (see example below in which a yellow beat was moved slightly to the left).



Before



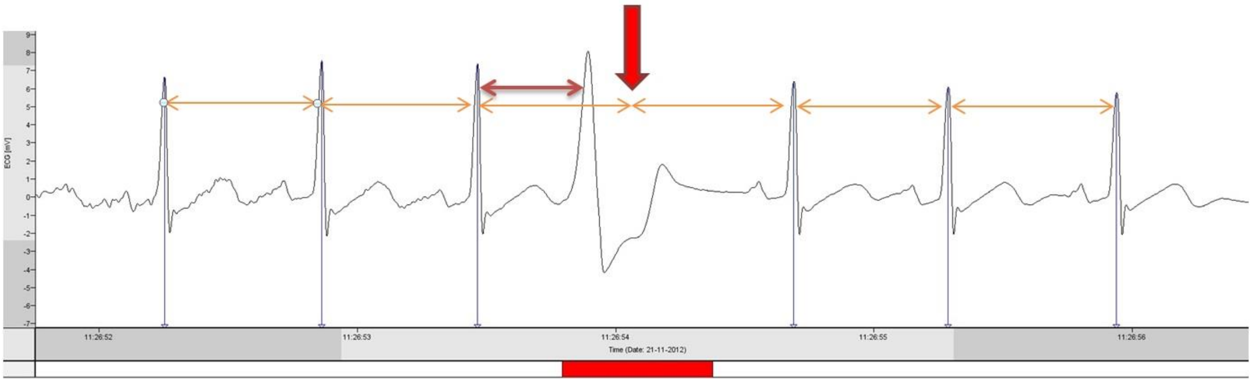
After

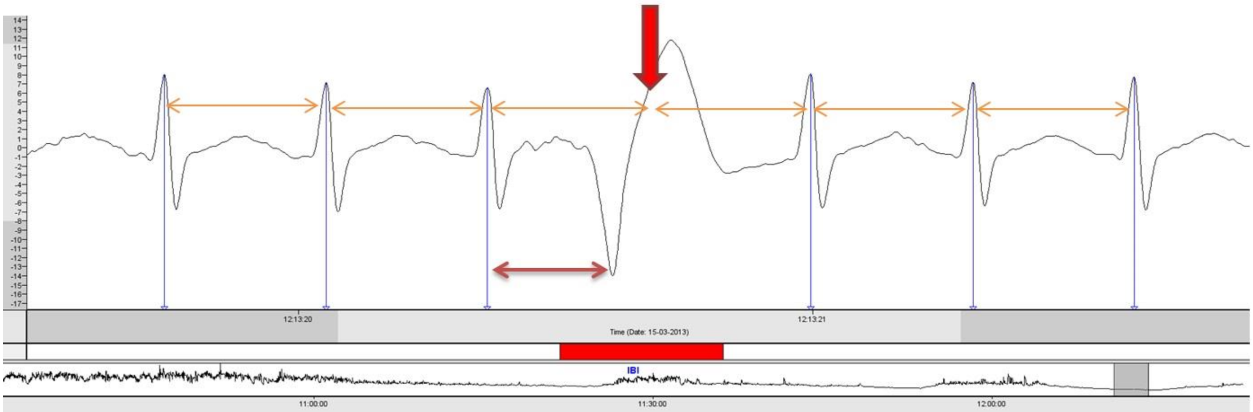
5.3.2.3 PHYSIOLOGICAL ARTEFACTS

In HRV analysis, the aim is to examine the sinus rhythm modulated by the autonomic nervous system. Hence, physiological artefacts should be excluded from analysis by marking them as artefacts. Physiological artefacts include ectopic beats and arrhythmic events or more generally: beats not originating from the sinus node.

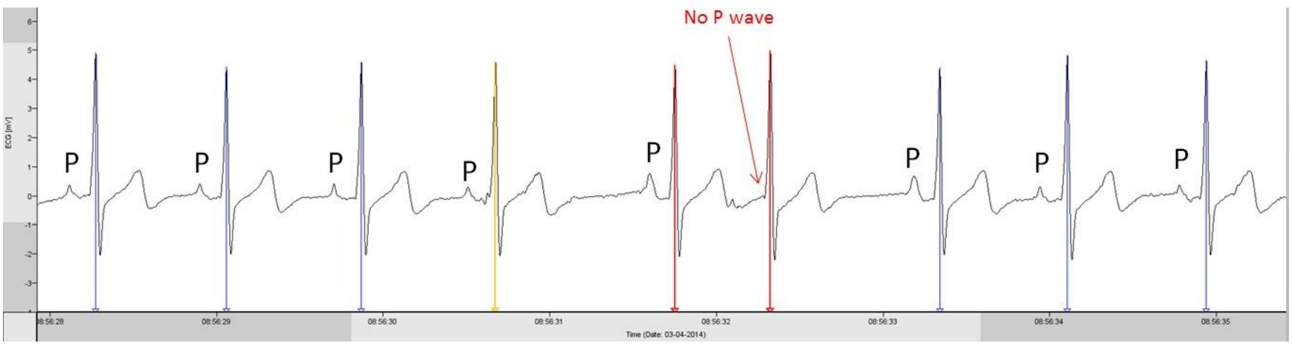
Normally, the electrical activity starts in the sinus node; the natural pacemaker of the heart. Electric current will travel through the atria and then is momentarily delayed in the AV node as a result of slow conduction. Thereafter, the current travels through the bundles of His to depolarize the ventricles in an organized manner (starting at the apex). This normal, organized electrical activity will result in the known P-QRS-T morphology.

Generally, there are two main types of ectopic beats; those originating from the ventricles and those originating somewhere else. An ectopic beat originating from the ventricles is called a premature ventricular contraction (PVC). You can recognize these beats by an early, broad QRS complex. Below, two examples of PVCs are shown (both are PVCs, the different morphology is due to a different site where the PVC originates in the ventricles). You can see the beat is early (premature) since, looking at the surrounding IBIs, it was expected at the big red arrow pointing downwards. Also, The QRS complex is clearly wider compared to the sinus beats.





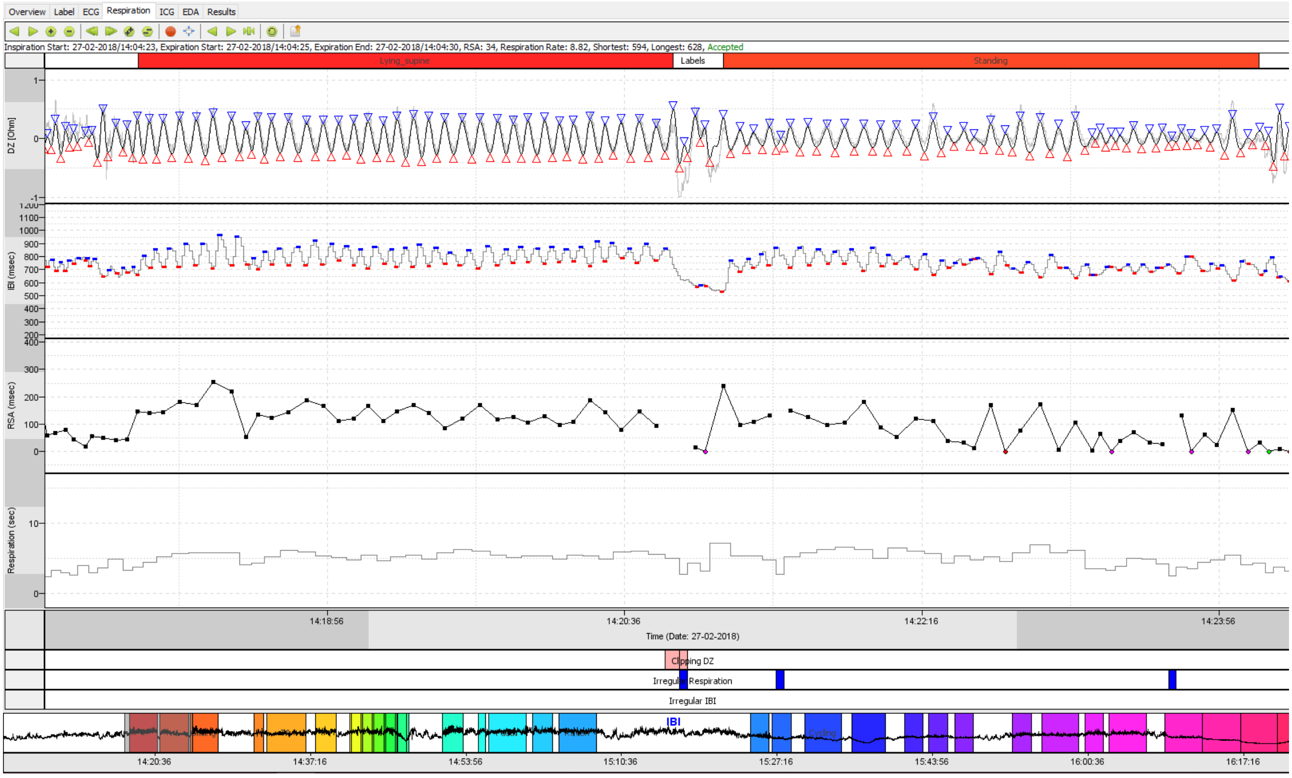
When an ectopic beat originates above the ventricle, the electrical current will travel through the bundles of His down the ventricle, thus giving a normal narrow QRS complex. When one sees a short IBI and the P wave (representing the depolarization of the atria) has a different morphology or is not present at all, this is most likely a premature atrial contraction (PAC). Below, an example is shown.



TIP14: *You can also choose to export the interbeat time interval series to a text file for use in different software packages like the CarSpan or Kubios by clicking on the menu buttons 'Export Beats To ASCII File'. Chose the appropriate directory and file name and save the IBI time series as a text file.*

VU-DAMS gives an in option in the artefact pop-up window option to mark these beats as ‘premature ventricular contraction’ or ‘premature atrial contraction’. The count of PVC’s or PAC’s per label (if present) are included in the final output.
Note that, when there is a period of a lot ectopy, you might have to delete the period entirely when computing HRV, instead of the single ectopic beats as the HRV variables cannot be calculated reliably.

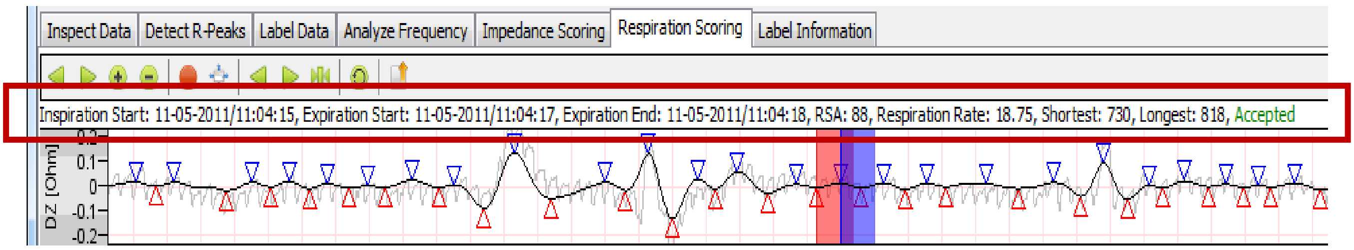
5.4 RESPIRATION



The windows of the *Respiration* tab shows 2 physiological and 2 derived signals for the time period indicated at the X-axis of the graph:

1. **Filtered Impedance** (dZ in Ohm, grey)
2. **CardioTachogram:** A beat-per-beat estimate for the IBI (msec).
3. **The time series of RSA values** across the consecutive breaths (in msec).
4. **Respiration rate** derived from the impedance signal in breaths per minute

The currently selected respiratory cycle is indicated by the combination of a red, purple and blue box. The purple box reflects the overlap of the inspiration phase which is extended by a dZ-HR shift (1000 msec) with the expiration phase. You can activate/de-activate the raw impedance signal in the settings screen (see appendix A).



When selecting a single breath (by right clicking on it with the mouse) you will see the following information per breath on top of the upper window: Inspiration start, expiration start, expiration end, RSA (in msec), respiration rate (in breath/pm), shortest IBI (highest heart rate) during inspiration on decelerating slope (msec), Longest IBI (lowest heart rate) during expiration on an accelerating slope (msec) and weather the breath is accepted or rejected. Rejection codes signal one of the following reasons why RSA was not accepted:

RSA : -1 undetectable 'shortest IBI' (RSA graph:)

RSA : -2 undetectable 'longest IBI' (RSA graph:)

RSA : -3 both 'longest IBI' and 'shortest IBI' were undetectable (RSA graph:)

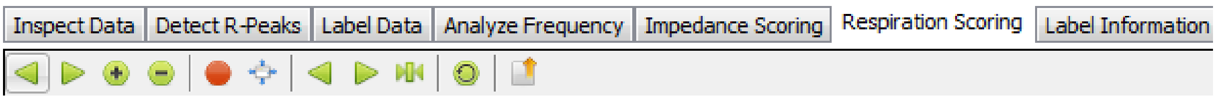
RSA : -4 'longest IBI' is shorter than the 'shortest IBI' (RSA graph:)

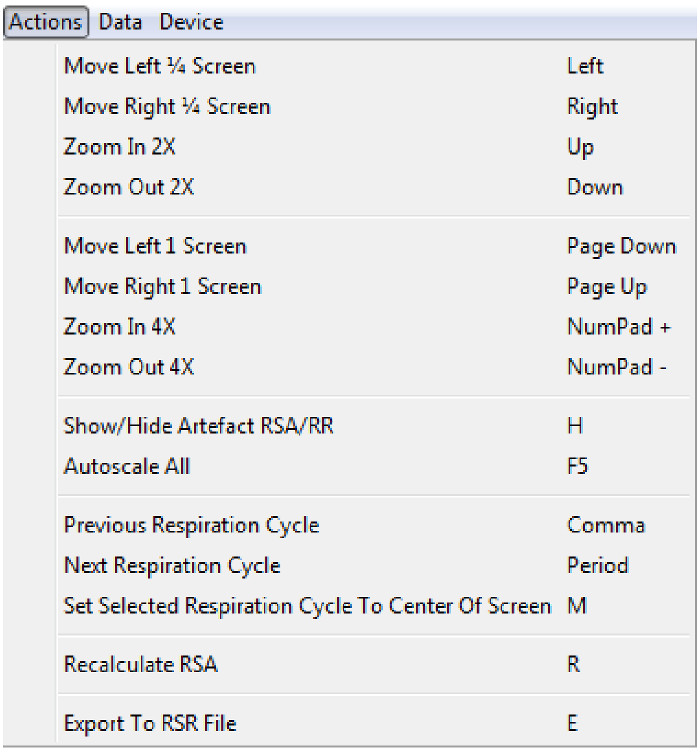
RSA : -5 'Irregular IBI detected' (RSA graph:)

RSA : -6 'Irregular respiration rate' (RSA graph:)

RSA : -7 'Clipping dZ' (RSA graph:)

All actions for the *Respiration Scoring* tab are presented in the form of buttons. The function of these buttons should be self-explanatory.





5.4.1 VARIABLES

From this signals shown in this tab the following variables are derived: respiration rate (breath per minute), mean RSA (msec) and mean RSA0 (msec). For each variable the mean per label is calculated (*see section 5.2.2 Labeling your data*). The values are displayed in the *Results tab (section 5.7).*

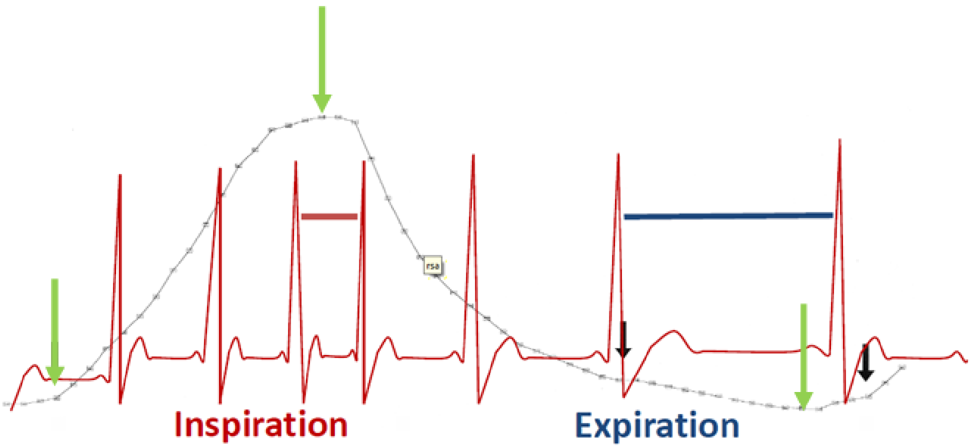
RESPIRATION RATE

For each respiratory cycle the total cycle time between begin of inspiration and end of expiration is extrapolated to a per-minute respiration rate.

RESPIRATORY SINUS ARRHYTHMIA (RSA)

RSA is heart rate variability in synchrony with respiration. RSA scoring by the DAMS program is based on the peak-valley method (Grossman, van Beek, & Wientjes, 1990 ; de Geus et al., 1995). This method computes per each respiratory cycle two IBIs: The shortest IBI during an interval starting at the begin of inspiration and ending 1000 msec (default) after the end of inspiration and the longest IBI during an interval starting at the beginning of expiration and ending 1000 msec (default) delay after the end of expiration. RSA is calculated by the subtraction of the shortest IBI from the longest IBI, provided that the shortest IBI (highest HR) is part of an accelerating series within the inspiratory interval and the longest IBI (lowest HR) of a decelerating series within the expiratory interval. This is illustrated in the Figure 3.

Figure 3. Peak valley method to calculate RSA



If either the decelerating longest or accelerating shortest IBI is missing for a breath cycle, or a negative RSA value is obtained on subtraction, these breaths are treated as missing in RSA calculation. This may lead to over-estimation of RSA, as breaths where RSA was near-zero may be rejected disproportionately for negative values due to measurement error. Therefore, two different mean RSA variables are calculated: the mean RSA only for those breaths in the label with a valid positive RSA value, and the recommended "RSA-zero" in which breaths with an 'invalid' negative RSA are included in the mean calculation with value zero. DAMS labels these variables in the results files as RSA and RSA0 respectively.

5.4.2 DATA INSPECTION AND MANUAL CORRECTION

5.4.2.1 IMPEDANCE

The top window shows the filtered impedance signal (0.1 – 0.4 Hz). Behind the filtered signal the raw impedance signal is given in light grey. The start and end of each breath is extracted from the filtered dZ signal by an algorithm which applies both amplitude and frequency modulation. A red triangle indicates the start of an inhale and the following blue triangle the end of that inhale. An exhale starts at this blue triangle and ends at the following red triangle. The derived respiration rate (breaths per minute) is depicted in the second bar.

In the bars called "Clipping DZ" and "Irregular respiration" at the bottom of the window the algorithm marks periods in which the signal goes out of bounds (red areas in the clipping DZ) and when respiration is irregular (based on a maximal percentage for deviation of the duration of consecutive breaths; blue areas in the irregular respiration bar). Breaths that are marked in the bars are removed from further analyses. Fortunately, automatic scoring of the respiration signal works quite well in most subjects. However, similar to the R-peak detection algorithm, visual inspection and manual corrections may be necessary. Mostly, it will suffice to just

browse through the impedance (DZ) signal after having set the time axis at a low temporal resolution (e.g. ten minutes per screen). While browsing through the respiration signal from the beginning to the end of the file check the following:

1) Did the program mark more than 1/10 of the recording as artefact in either the "clipping dZ" or "Irregular respiration" bars (red and blue areas respectively)? No continue to point 2, yes:

- → Check whether setting the parameters of the scoring algorithm to a different threshold range (see TIP16) leads to less data loss.
- → Zoom in on large areas that are marked blue in the irregular respiration tab and check the raw impendence signal. If the signal looks indeed irregular you do nothing. However, if the respiration signal looks good when zooming in but the program deletes a lot, try unchecking ''automatic impedance range check'' via *Edit → Settings → Respiration Scoring.*

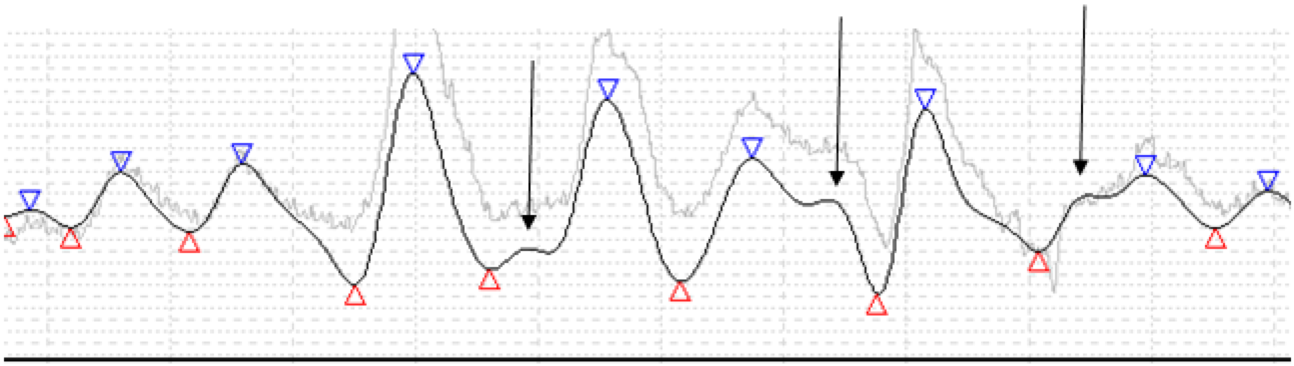
TIP15: *Pay special attention to the breath cycles measured during the night. Some subjects show strong abdominal breathing which seriously affects detection of the respiration signal by thorax impedance. This can often be repaired by 'rescoring the cycles' (under the main menu item Edit → Settings → Respiration Scoring) and changing the 'Relative Threshold' parameter of the scoring algorithm*

TIP16: *When measuring (small) children, you might need to change some settings for respiration, because of the higher heart rate in young children, you might benefit from changing settings for lag time as well. Change phase shift for shortest/longest IBI down.*

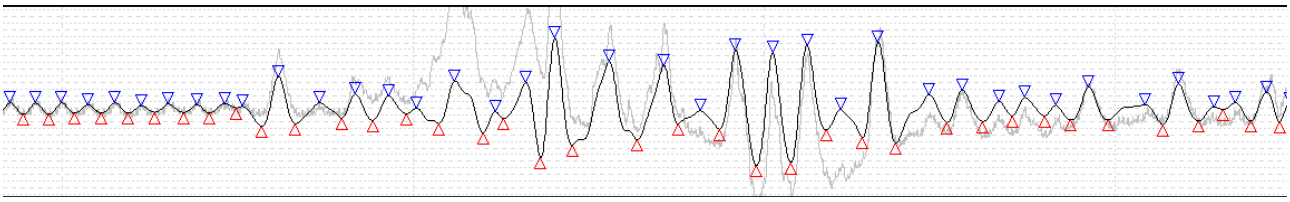
NOTE CAREFULLY: *After changing the settings, the respiration signal and RSA are recalculated on the original signal. This means that manually rejected fragments will be restored, so first make sure you have the chosen the optimal settings before manually rejecting fragments of the recording. It will warn you before recalculation. More information on changing the setting can be found in Appendix A: Changing respiration scoring settings.*

2) Do all inspirations and expirations appear to be appropriately scored (indicated by blue and red triangles)? Yes continue to point 3, no:

- → Manually delete an inappropriately scored fragment of the signal by clicking and dragging the mouse in this window irregular respiration window.



3) Is there noisy signal that is correctly scored? No continue to point 4, yes:

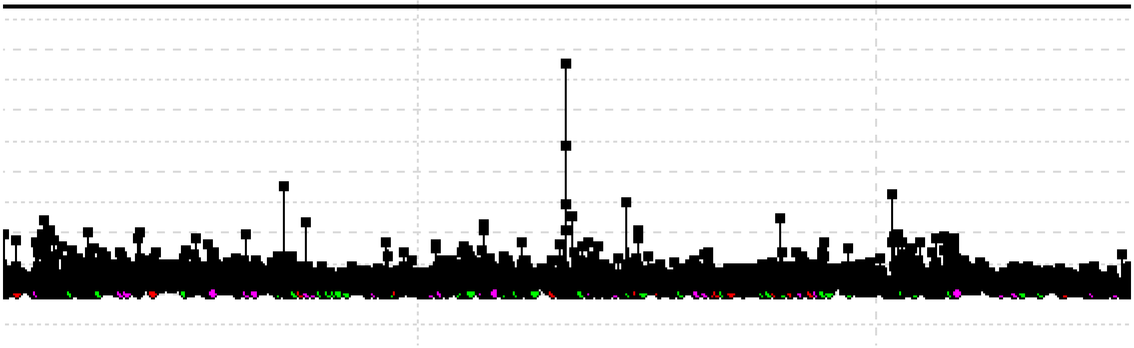


➔ Manually delete the noisy fragment of the signal by clicking and dragging the mouse in this window irregular respiration window

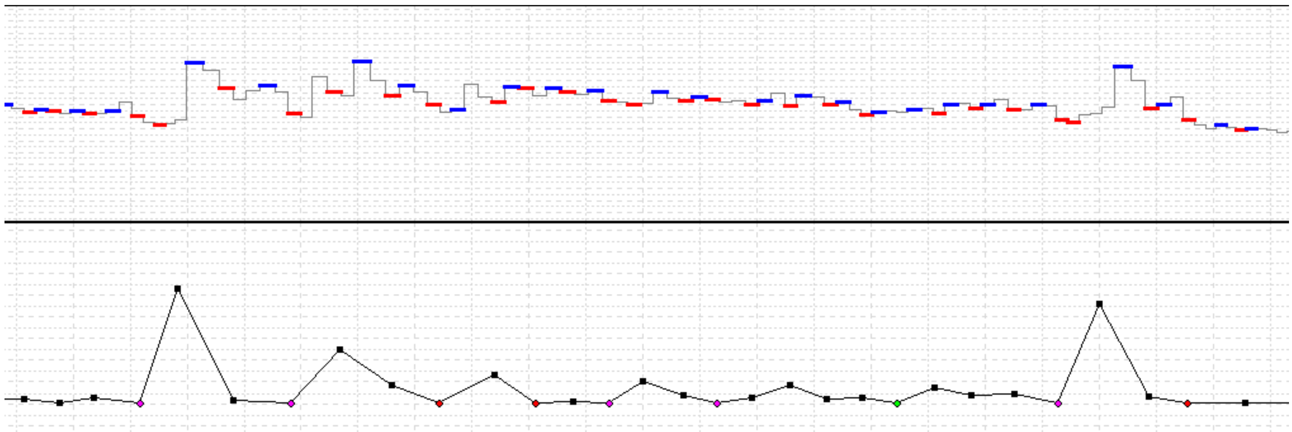
4) Check whether the per breath estimate of the respiration rate (in the Respiration window) takes on expected values between 7-14 at night, 12-22 across most daily activities except moderate to high physical activity where respiration rate can increase to 30 breath per minute.

5.4.2.2 RSA AND TACHOCARDIOGRAM

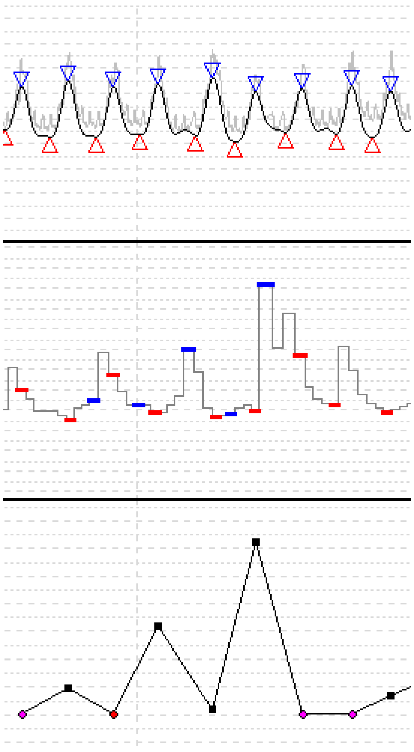
The IBI values used to calculate the RSA are given in the tachocardiogram (third bar). The tachocardiogram shows the shortest IBI during inspiration, provided that it is part of an accelerating IBI series, is indicated by a fat red mark. The longest IBI during expiration, provided it was part of a decelerating IBI series, is indicated by a fat blue mark. Based on a maximal percentage for deviation of the duration of consecutive beats periods, deviant beats are marked yellow in the "Irregular IBI" bar. To change these setting see Appendix A.
The third bar gives the calculated RSA values for each breathing cycle. To check for deviant RSA, browse through the signal with 60 minute (or longer) time frame and check for extreme outliers, either really high RSA or really low RSA values.



When encountering a deviant value zoom in and inspect the respiration signal and tachocardiogram. If a peak in the RSA is accompanied by a stair-case increase in the tachocardiogram this increase in RSA is correct.



However, when the increase is accompanied by a sudden increase in the tachocardiogram while the impedance signal shows no change (see below) it should be manually removed. This can be done by clicking and dragging in the “Irregular IBI” bar. The RSA value will be removed from the screen and any further analyses.



5.4.3 EXPORTING RAW BREATH TO BREATH DATA

You have the option to export breath to breath results to a tab delimited text output file using the button "Export To RSR File". These *.rsr* files give the following information on each respiratory cycle on a single line:

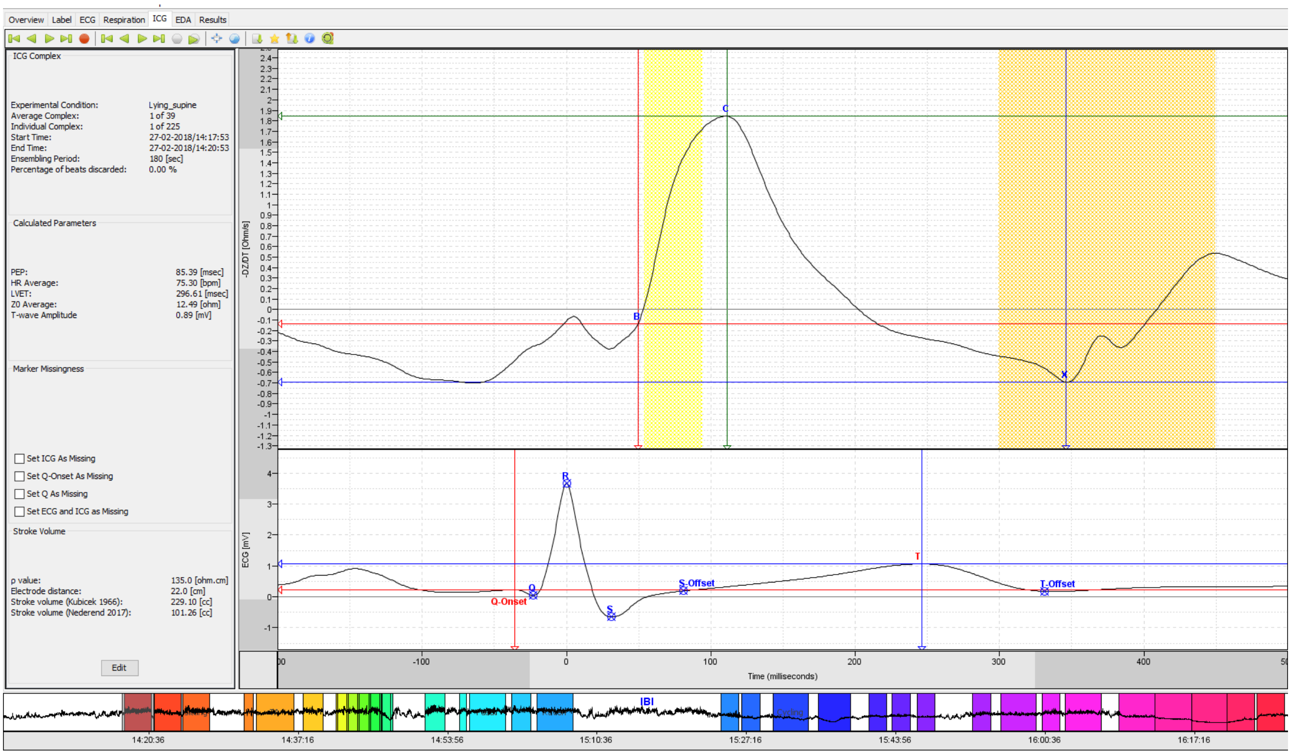
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1105016	16	11-05-2011	09:04:13	1800	1800	754	755	16.67	1	768.5	-34.16	114.64	148.81	A	10
1105016	17	11-05-2011	09:04:16	1700	1200	748	-1	20.69	-2	751.67	-99.87	-1.6	98.27	A	10
1105016	18	11-05-2011	09:04:19	2000	3200	716	812	11.54	96	742.17	-35.45	149.61	185.06	A	10
1105016	19	11-05-2011	09:04:24	1900	1200	695	-1	19.35	-2	737.33	-137.62	56.78	194.4	A	10
1105016	20	11-05-2011	09:04:27	2600	1900	683	748	13.33	65	717.2	19.24	74.63	55.39	A	10

column 1 : Subject ID
column 2 : Respiratory cycle number
column 3 : Date (dd-mm-yy)
column 4 : Start of respiratory cycle (hh:mm:ss)
column 5 : Inspiration time [msec]
column 6 : Expiration time [msec]

column 7 : Shortest accelerating IBI in inspiration [msec]
column 8 : Longest decelerating IBI in expiration [msec]
column 9: RR [breath per min]
column 10: RSA [msec]
column 11: Mean IBI across the cycle [msec]
column 12: Amplitude dZ at start inspiration [milliOhm/sec]
column 13: Amplitude dZ at start expiration [milliOhm/sec]
column 14: Tidal volume [milliOhm/sec] - calibration is needed to translate this to ml
column 15: Rejected (R) as artefact or accepted (A)
column 16+: Labels (-9999 = not available)

You can import this text file in e.g. SPSS for more fine grained analyses that use breath-to-breath information rather than the averaged values per label that are typically produced under the *Results tab (section 5.7)*. Please note that amplitude is in milliOhms/sec and needs calibration before volumes have physiological meaning. Careful outlier detection is needed before you do further statistical analysis on these breath-to-breath data. In view of the huge number of breath cycles to be quality controlled in 24-hour ambulatory monitoring, some automation of these checks is desirable, for instance by scripting in MATLAB, R or even SPSS.

5.5 IMPEDANCE SCORING

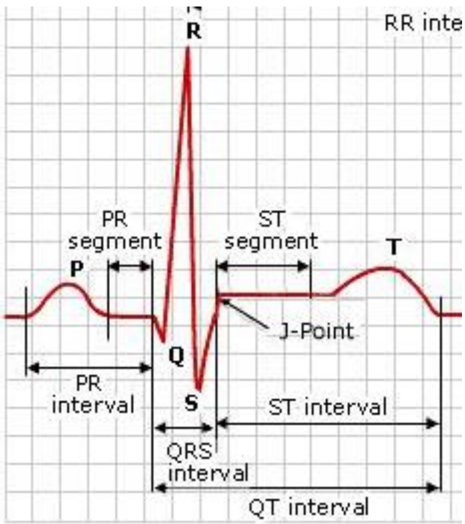


The impedance cardiogram (ICG) is the change (d) in thorax impedance (Z) over time (t), (dZ/dt). This characteristic ICG waveform derives from the change in thorax impedance caused by left ventricular ejection of blood into the descending aorta during the systolic phase of the cardiac cycle.

The top right window shows the ICG complex. In this window you can see a red, green and blue line, accompanied bij the letter B, C and X respectively. The red line depicts the time point and ohm of the B-point. The **B-point** reflects the opening of the aortic valves, marking the end of the electromechanical systole and the beginning of the left ventricular ejection time. The green line depicts the time point and ohm of the C-point. The **C-point** is the point where the velocity of ejection is at its maximum and impedance at its minimum (in the graph, it is drawn in reverse polarity, so the dZ/dt minimum is shown as a maximum by the program). The blue line depicts the time point and ohm of the X-point. The **X-point** reflect the closing of the aortic valves.

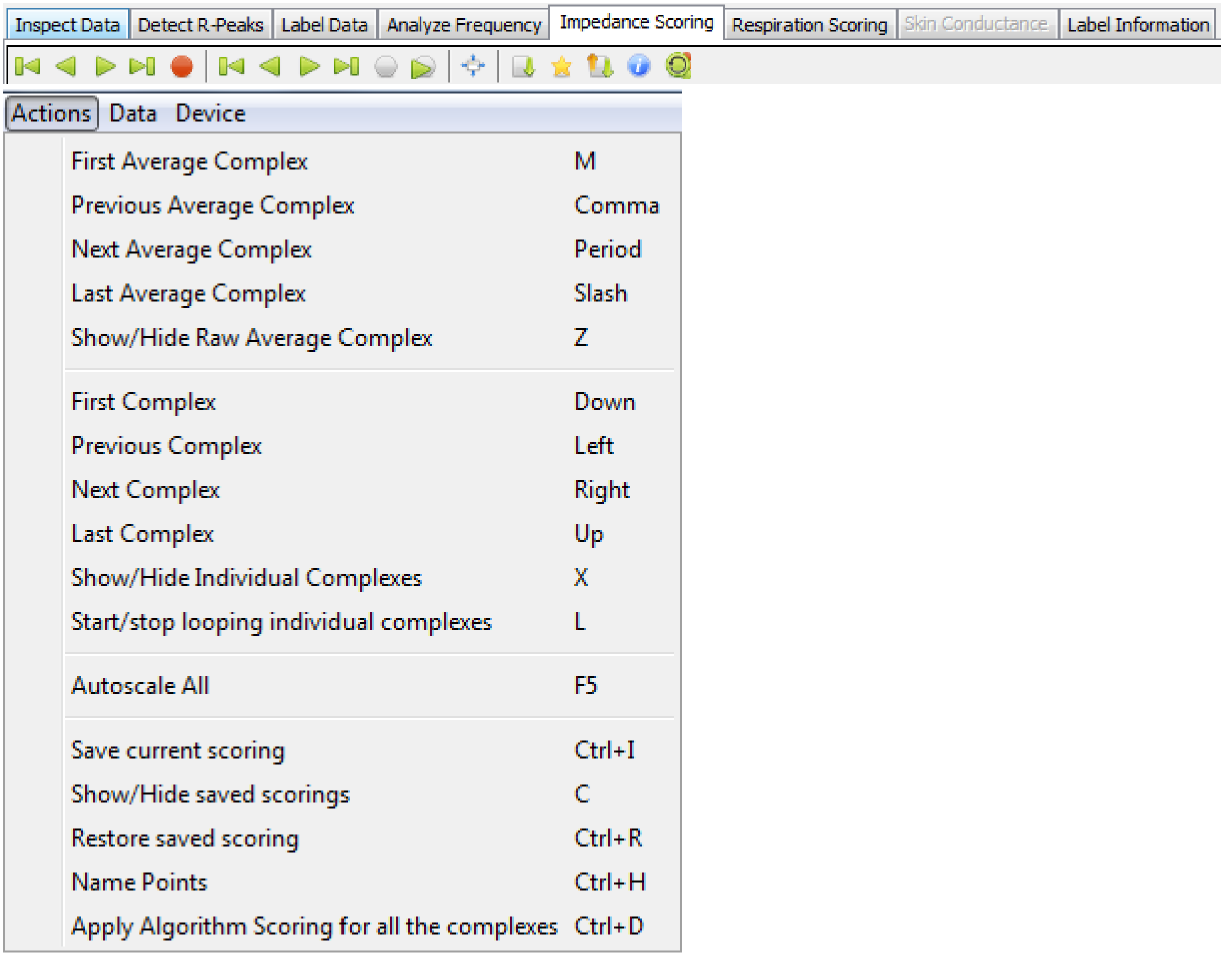
In de window below the ICG complex the averaged ECG is presented. This window also contains a red and a blue line with additional blue crosses. Similar to the ICG complex window these lines and crosses display important cardiac events (Figure 4). Each line or cross is accompanied by a letter to indicate which event it refers to (see image below for the major ECG time points and intervals). The red line indicates the time point and voltage (mV) of the Q-wave onset. The **Q-wave** onset in the ECG marks the start of the ventricular depolarization and represents depolarization of the interventricular septum. The first blue cross following the red Q-wave line is the Q-point. The **Q-point** marks the beginning of the depolarization of the main mass of the ventricles. The Q-point cross is followed by a cross marking the **R-peak**. The beginning of the final depolarization of the ventricles, at the base of the heart, is marked by the **S-point**. Following the S-point is the S-offset. The **S-offset** marks the end of ventricular depolarization (S-offset is often called J point in literature). Ventricular depolarization is followed by repolarization of the ventricles, the maximum voltage during repolarization is indicated by the blue **T-point** line. The **T-offset** indicates the ending of repolarization.

Figure 4. ECG points and intervals



On the left side of the window information is given on various variables which will be discussed in section 5.5.2.2.

All actions for the *Impedance Scoring* tab are presented in the form of buttons. The function of these buttons should be self-explanatory.



5.5.1 VARIABLES

From this tab the following variables are derived: Pre-Ejection-Period (PEP) in msec, Left-ventricular ejection time (LVET) in msec, T-wave amplitude in mV and the Stroke volume (SV) in cm^3. For each variable the mean per label is calculated (see section 5.2.2 Labeling your data). The values are displayed in the “Results” tab (see section 5.8).

PEP

The PEP is an index of contractility which is only influenced by sympathetic but not parasympathetic activity in humans. This quality makes the PEP the measure of choice to monitor changes in cardiac sympathetic activity non-invasively. The PEP is defined as the interval from the onset of left ventricular depolarization, reflected by the Q-wave onset in the ECG, to the opening of the aortic valves, reflected by the B-point in the ICG signal (Nederend et al., 2017 ; Lozano et al., 2007; Sherwood et al., 1990; Willemsen et al., 1996). Increased SNS activity leads to a shorter PEP.

LVET

As the name indicated the left ventricular ejection time (LVET) is the systolic time interval between the opening (B-point) and closing of the aortic valves (X-point). Simply said the time it takes to eject the blood from the left ventricle into the aorta. LVET is considered a measure of sympathetic chronotropic influence on the heart. Chronotropic refers to the fact that the SNS influences the activity of the SA node, rather than the myocardium as is the case for the PEP. With increased SNS activity the LVET is expected to decrease.

TWA

The T wave is the asymmetrical wave in the ECG that comes after the QRS complex and typically lasts approximately 150 ms. It reflects ventricular repolarization in which the sympathetic nerves play an important role. In DAMS TWA is defined as the difference between the peak of the T wave (T) and the value at the end of repolarization (T-offset). When SNS activity increases (either due to pharmacological interventions or due to stress) the TWA decreases and is sometimes even reversed.

SV

SV is the average amount of blood ejected during the cardiac cycle. When multiplied by heart rate this yields the cardiac output (CO), the total amount of blood circulated through the body per minute. SV is computed from the ICG by using the product of the maximal amplitude of the dZ/dt and the ejection time, weighing for blood resistivity, baseline thorax impedance and the total volume enclosed by the measuring electrodes.

The amount of blood ejected per beat in [cm^3]. Calculated with the Kubicek equation:

$$SV = \frac{-\rho_{blood} Le^2 \left(\frac{dZ}{dt}\right)_{min} t_{lve}}{Z_0^2}$$

where $t_{lve} = t_{incisura} - t_{upstroke}$

The Kubicek equation for SV was originally derived using band electrodes and a correction on the baseline thorax impedance (Z_0) is needed for spot electrodes. We refer to Nederend et al., 2017 for the appropriate correction when using the VU-AMS: $Z_0 = 7.337 - 6.208 * dZ/dt_{max}$, where dZ/dt_{max} is the amplitude of the ICG signal at the C-point. Both the SV calculated by the original Kubicek formula and the corrected Nederend version are given in the *Results tab (section 5.7)*.

N.B.: The reliability of between-individual differences in Stroke Volume (SV) computed by impedance cardiography remains a heavily debated issue. With ambulatory SV the concerns are even more valid, because movement artefacts and the lack of a phonocardiogram do NOT increase SV reliability. In addition, spot electrodes pick up only half of the impedance measured by band electrodes, and without correction for this, the Kubicek formula yields supraphysiological SV's. If you use percentual changes in SV strictly in a within subject design all these concerns are greatly reduced.

5.5.2 VISUAL INSPECTION AND MANUAL CORRECTION

To improve signal quality, the ICG waveform is obtained by ensemble averaging over beats within in a fixed time period, time locked to the R-wave peak. The typical period for ensemble averaging is one minute. In DAMS we deviate from this practice and instead compute a Large Scale Ensemble Average across the entire label (see: Riese et al., 2004 for the rationale).

Because the ICG signal is highly susceptible to even subtle movement artefacts, it is necessary to filter the signal to overcome the noise confound and assure reliable detection of the inflection points in the ICG curve. The ICG signal is passed through a low pass filter with a cut-off frequency of 60 Hz. By default, this filter is set to "ON"

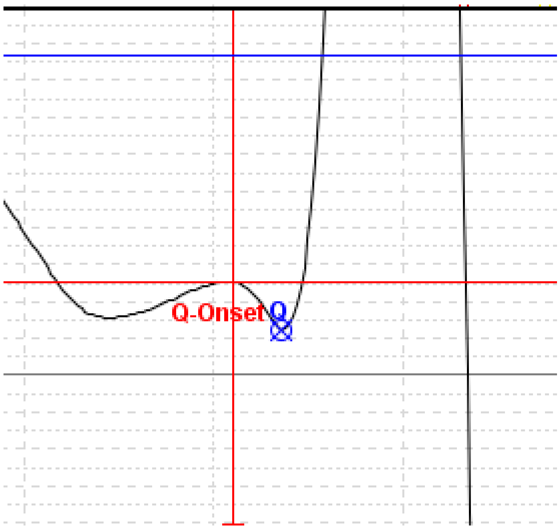
A text is displayed next to the ICG curve in the Impedance Tab indicating that the filtering is "ON". However, the filter can be disabled (see Appendix A).

DAMS runs an automatic scoring algorithm, which tries to detect the three specific locations in each ICG waveform (termed 'ICG complex'): The B-point, C-point, X-point. In the ECG window the locations of the QRST complex are given.

Though ensemble averaging improves automated detection of the crucial landmarks in the ECG and in the ICG, substantial errors in positioning of the B-point remain (Nederend et al., 2017 ; Lozano et al., 2007; Willemsen et al., 1996; Berntson et al., 2004). The number of algorithms proposed to score the impedance cardiogram is countless. We have tried quite a few at the Vrije Universiteit. Our current stance is that automatic scoring simply will not work for all signals. We therefore visually inspect every ensemble averaged ICG complex and manually correct the locations of the 3 key time points when needed. Scoring of the Q-point in the ECG also always needs to be inspected and, when necessary, corrected. Manual scoring is inherently subjective but it does lead to reliable and valid results. To safeguard reliability, scoring should ideally be repeated by multiple raters.

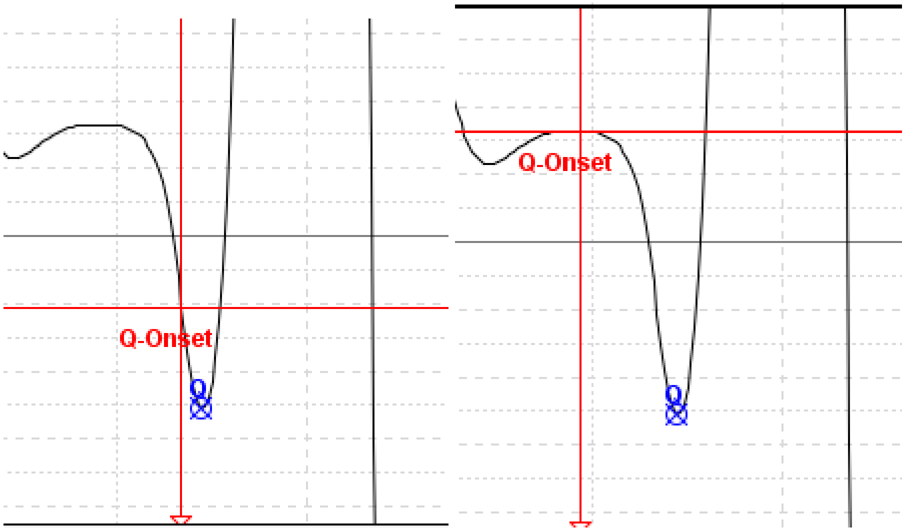
5.5.2.1 Q-POINT& T-WAVE

The ensemble averaged ECG allows for easy Q-point scoring. The Q-point might look obscured but will appear more clearly when zooming in on the Y-axis.



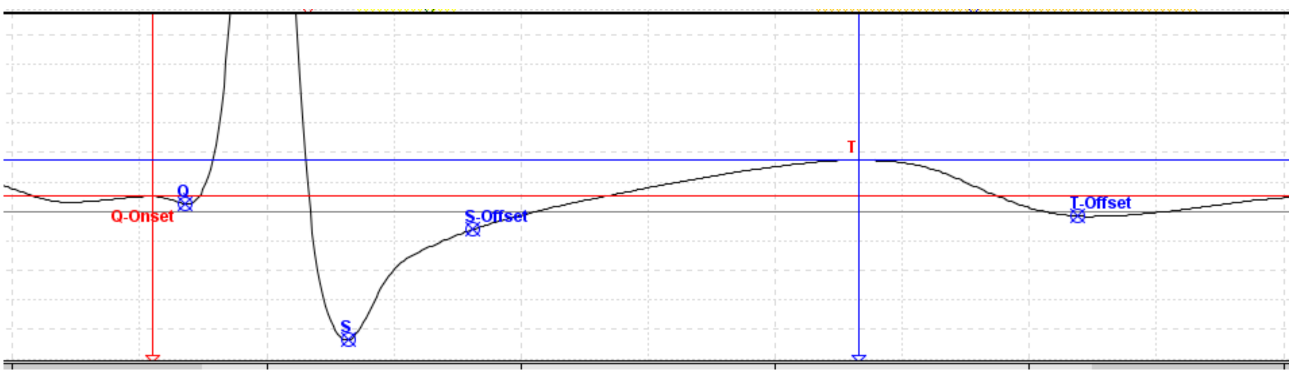
On occasion however, the Q-point will be set at the wrong place. Since the Q-point marks the start of the PEP a wrongly scored Q-point will influence the value of the PEP.

Example of a Q-point set at the wrong place by the algorithm and how it should be manually adjusted.



> ***TIP17:*** *In some subjects, no Q wave is visible. Check the ''set Q-Onset as missing'' box in the left panel. VU-DAMS will now estimate this point by subtracting 12 ms from the Q point.*

5.5.2.2 T-WAVE AMPLITUDE



The ensemble averaged ECG allows for easy T-wave scoring. Usually no adjustments are necessary. However, during ambulatory recordings the shape of the ECG can be influenced by vigorous body movements. This can lead to a deviant T-wave which cannot be scored correctly and should be set to missing.

5.5.2.3 ICG COMPLEX

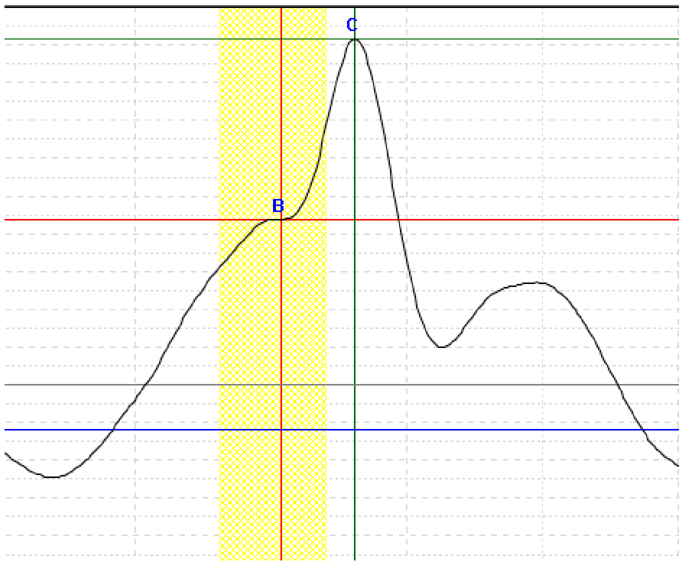
Two factors are of importance when scoring the ICG complex, morphology and consistency. These will be discussed below followed by some best practices.

MORPHOLOGY

<u>The B-point</u>

The B-point is in any case after the R peak, as blood outflow will not start before de ventricles are depolarized. It should be close to light grey the dZ/dt = 0 line, and be the starting point of the longest uphill slope before the dZ/dt_{min} point. Sometimes the starting point is located clearly at the beginning of the uphill slope in a through (image 1 below), however it also frequently present as an inflexion on the uphill slope (image 2 below).

The B-point is located in a small through before the longest uphill slope or as an inflexion on the uphill stroke.

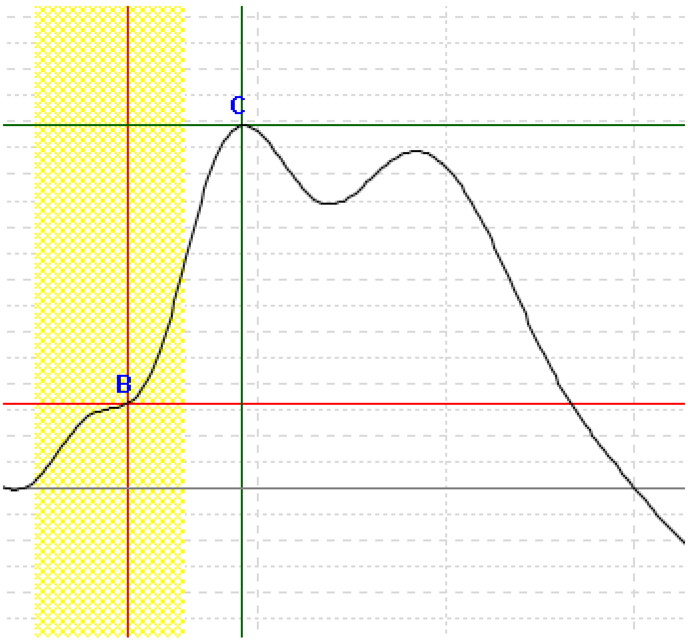


To help determine the location of the B-point it is surrounded by a yellow block. The yellow block indicates the most likely window for the placement of the B-point. However keep in mind that large individual differences in physiology exist and the general rules may not always apply. If your signal B-point lies outside of this yellow block, this does not at all mean it's incorrect. Occasionally, there is simply no clearly identifiable point that can be chosen to fit the above description of the B-point. In that case the point of the dZ/dt = 0 crossing may be appropriate (see also Sherwood et al., 1990).

C-point

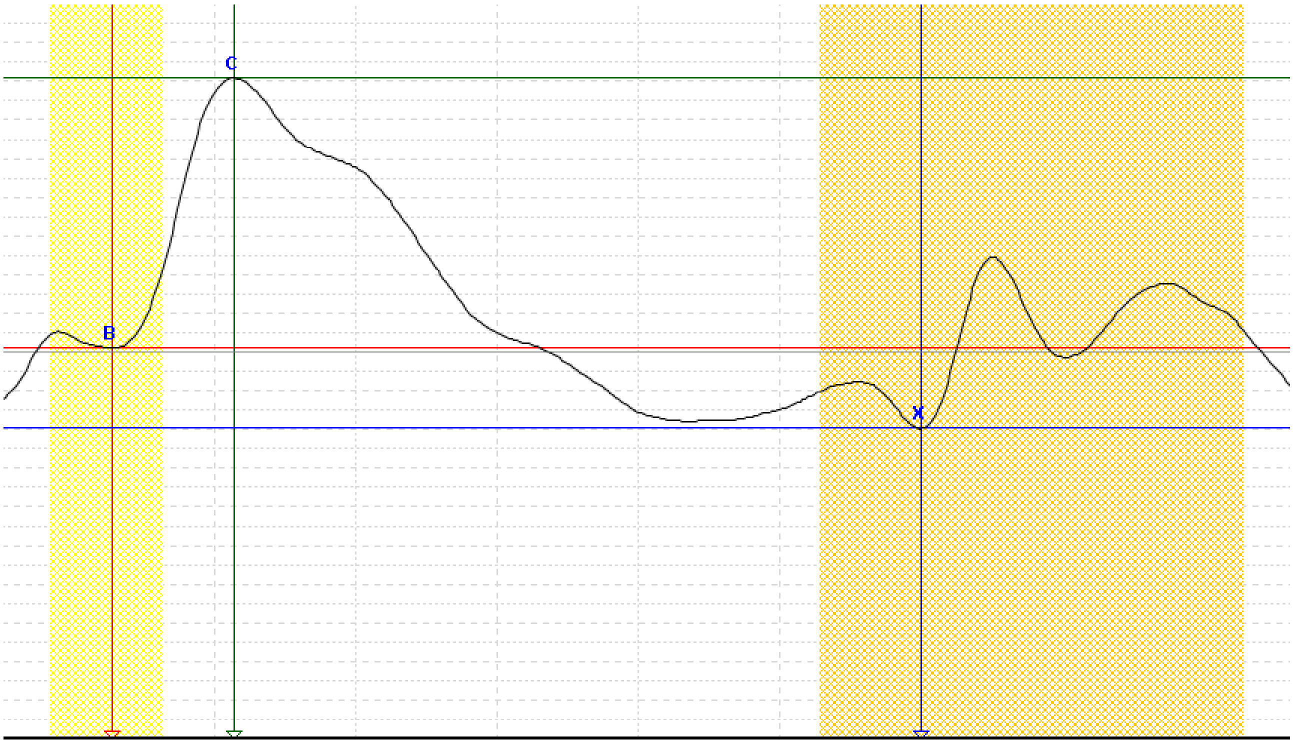
The C-point is normally visible as a clear peak in the window between the B- and the X-point. In some cases the dZ/dt signal shows a double peak, a bit like rabbit ears. If one of the peaks is clearly (40%) higher then this peak is chosen. If the peaks are of comparable magnitude, choose the first peak.

.



X-point

The X-point is always a local minimum after the dZ/dt_{min} and will never precede the T-wave peak. Often it is the lowest point in the entire signal, but not necessarily. In the ideal situation it can be seen as a sharp trough in the ICG signal. This is the most clearly identifiable choice for the X-point. It may be that two or more troughs lie in close proximity without one being clearly the lowest point in all complexes. The latter part of the ICG waveform then looks like a "W". In this case choose the first trough.



To help determine the location of the X-point it is surrounded by an orange block. The orange block indicates the most likely window for the placement of the X-point. However, keep in mind that large individual differences in physiology exist and the general rules may not always apply. If your signal X-point lies outside of this orange block, this does not at all mean it’s incorrect.

CONSISTENCY

Whatever point you choose, choose that point consistently. If a "less-than-ideal" upstroke is present in all complexes, but an "ideal" upstroke is present in some, choose the less-than-ideal one in all complexes, even those featuring a more "ideal" upstroke.

However, for laboratory recordings which include changes in posture or activity level and ambulatory recordings in general, one should take the posture and physical activity level into account with regard to consistency. Changes in posture and physical activity can impact the shape of the ICG complex within a participant.

In these situations, we advise the be consistent for labels which represent similar postural and activity conditions. Score all complexes that fall within the same motility range consistently. This also goes for posture. When you know the posture, score all conditions with that posture consistently (for example, when participants are doing desk work or sleeping).

TIP18: *Before starting to score the ICG, try browsing through the entire ICG signal first. You can then decide which points can be most consistently identified, and this holds for both for the B-point and the X-point.*

If it is not possible to score the ICG complex based on morphology and/or consistency (for example 1 complex does not look like any other complex in this participant) you should set either the PEP or the LVET to missing in the left window.

BEST PRACTICES

When learning to score the ICG complex we greatly advise multiple raters scoring the same file. Reliability increases if two (or more) raters score the same data set independently. After interrater reliability is established, the various raters should ideally compare their deviant scoring to converge on a single solution, in view of the consistency principle. Mostly one will have picked a different B-point than the other(s). In such situations averaging the B-point location is an attractive but invalid solution. The raters should reach consensus on the correct B-point location to satisfy the consistency criteria.

Sometimes scoring is difficult and doubtful, at other times you feel pretty sure. After scoring you might want to generate three parameters for "scoring-quality". Make separate judgments for B-point scoring, X-point scoring and general signal quality on a scale from 0 (yuk!) to 10 (excellent!). Later on, during statistical analysis, request to see the mean of all parameters as a function of your quality rating.

5.6 EDA

EDA is a measure of the electrodermal activity of the skin regulated by the sweat glands of the body. EDA can be used an indication of psychological or physiological arousal since sweat glands are only innervated by the sympathetic nervous system. The EDA signal is measured using direct current (DC) utilizing a 16 bit A/D converter. The sampling rate is 10 Hz with a signal range of 0-95 micro Siemens (µS).

EDA measurement can be characterized into two types namely tonic and phasic components.

TONIC EDA

Tonic EDA is the level of skin conductance obtained over longer periods of minutes to hours. It is a slow changing signal and it is often referred as Skin Conductance Level (SCL). It is important to note that SCL does not exclusively measure the person's arousal state. It is very sensitive to the environmental temperature and can change over time due to changes in the skin-electrode interface.

PHASIC EDA

Phasic EDA refers to rapid changes in skin conductance, traditionally named the Galvanic Skin Response (GSR) and now commonly called 'Skin Conductance Responses (SCRs)'. These changes occur:

- in the presence of discrete external stimuli which can be sound, cognitive processes, decision making, shock, etc.
- spontaneously, without a clear relation to discrete external stimuli, and are then called non-specific SCRs (ns-SCRs).

The traditional application of EDA measurement was quantifying and classifying the response to single discrete stimuli. The VU-AMS was primarily designed to conduct research where the interest lies in physiology across time intervals rather than in responses to single stimuli. Therefore the VU-DAMS software works with labels to define intervals of interest for all signals, including EDA, and calculates parameters over those intervals (label-based design). To accommodate researchers who want to use the VU-AMS for traditional, lab-centered, stimulus-based designs there is a legacy mode (event-based design).

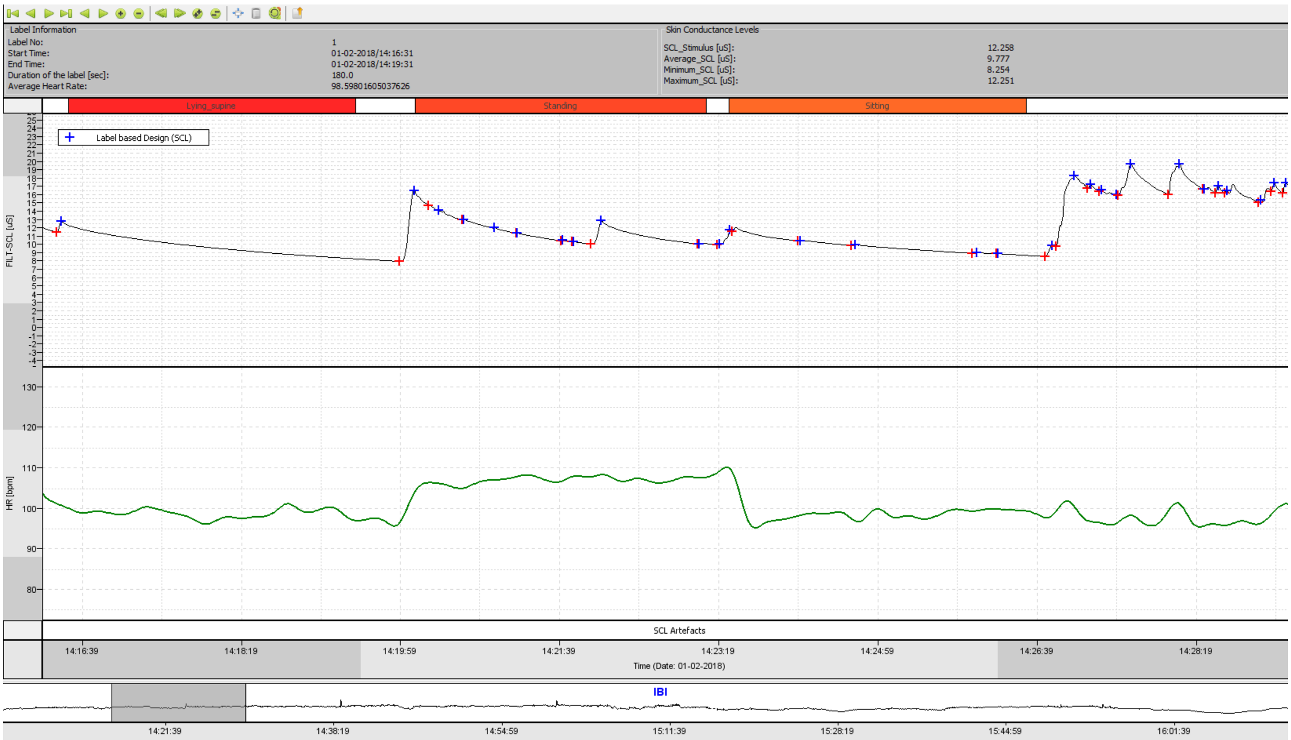
Depending on the type of study conducted, the electrodermal activity analysis method can be chosen from the main menu by clicking *Edit → Settings → SCL Data → Analyze Options* (see appendix A).

The default setting for the analyzing method is the Label based Design (SCL). The window lay out, variables and data cleaning of the event based or label based design will be discussed separately.

NOTE: While Electrodermal Activity (EDA) recording on the VU-AMS is reliable and fully supported, the analysis options for electrodermal activity in VU-DAMS are still in beta. For anything other than visual inspection of the data it is currently recommended to export your raw SCL signal to ASCII or EDF and process it in an external package.

5.6.1 LABEL BASED DESIGN

In this analysis mode, EDA offers an index of the general level of arousal, driven by sympathetic nerve activity.

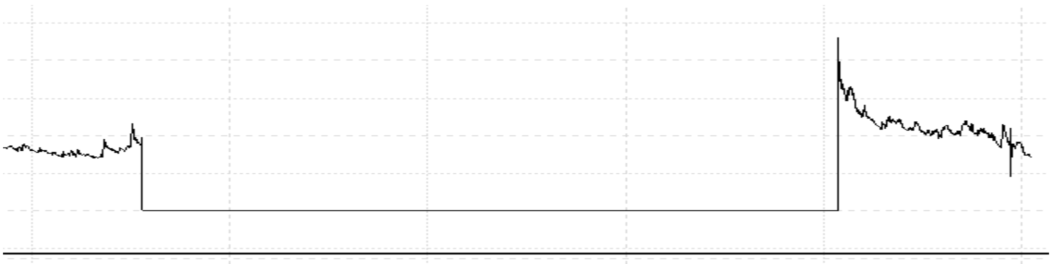


When label-based design mode is active (see Appendix A to switch modes), the top window provides label information. On the left side information is given on each label in your file with regard to the label number, start and end time of the label, the duration in seconds and the average HR. On the right side SCL information for the label is provided, such as the average, minimum, and maximum SCL.

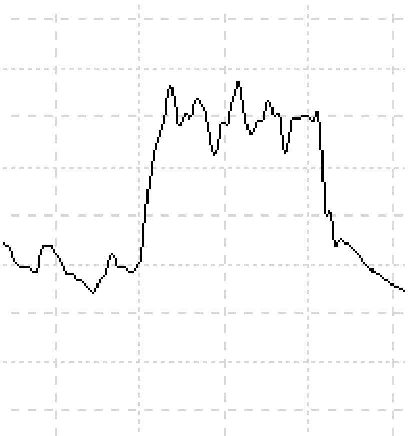
The second window shows the filtered EDA signal. Above the signal colored bars with label information is presented. In the example image above the conditions were "Lying_supine", "Standing", and "Sitting". The start of a detected skin conductance response (SCR) is indicated by a red + sign, and the peak of the response is marked by a blue + sign. The window below the EDA signal shows the filtered HR signal.

Below the HR signal you can find the Artefacts bar. In the Artefacts bar, clipping of SCL is automatically detected and marked as an artefact. These artefacts are labeled by a red bar. Users can mark low-quality signal as an artefact by clicking and dragging in the artefact bar (similar to the ECG artefact marking).

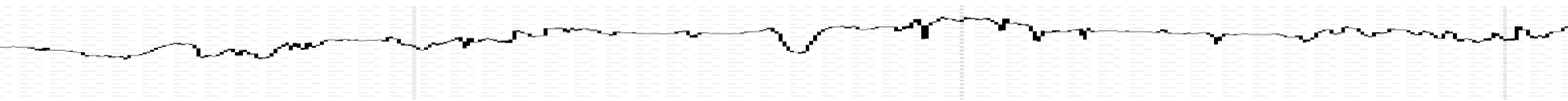
EDA artefact: lost connection



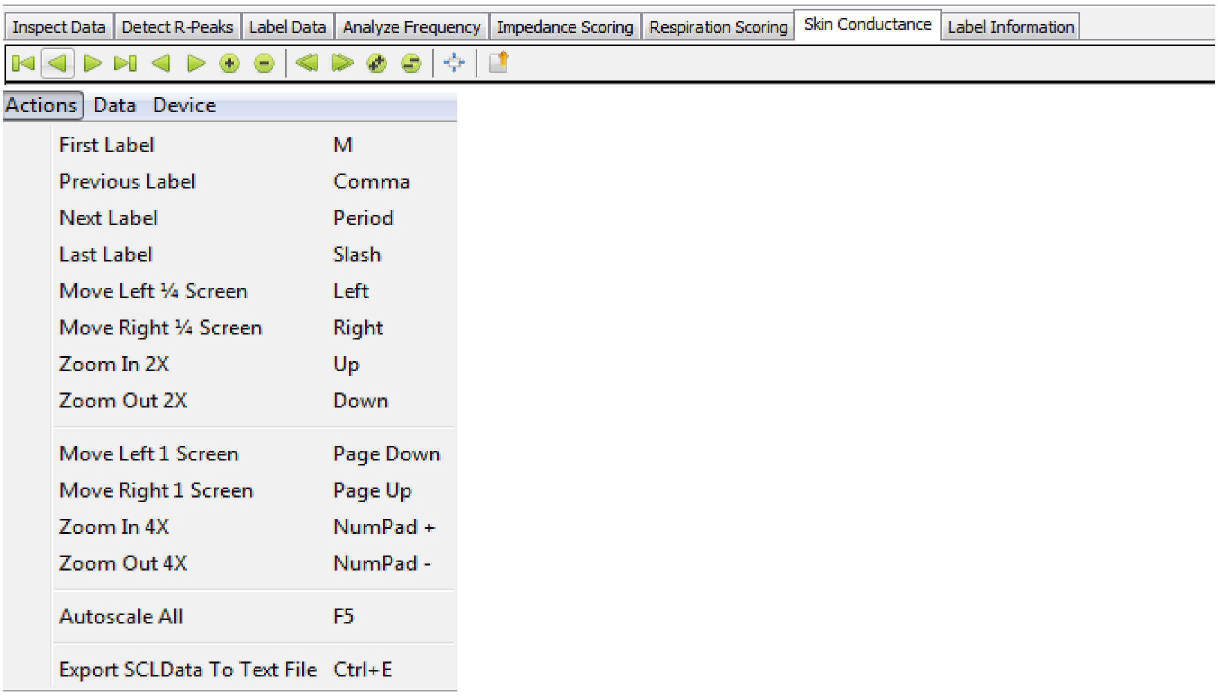
EDA artefact: electrode was touched



EDA artefact: period with lots of hand movement



All actions for *the Skin Conductance* tab for **Label Based Design (SCL)** are presented in the form of buttons:



LABEL BASED VARIABLES

From this tab the following variables are derived: the tonic variables average SCL (uS), minimal SCL (uS), maximal SCL (uS) and phasic variable number of ns.SCRs (p/m). Lastly the percentage artefact free EDA (%) is provided. Each variable is calculated per label (see section 5.2.2 Labeling your data). The values are displayed in the "Results" tab (see section 5.8).

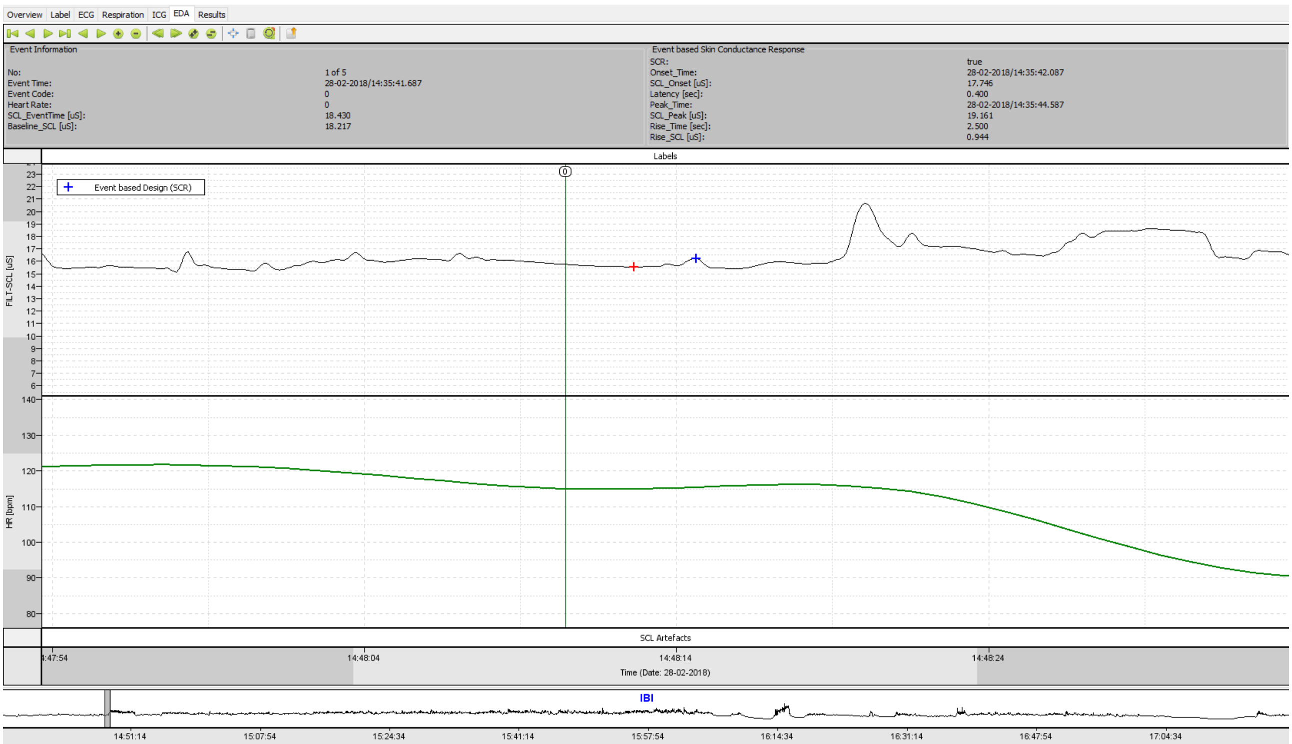
5.6.2 EVENT BASED DESIGN

Event based designs are experiments designed to quantify responses to discrete stimuli, the Skin Conductance Response (SCR). Some of the examples of event-based design experiments are risky decision-making, affective picture presentation, lie detector paradigms etc.

An experiment can be classified as “Event Based Design” when the events are clearly defined by discrete stimuli which may evoke an individual skin conductance response. These responses usually last for few seconds and they follow a specific pattern of Steep Rise- Sharp Peak – Slower return to Baseline. There are two main aspects taken into account while designing event based experiments:

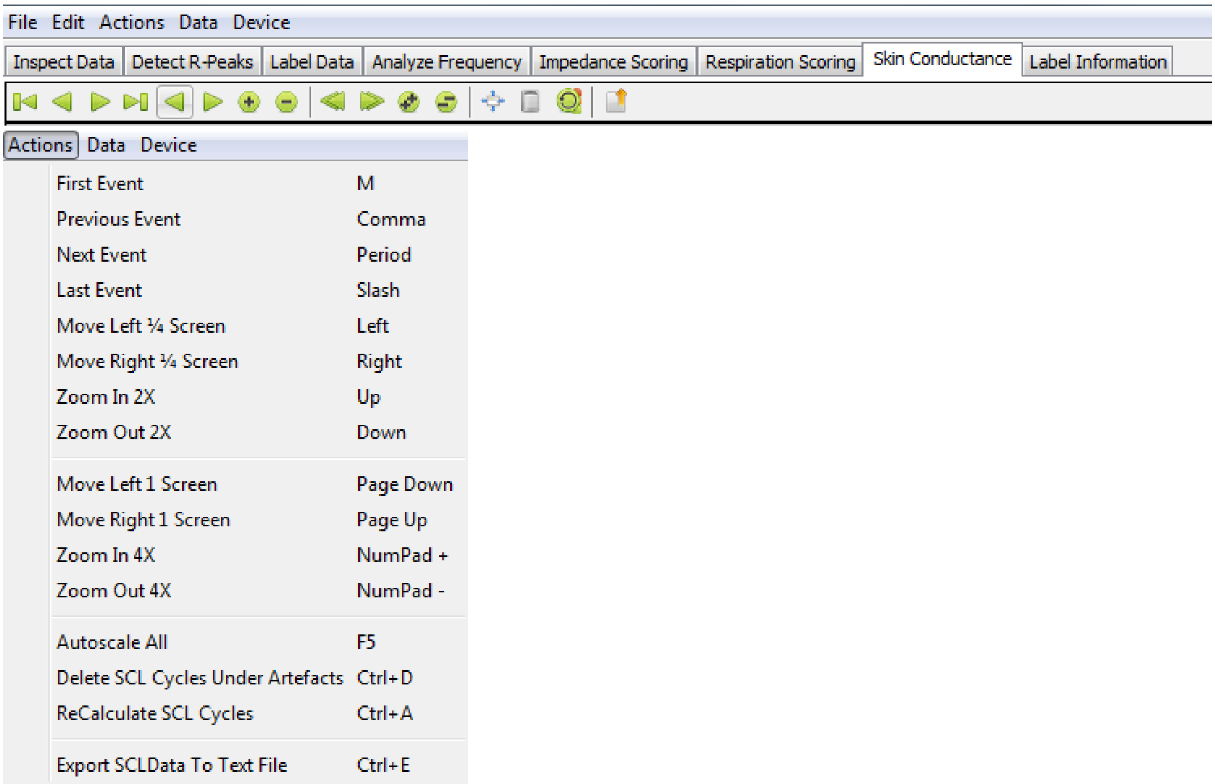
- Clear events
- Sufficient Inter- Event time

The inter-event time should be sufficient enough to make sure that the skin conductance is back to baseline before the start of a new event. In general, the onset might be noticed within 4 seconds upon application of a stimulus. The skin conductance response (SCR) can take 4-6 seconds to complete. So, the inter-event time assumed while designing these type of experiments is at least 8 seconds.

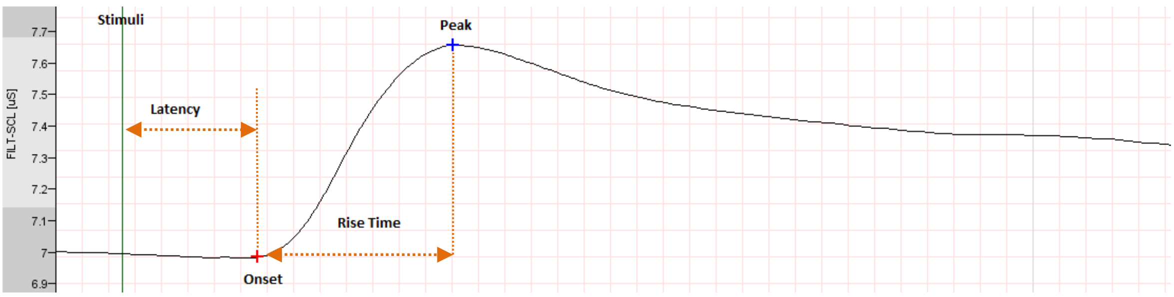


When choosing this design the window shows many similarities with the label based design. However, with an event-based design only the first SCR after an event (indicated by the green marker line) is scored. The top window also shows different information. At the left side event information is given on the number of the current event, the timepoint, the code, the heart rate and SCL at the occurrence of the events and the SCL prior to the event (baseline). On the right side several EDA parameters are given for the event. When investigating a SCR to an event/stimulus we are om much more variables compared to the label-based design. These variables will be discussed in section 2.7.2.1 Event based variables. Artefact detecting works the same as in the label-based design.

All actions for the *Skin Conductance* tab for **Event Based Design (SCR)** are presented in the form of buttons:



EVENT BASED VARIABLES



The immediate first peak occurring after the specific stimuli is the point of interest. In these experiments, it is uncommon to see second or late responses due to habituation. This is because of sufficient inter-event time between the events. The following information can be derived from the skin conductance signal.

For each event the following variables are calculated:

1. Baseline_SCL [uS] – is the average SCL in a window of 1 sec before the stimulus.
2. SCL_EventTime [uS] – is the SCL measured when stimulus is applied.
3. Onset _Time – is the start time of a skin conductance response (SCR).
4. SCL _Onset [uS] – is the skin conductance value measured at onset time.
5. Latency [sec] – is the time between the stimulus time and the onset time. It is usually between 1-4 seconds.
6. Peak_Time – is the time at which a max SCR is recorded within 7 seconds upon application of stimulus.
7. SCR_Peak [uS] – is the maximum SCR measured after a stimulus.
8. Rise time [sec] – is the time between the onset time and the time at which a peak is found. It is usually between 2- 4 seconds.
9. Rise_SCL [uS] – is the amplitude difference between the SCL baseline and the SCL measured at peak.

5.6.3 EXPORT SCL DATA

You have the option to export SCL data (label based or marker based) results to a tab delimited text output file using the button "Export SCL Data to text file". These *.scl* files give the following information on each SCL cycle. The output columns of the marker based .scl file is given below:

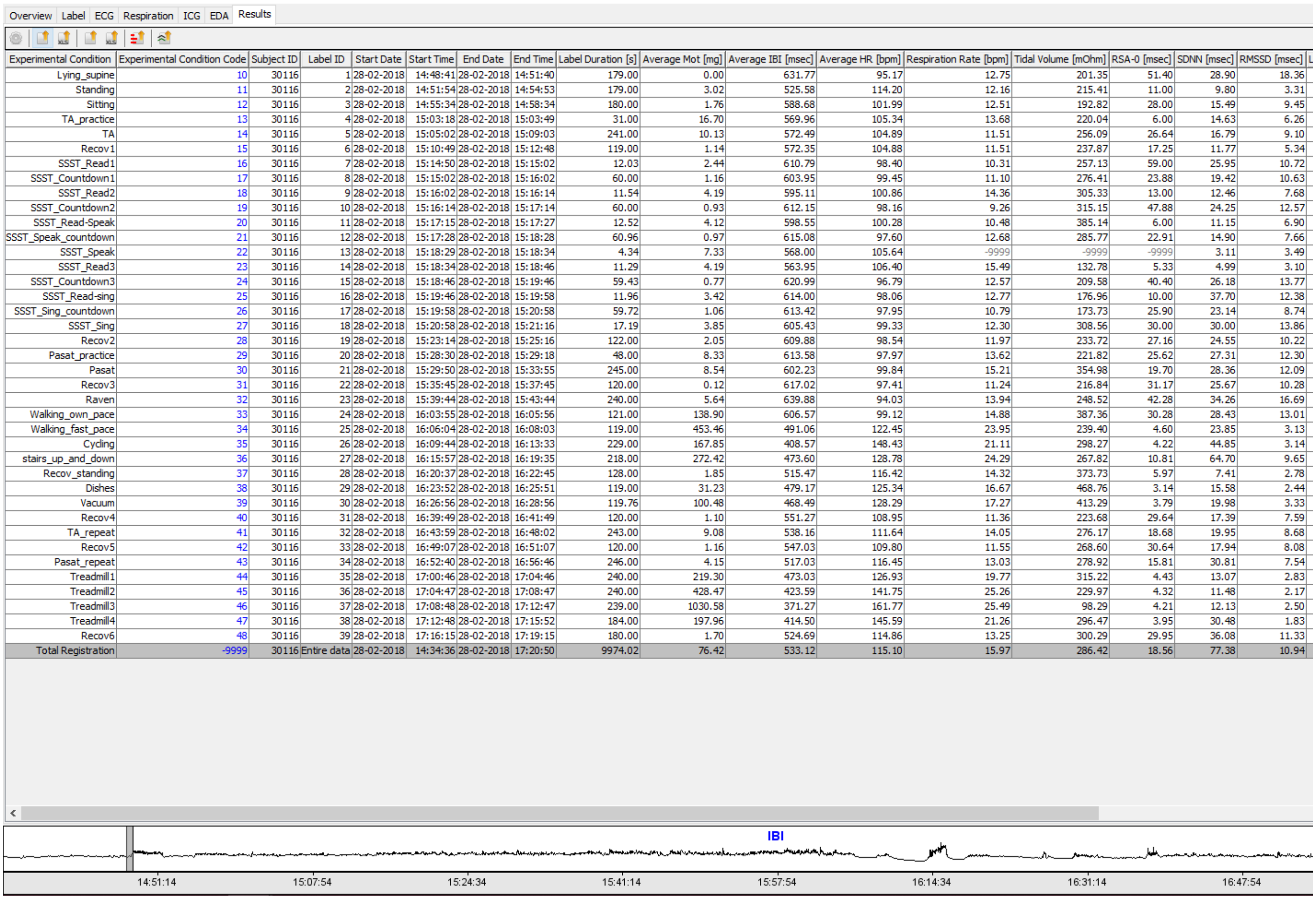
```
Skin Conductance Response Per Event
C43_SCR.scl 07-01-2015/09:19:53 07-01-2015/12:01:02
No  Event date/time  EventCode  HeartRate  SCL_EventTime[uS] Baseline_SCL[uS] SCR_Present Onset_Time SCL_Onset[uS]  Latency[sec] Peak_Time  SCL_Peak[uS] Rise_Time[sec] Rise_SCL[uS]
```

The output columns of the label based .scl file is given below:

```
Skin Conductance Response Per Label
C43_SCR.scl 07-01-2015/09:19:53 07-01-2015/12:01:02
Begin date/time    End date/time    Labelcode    Label_Duration    Average_HR    SCL_LabelStart    Average_SCL    MinumumSCL    MaximumSCL
```

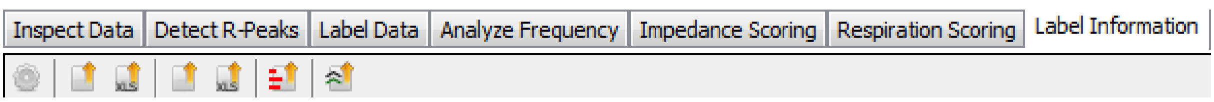
5.7 RESULTS

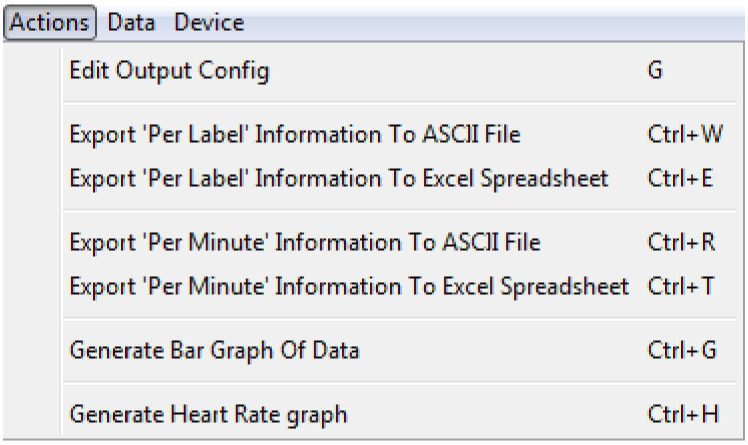
After clicking on the *Results* tab wait until all cells are finished calculating. A default set of variables is the 26 most used variables is displayed in the window. Each row represents a single labeled experimental/ambulatory condition.



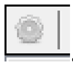
Experimental Condition	Experimental Condition Code	Subject ID	Label ID	Start Date	Start Time	End Date	End Time	Label Duration [s]	Average Mot [mg]	Average IBI [msec]	Average HR [bpm]	Respiration Rate [bpm]	Tidal Volume [mOhm]	RSA-0 [msec]	SDNN [msec]	RMSSD [msec]
Lying_supine	10	30116	1	28-02-2018	14:48:41	28-02-2018	14:51:40	179.00	0.00	631.77	95.17	12.75	201.35	51.40	28.90	18.36
Standing	11	30116	2	28-02-2018	14:51:54	28-02-2018	14:54:53	179.00	3.02	525.58	114.20	12.16	215.41	11.00	9.80	3.31
Sitting	12	30116	3	28-02-2018	14:55:34	28-02-2018	14:58:34	180.00	1.76	588.68	101.99	12.51	192.82	28.00	15.49	9.45
TA_practice	13	30116	4	28-02-2018	15:03:18	28-02-2018	15:03:49	31.00	16.70	569.96	105.34	13.68	220.04	6.00	14.63	6.26
TA	14	30116	5	28-02-2018	15:05:02	28-02-2018	15:09:03	241.00	10.13	572.49	104.89	11.51	256.09	26.64	16.79	9.10
Recov1	15	30116	6	28-02-2018	15:10:49	28-02-2018	15:12:48	119.00	1.14	572.35	104.88	11.51	237.87	17.25	11.77	5.34
SSST_Read1	16	30116	7	28-02-2018	15:14:50	28-02-2018	15:15:02	12.03	2.44	610.79	98.40	10.31	257.13	59.00	25.95	10.72
SSST_Countdown1	17	30116	8	28-02-2018	15:15:02	28-02-2018	15:16:02	60.00	1.16	603.95	99.45	11.10	276.41	23.88	19.42	10.63
SSST_Read2	18	30116	9	28-02-2018	15:16:02	28-02-2018	15:16:14	11.54	4.19	595.11	100.86	14.36	305.33	13.00	12.46	7.68
SSST_Countdown2	19	30116	10	28-02-2018	15:16:14	28-02-2018	15:17:14	60.00	0.93	612.15	98.16	9.26	315.15	47.88	24.25	12.57
SSST_Read-Speak	20	30116	11	28-02-2018	15:17:15	28-02-2018	15:17:27	12.52	4.12	598.55	100.28	10.48	385.14	6.00	11.15	6.90
SSST_Speak_countdown	21	30116	12	28-02-2018	15:17:28	28-02-2018	15:18:28	60.96	0.97	615.08	97.60	12.68	285.77	22.91	14.90	7.66
SSST_Speak	22	30116	13	28-02-2018	15:18:29	28-02-2018	15:18:34	4.34	7.33	568.00	105.64	-9999	-9999	-9999	3.11	3.49
SSST_Read3	23	30116	14	28-02-2018	15:18:34	28-02-2018	15:18:46	11.29	4.19	563.95	106.40	15.49	132.78	5.33	4.99	3.10
SSST_Countdown3	24	30116	15	28-02-2018	15:18:46	28-02-2018	15:19:46	59.43	0.77	620.99	96.79	12.57	209.58	40.40	26.18	13.77
SSST_Read-sing	25	30116	16	28-02-2018	15:19:46	28-02-2018	15:19:58	11.96	3.42	614.00	98.06	12.77	176.96	10.00	37.70	12.38
SSST_Sing_countdown	26	30116	17	28-02-2018	15:19:58	28-02-2018	15:20:58	59.72	1.06	613.42	97.95	10.79	173.73	25.90	23.14	8.74
SSST_Sing	27	30116	18	28-02-2018	15:20:58	28-02-2018	15:21:16	17.19	3.85	605.43	99.33	12.30	308.56	30.00	30.00	13.86
Recov2	28	30116	19	28-02-2018	15:23:14	28-02-2018	15:25:16	122.00	2.05	609.88	98.54	11.97	233.72	27.16	24.55	10.22
Pasat_practice	29	30116	20	28-02-2018	15:28:30	28-02-2018	15:29:18	48.00	8.33	613.58	97.97	13.62	221.82	25.62	27.31	12.30
Pasat	30	30116	21	28-02-2018	15:29:50	28-02-2018	15:33:55	245.00	8.54	602.23	99.84	15.21	354.98	19.70	28.36	12.09
Recov3	31	30116	22	28-02-2018	15:35:45	28-02-2018	15:37:45	120.00	0.12	617.02	97.41	11.24	216.84	31.17	25.67	10.28
Raven	32	30116	23	28-02-2018	15:39:44	28-02-2018	15:43:44	240.00	5.64	639.88	94.03	13.94	248.52	42.28	34.26	16.69
Walking_own_pace	33	30116	24	28-02-2018	16:03:55	28-02-2018	16:05:56	121.00	138.90	606.57	99.12	14.88	387.36	30.28	28.43	13.01
Walking_fast_pace	34	30116	25	28-02-2018	16:06:04	28-02-2018	16:08:03	119.00	453.46	491.06	122.45	23.95	239.40	4.60	23.85	3.13
Cycling	35	30116	26	28-02-2018	16:09:44	28-02-2018	16:13:33	229.00	167.85	408.57	148.43	21.11	298.27	4.22	44.85	3.14
stairs_up_and_down	36	30116	27	28-02-2018	16:15:57	28-02-2018	16:19:35	218.00	272.42	473.60	128.78	24.29	267.82	10.81	64.70	9.65
Recov_standing	37	30116	28	28-02-2018	16:20:37	28-02-2018	16:22:45	128.00	1.85	515.47	116.42	14.32	373.73	5.97	7.41	2.78
Dishes	38	30116	29	28-02-2018	16:23:52	28-02-2018	16:25:51	119.00	31.23	479.17	125.34	16.67	468.76	3.14	15.58	2.44
Vacuum	39	30116	30	28-02-2018	16:26:56	28-02-2018	16:28:56	119.76	100.48	468.49	128.29	17.27	413.29	3.79	19.98	3.33
Recov4	40	30116	31	28-02-2018	16:39:49	28-02-2018	16:41:49	120.00	1.10	551.27	108.95	11.36	223.68	29.64	17.39	7.59
TA_repeat	41	30116	32	28-02-2018	16:43:59	28-02-2018	16:48:02	243.00	9.08	538.16	111.64	14.05	276.17	18.68	19.95	8.68
Recov5	42	30116	33	28-02-2018	16:49:07	28-02-2018	16:51:07	120.00	1.16	547.03	109.80	11.55	268.60	30.64	17.94	8.08
Pasat_repeat	43	30116	34	28-02-2018	16:52:40	28-02-2018	16:56:46	246.00	4.15	517.03	116.45	13.03	278.92	15.81	30.81	7.54
Treadmill1	44	30116	35	28-02-2018	17:00:46	28-02-2018	17:04:46	240.00	219.30	473.03	126.93	19.77	315.22	4.43	13.07	2.83
Treadmill2	45	30116	36	28-02-2018	17:04:47	28-02-2018	17:08:47	240.00	428.47	423.59	141.75	25.26	229.97	4.32	11.48	2.17
Treadmill3	46	30116	37	28-02-2018	17:08:48	28-02-2018	17:12:47	239.00	1030.58	371.27	161.77	25.49	98.29	4.21	12.13	2.50
Treadmill4	47	30116	38	28-02-2018	17:12:48	28-02-2018	17:15:52	184.00	197.96	414.50	145.59	21.26	296.47	3.95	30.48	1.83
Recov6	48	30116	39	28-02-2018	17:16:15	28-02-2018	17:19:15	180.00	1.70	524.69	114.86	13.25	300.29	29.95	36.08	11.33
Total Registration	-9999	30116	Entire data	28-02-2018	14:34:36	28-02-2018	17:20:50	9974.02	76.42	533.12	115.10	15.97	286.42	18.56	77.38	10.94

All actions for the *Results* tab are presented in the form of buttons:





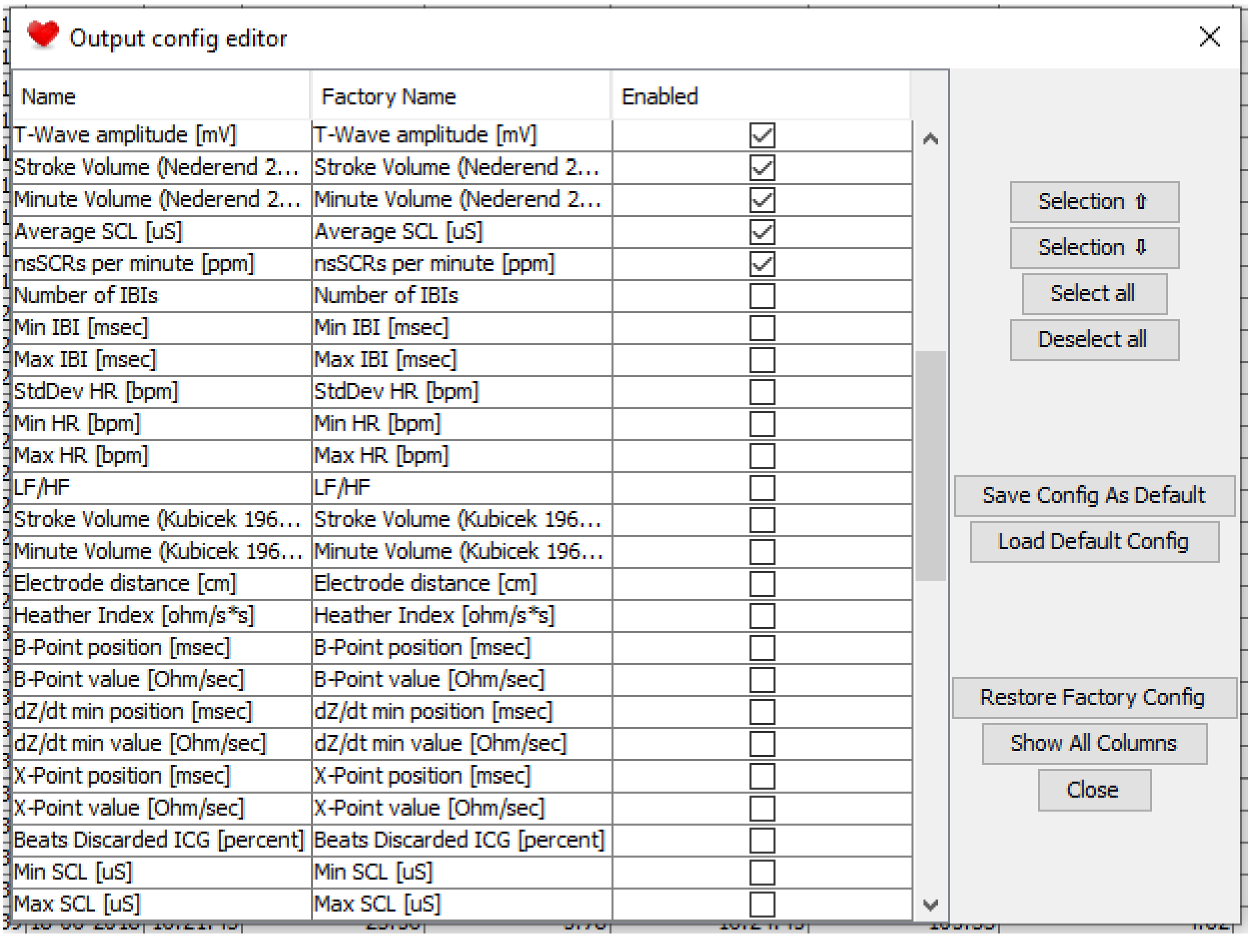
5.7.1 THE OUTPUT CONFIGURATION EDITOR

To display additional variables that are of your interest you can adjust the output configuration by clicking the Edit Output Config button: . This is the first button that is present in the actions bar at the top of the screen. When pressing this button the following window pops up, called the Output config editor.

In this window you can select the variables you would like information on by pressing the white box under “Enabled”. To omit variables simply uncheck the box behind the variable. An overview of all variables calculated by DAMS is given in Appendix B. When you are done selecting your variables of interest you press the button “Closed”. You will return to the Results tab which will now contain information on the added variables.

In the Output config editor you can also change variable names. To change a variable name click in the left column and type in the new name. The second column will show the factory configured name of that particular variable. To restore the output configuration to the factory configuration, click on *Restore Factory Config*.

To change the order of variables in the output, use *Selection up* and *Selection down* buttons. To restore the output order to the factory configuration, click on *Restore Factory Config*.



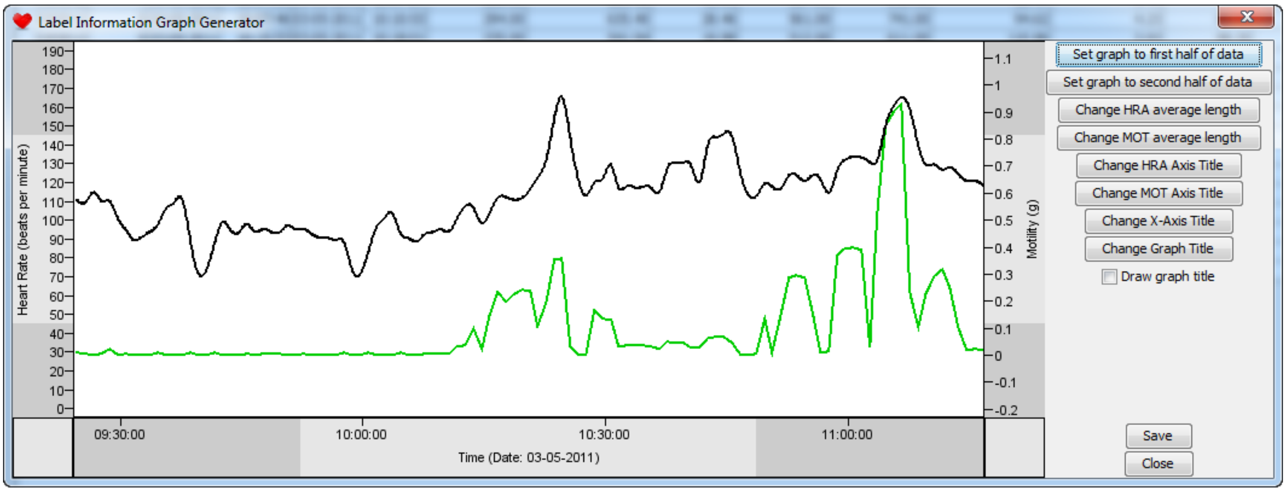
TIP19: *After changing the output configuration, you can Save Config As Default. This output configuration will then be applied to all .amsdata files that are opened / processed with this DAMS version by this user on this particular computer. You can switch between factory configuration and default configuration by using Load Default Config and Restore Factory Config.*

5.7.2 GRAPHS

Within the Results tab you can make line and bar graphs of your data.

5.7.2.1 LINE GRAPHS

Click on the *Generate Heart Rate graph* button. You will see the following screen.



The graph will; show the heart rate and motility signal of the entire recording. You can change names of the x-axis, y-axis of the HR and the y-axis of the motility as well as the graph title itself (or hide it by unchecking the *`Draw graph title'* box).

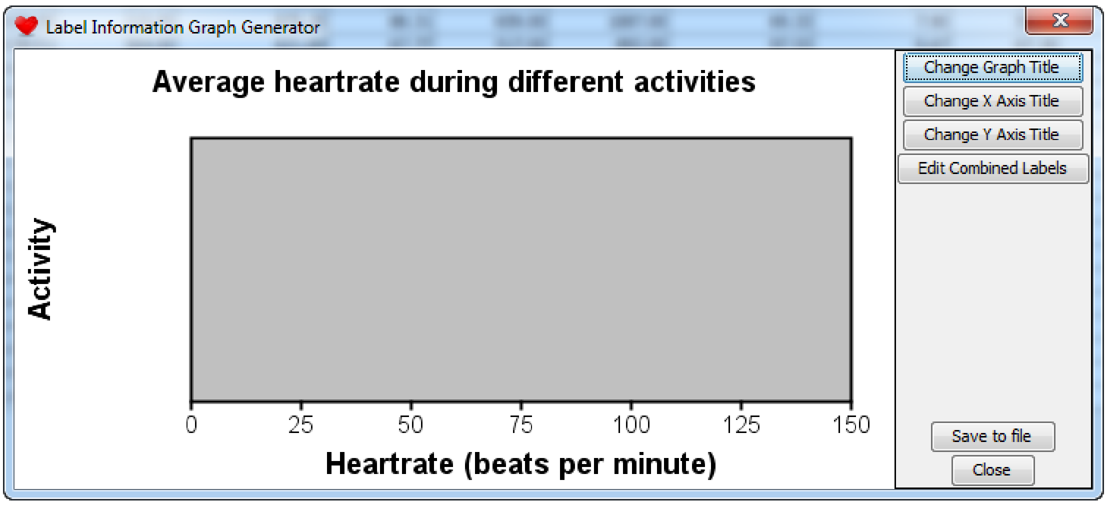
You can adjust the degree of smoothing of the signals by *Change HRA average length* or *Change MOT average length.* Increasing the length will give a more smooth signal.

For long recordings (e.g. > 24H) you have the option to display the first or second half of the data by using the buttons of the heart rate graph menu.

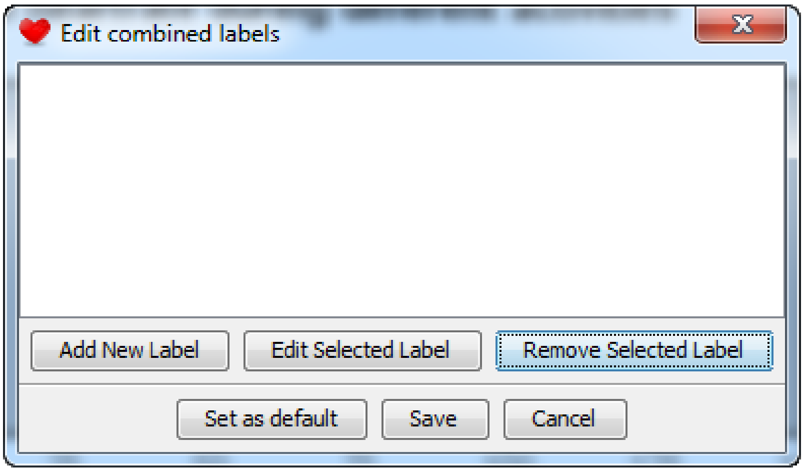
When finished, you can save the graph to a .png file by clicking on *Save.*

5.7.2.1 BAR GRAPH

With the heart rate graph it is possible to display the average heart rate across single activities or sets of combined activities. Select or combine various labels into a single bar. Click on the *Generate Bar Graph Of Data* button. You will see the following screen:

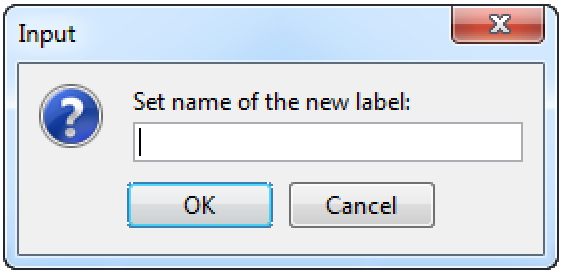


Now click on *Edit Combined Labels*, you will see the following screen:

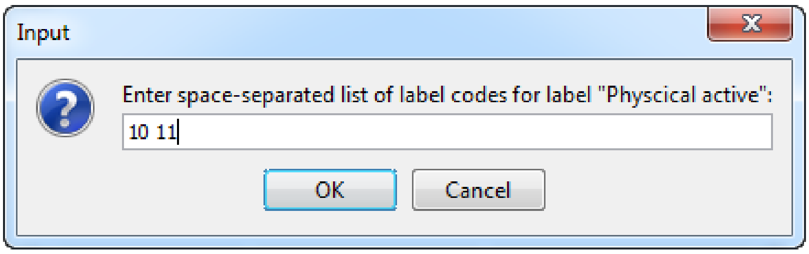


To add bars to the bar graph you need to define what activities should be averaged into a single bar. You can either choose to have a bar represent a single level of a labeling category for example Cycling, or you can combine multiple levels of a category, i.e. cycling and walking into one bar that represents 'physical active' periods.

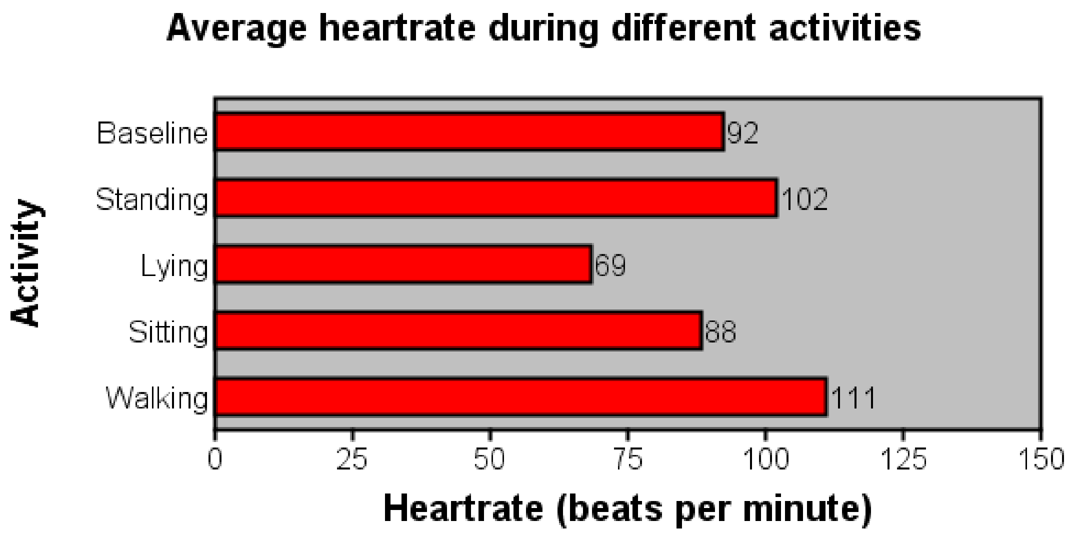
To do this click on *Add New Label* and enter the category name you want to give to this specific bar (e.g. Active).



Click *OK*. Then enter the space separate list of level codes that need to be averaged for this bar. e.g. 10 for cycling and 11 for walking would be entered like this:



When you have entered all desired bars, you can save the bar graph setting as default. You can change the name of the X-axis, Y-axis and Graph by clicking on the *Change … Title* buttons of the bar graph menu. The end result might look like this:

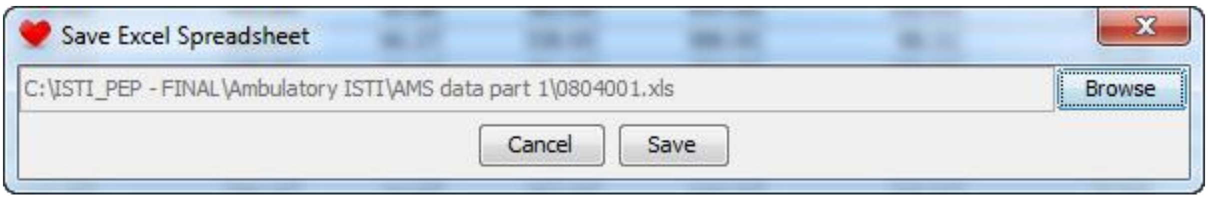


5.7.3 EXPORTING DATA

Data can be exported from DAMS in various ways. The variables calculated per label can be exported to either Excel or a text file. However, it is also possible to export the data divided into fixed time periods, and to export the raw signals.

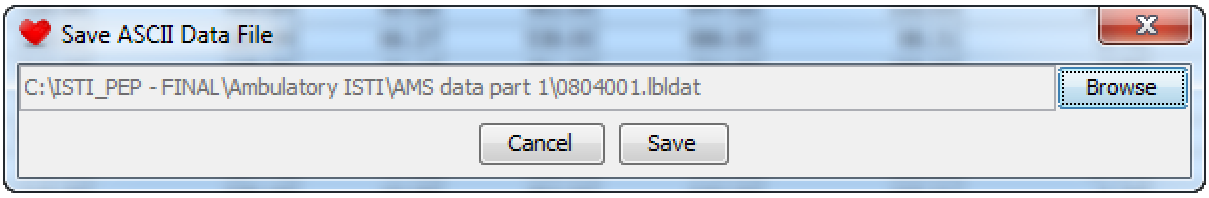
5.7.3.1 EXCEL

To export your data to an excel file you click on the third button in the selection panel. When clicked a window will appear asking you to enter the location to the output file by clicking on the browse button. When you clock save, the file will be saved as a .xls excel file.



5.7.3.2 TEXT FILE

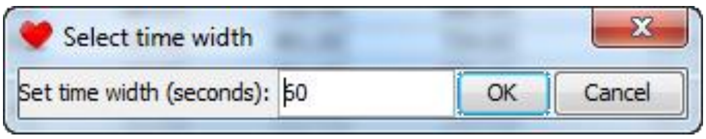
To export your data to a text file you click on the second button in the selection panel. When clicked a window will appear asking you to enter the location to the output file by clicking on the browse button. When you clock save, the file will be saved as a .lbldat text file.

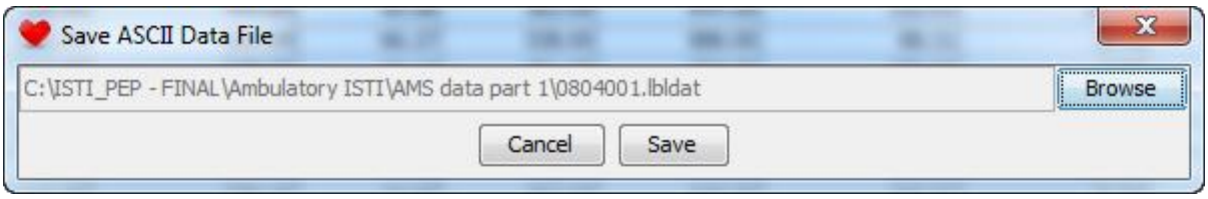


5.7.3.3 FIXED TIME-BASED LABELS

To export data across fixed periods of time. Click on either the *Export 'Per Minute' Information to ASCII File* or the *Export 'Per Minute' Information to Excel Spreadsheet* button. You will be prompted to enter the duration of the fixed time periods. The output will give averages across consecutive periods of this length in the chronological order of recording. The start and stop times of the fixed time period 'labels' will help you link the generated data to the real time of the experiment.

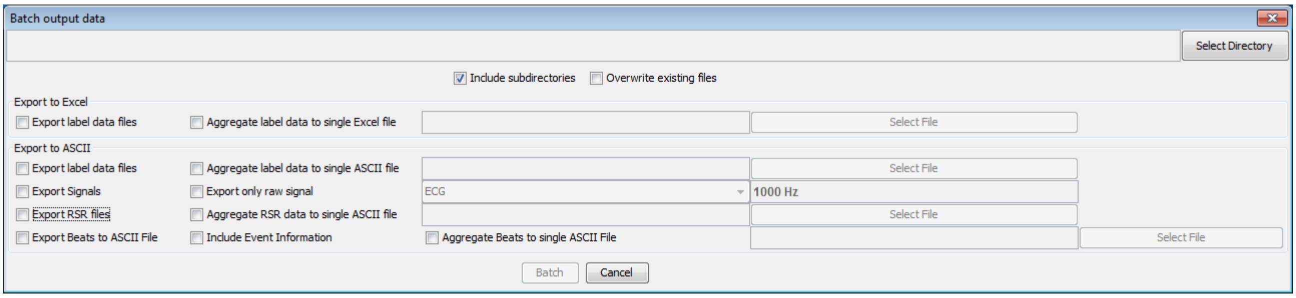
NOTE: Spectral analysis will only output values for LF and HF labels if the fixed periods are chosen longer than 3 minutes.





5.7.3.4 BATCH EXPORT

You can export text files in batch mode. This function will generate output files per subject or a merged output file of many subjects from single .amsdata files as long as they are placed in the same directory (or a subdirectory in that directory). So place all files that need to be exported for a certain project in one directory and click on *Batch export data* under *File* in the menu bar.

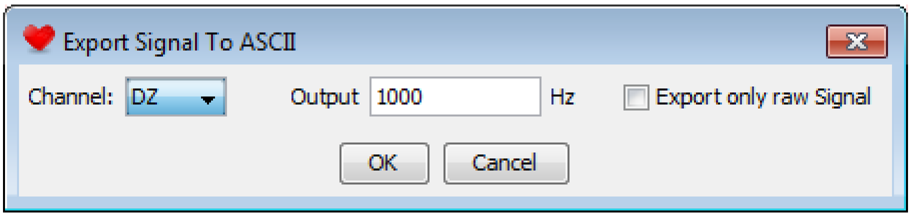


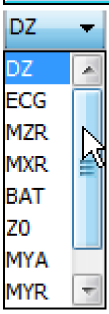
Simply select to export to XLS or to ASCII in single mode or merged mode. Then select the output directory and name the output file.

5.7.3.5 RAW SIGNALS

A raw data dump can be obtained from each recorded signal into a text file.

Click on *Data* in the main menu and select to *Export Signal To ASCII*. You can choose to export any signal to an ASCII file and choose the resolution of the output. Checking the 'Export only raw Signal' option will produce a file with only raw values (one per line)





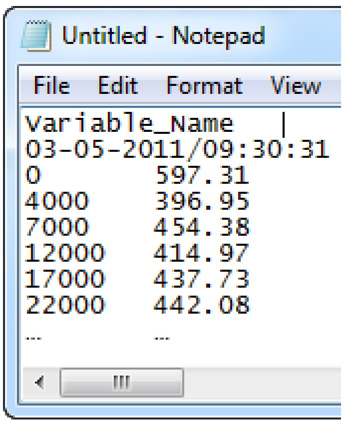
When not checked the file has a fixed format that includes the raw signal but also the cumulative time index and the label codes. Each line represents a single sample from the raw data, with samples spaced in time by the specified resolution. N.B.: This can generate very large data files in 24-hour recordings for signals with high sampling rates!

To avoid generating such large files, you can use the *Export Signal To EDF* functionality in the *Data* menu. When selecting multiple signals, those will be combined into a single file in the European Data Format (EDF; https://www.edfplus.info/) .

5.7.4 IMPORTING EXTERNAL SIGNALS

Click on *Data* in the menu and select to (re)import a raw signal dump from a text file. This option allows you to use downsampled data or to import an entirely different signal (as long as the format complies with the DAMS raw data dump format).

Click on *Data* in the menu and select *Load external signal*. In the pop-up screen select the ASCII file containing the external signal. This ASCII file should be structured as follows:



Variable_Name		→ Name of the external signal
03-05-2011/09:30:31		→ Start date and start time of the recording
0	597.31	→ The first column is the time in msec (starting at zero) and the second column is the corresponding value of the external signal
4000	396.95	
7000	454.38	
12000	414.97	
17000	437.73	
22000	442.08	
…	…	

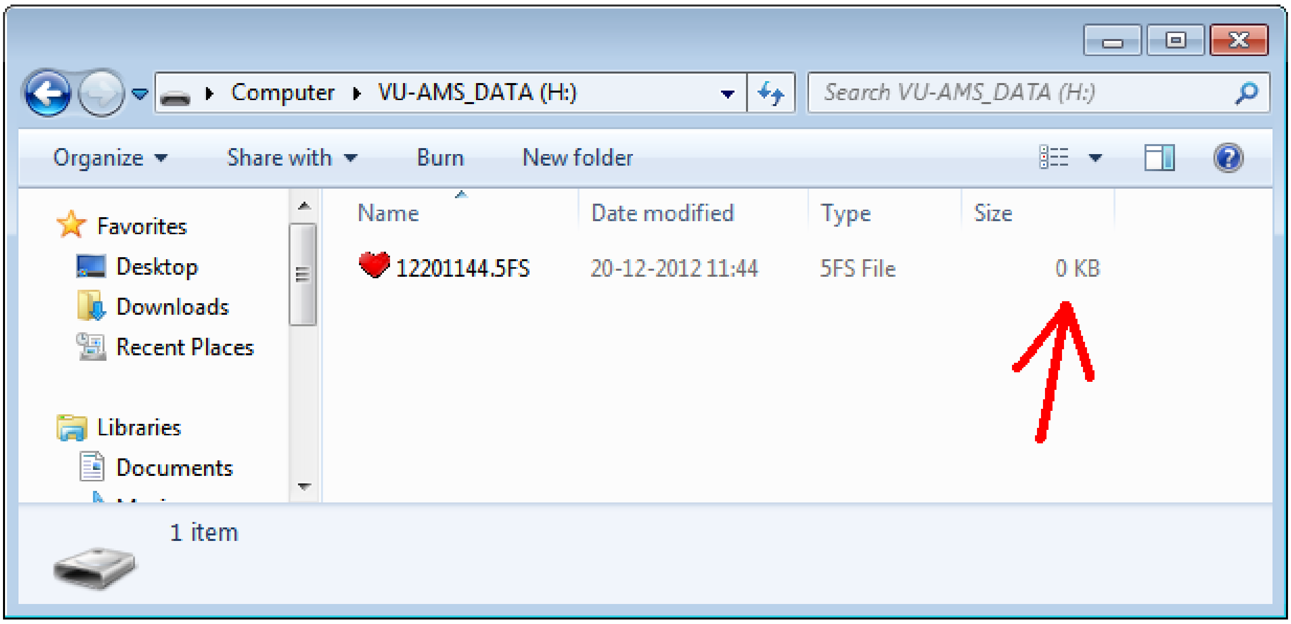
The external signal will appear above the whole IBI recording panel. The mean value of the external signal over each condition will appear in the Label Information tab under 'External Signal Average'. A maximum of three signals can be imported into VU-DAMS.

Click on *Data* in the menu and select *Clear external signal* to remove the external signal.

5.8 TROUBLESHOOTING

In the Help menu you find the *report a problem or request a feature* action, which you can use to compose an e-mail to the VU-AMS team.
The help menu also provides shortcuts to the VU-AMS website, this manual, and version information.

5.8.1 VU-AMS FILE HAS ZERO BYTES.



Cause: the VU-AMS has made a recording, but it was not stopped in the normal way. This usually happens when the batteries are removed during recording. Probably all data are on the card but an end-of-file summary has not been placed and the FAT table isn't updated.

Solution: This can be repaired by closely following the instructions on www.vu-ams.nl > Support> Tutorials > Troubleshooting > [Video manual: How to repair a 0KB file recorded with the VU-AMS5fs](#).

5.8.2 DEVIANT FLASHING

When on standby (meaning batteries inserted but not recording), the VU-AMS device flashes once every 10 seconds, when recording the VU-AMS device flashes once every 3 seconds. Faster flashing signals problems:

- The green light is flashing very rapidly
  Cause: The Compact Flash card is not (properly) installed.
  Solution: Install the Compact Flash card in the proper way.

- The green light is flashing rapidly
  Cause: The battery lid is not (properly) fastened.
  Solution: Fasten the battery lid in the proper way.

5.8.3 WARNING BEEPS

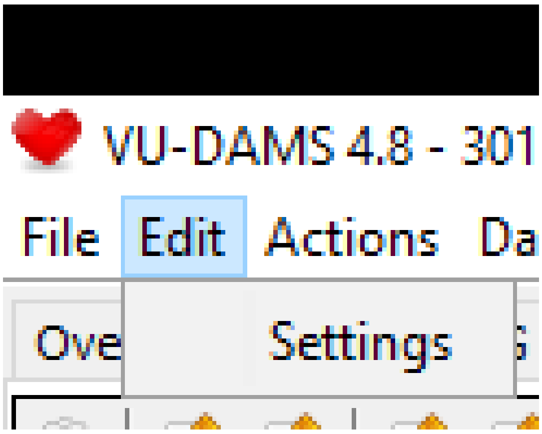
When the VU-AMS detects that something is amiss it can generate various warning beeps:

- You hear a double beep (the 'alert beep'), which is repeated after increasingly shorter intervals (from 30 to 10 seconds).
  Cause: The battery voltage is becoming low.
  Solution: Replace the batteries with fresh ones.

- You hear a triple beep (the 'warning beep').
  Cause: An electrode comes off, a lead wire gets detached, or the lead wire connector is pulled out by accident.
  Solution: No worries. Just attach the electrode again (use a spare one if necessary), reattach the lead wire, or plug the connector back into the socket.

APPENDIX A. ADJUSTING SETTINGS

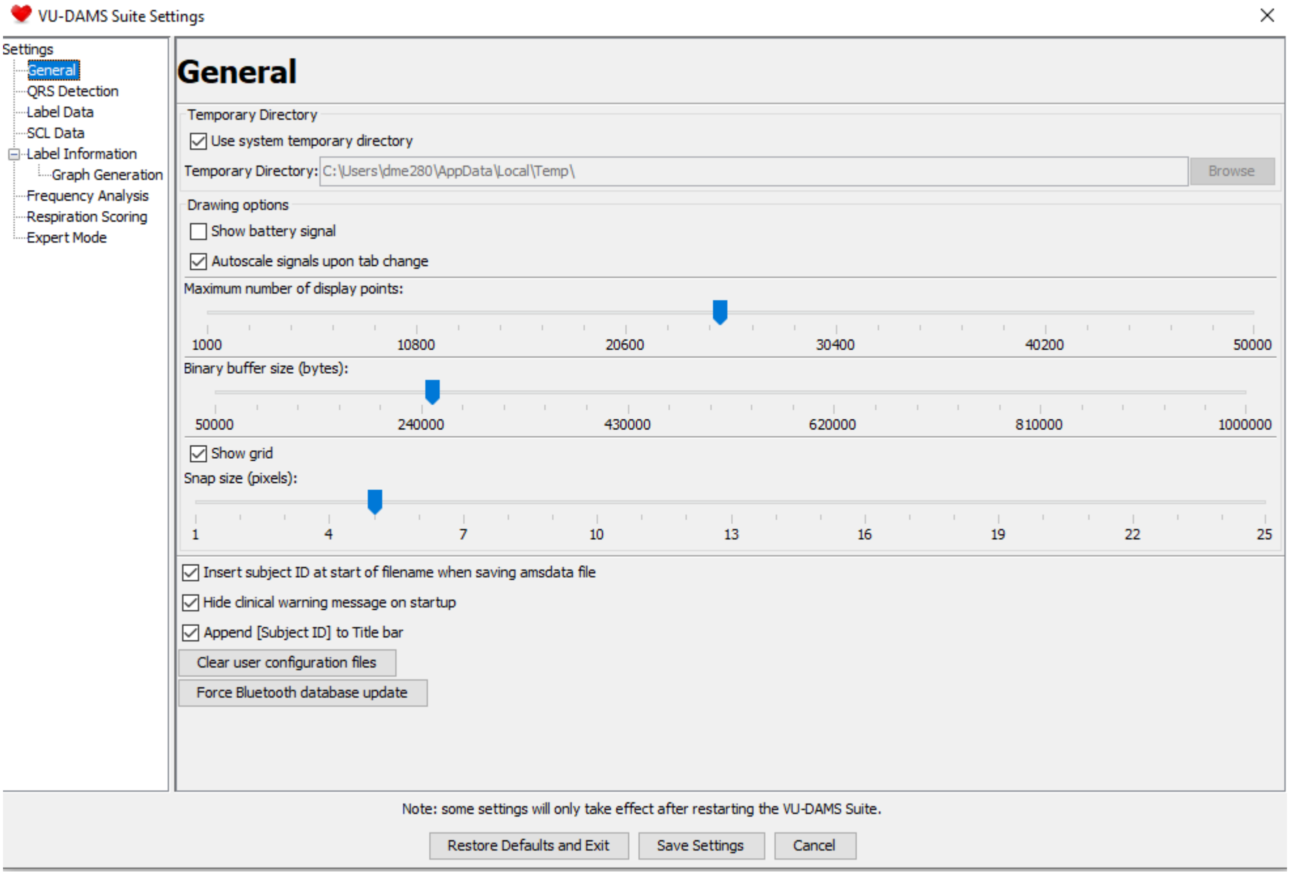
Various settings can be changed by clicking on the Edit > Settings button. This button is available irrespective of which tab you are currently working in.



Clicking on setting will open a new window calles the VU-DAMS Suite Settings. On the left side is a menu which will guide you to setting you would like to change.

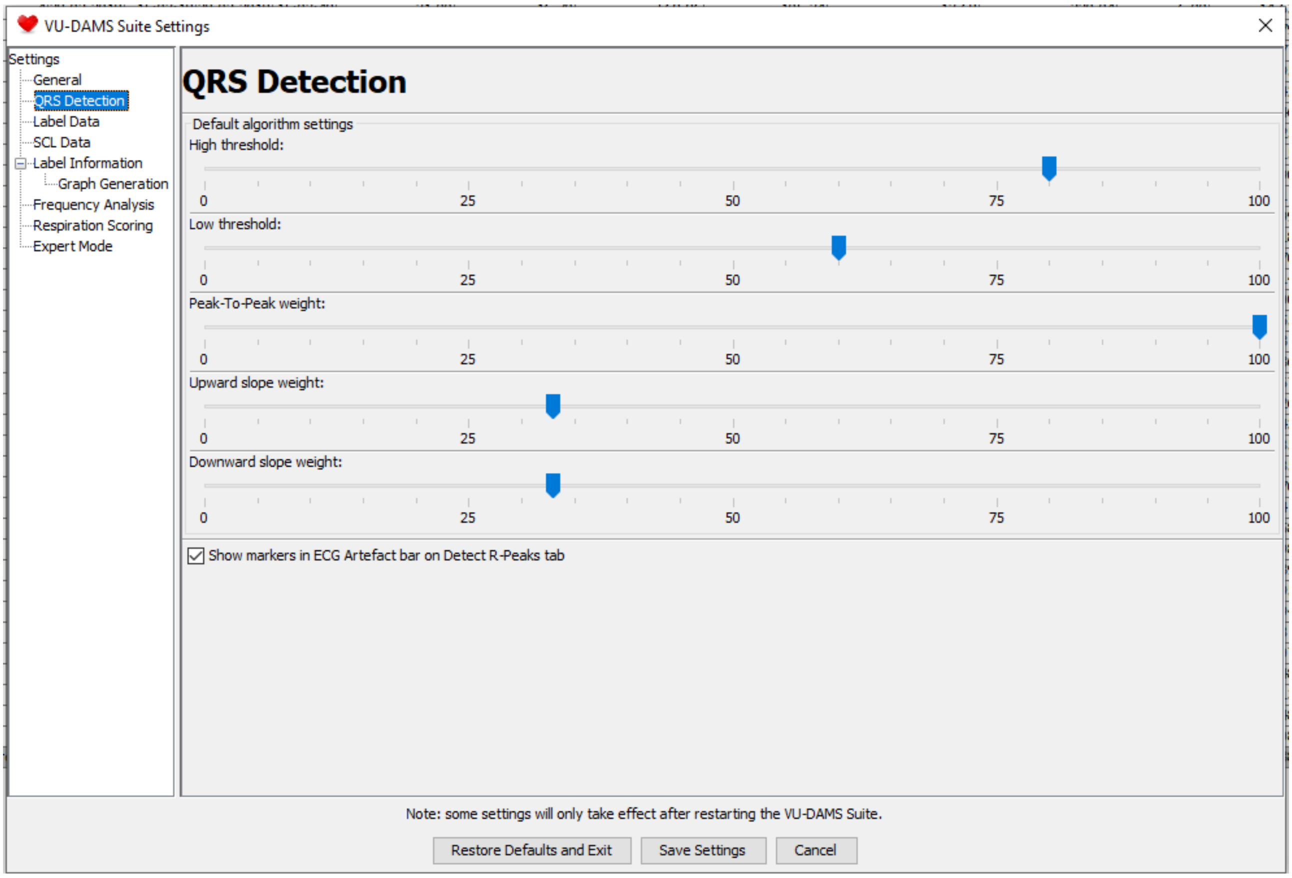
GENERAL

the general setting you can change display options like pixels and number of display points.



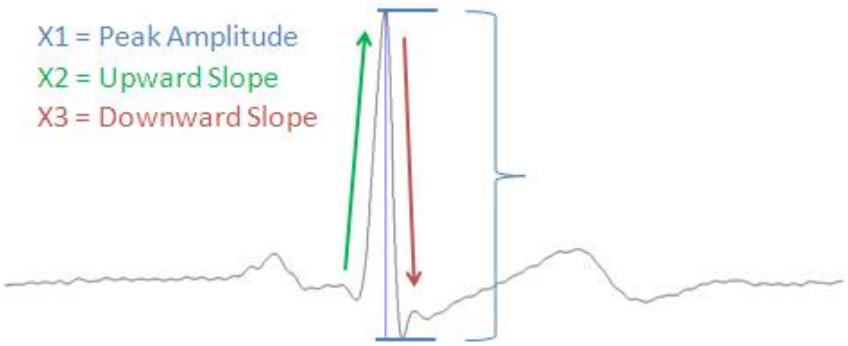
QRS DETECTION

When an ECG signal is of good/reasonable quality but you can clearly see that the automatic detector placed the beats anywhere but on top of the R-peak, you should rescan the complete ECG signal with different settings for the detection algorithms. The default settings for the algorithm upon opening the *.amsdata* file can be changed in the main menu by selecting *Edit* → *Settings* → *QRS Detection*.

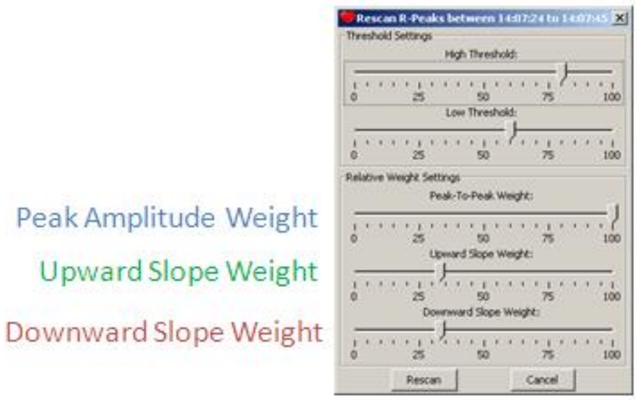


A pop up screen with 5 sliders (the same as in the settings screen) appears. You see a high and a low Threshold slider and three Relative weight sliders that change the weight of the peak amplitude, the downward slope and the upward slope. The algorithm calculates a Peak score based on the sum of these parameters multiplied by their weights and then divides this by the sum of the weights. The R-peaks of the selected area will be rescored when clicking on Rescan. This peak score will be used in combination with the threshold sliders/settings, such that all peaks with a score between the low and the high thresholds will be considered an R-peak. Set the sliders as desired and press Rescan to apply the new settings on the selected part of the ECG signal. Adjust the settings until satisfied.

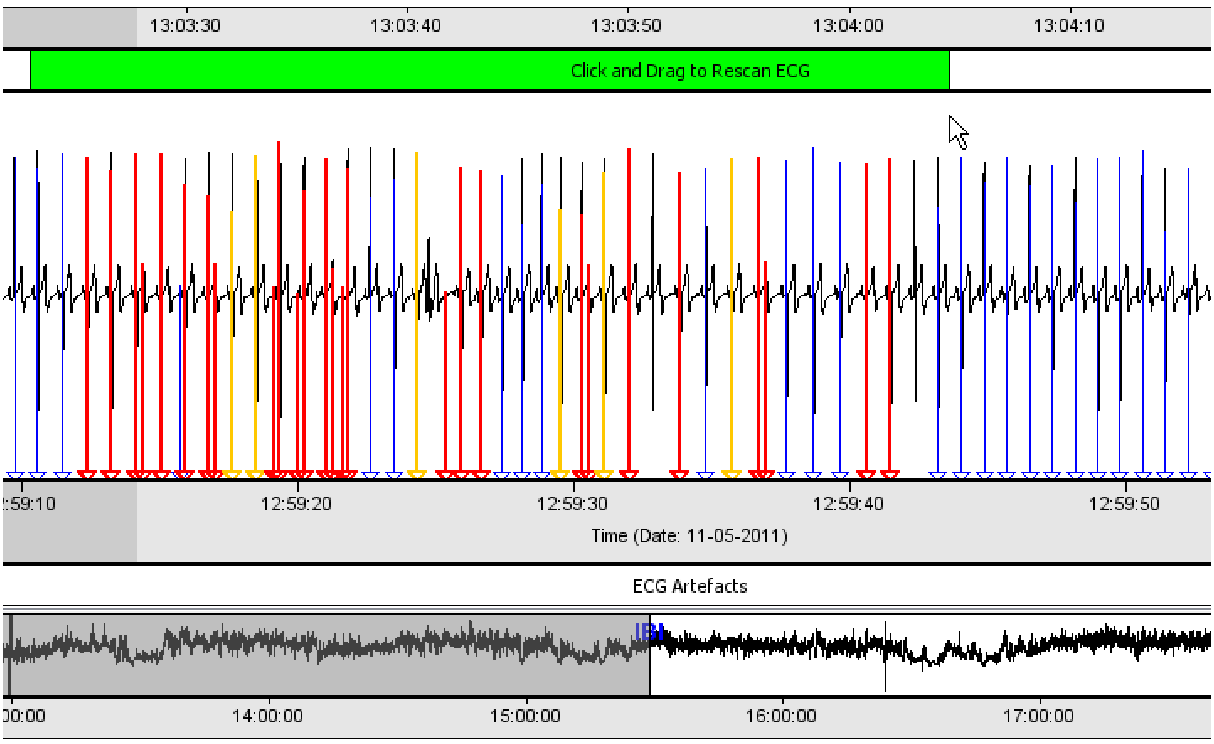
The default values can be changed in the settings screen:



Peak score =((Weight 1 * X1) + (Weight 2 * X2) + (Weight 3 * X3)) / (W1 +W2+W3)

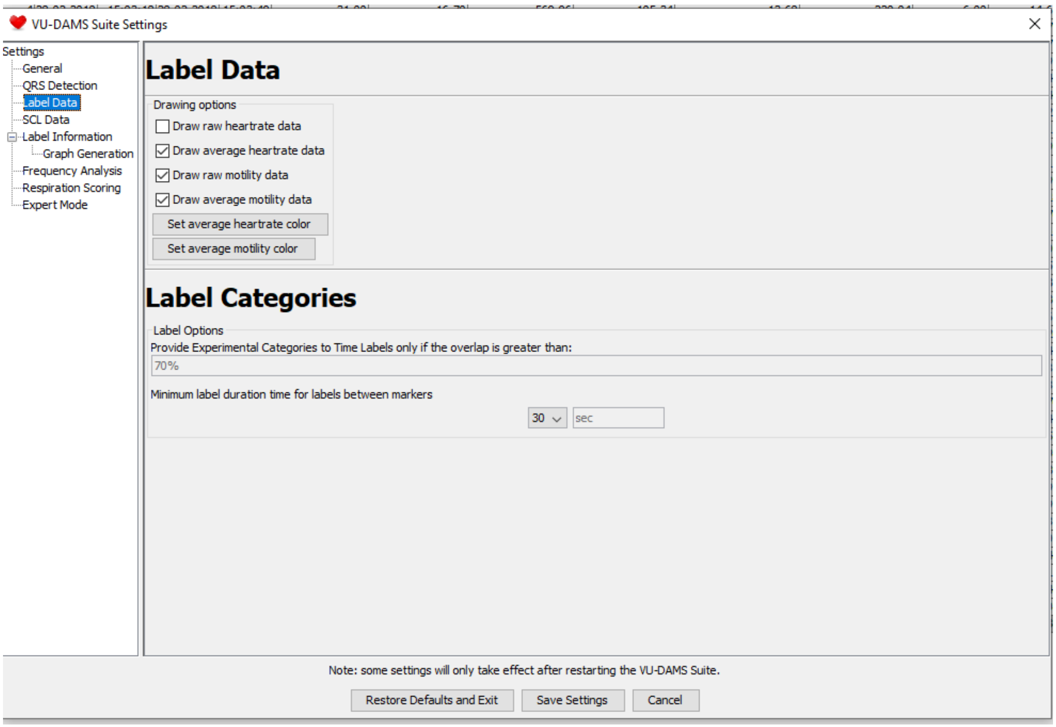


You can also use the Rescan bar to select only a certain part of the recording. Drag your mouse across the incorrectly scored part of the ECG (minimum rescan length is 10 seconds) in the Rescan bar while holding the left mouse button.



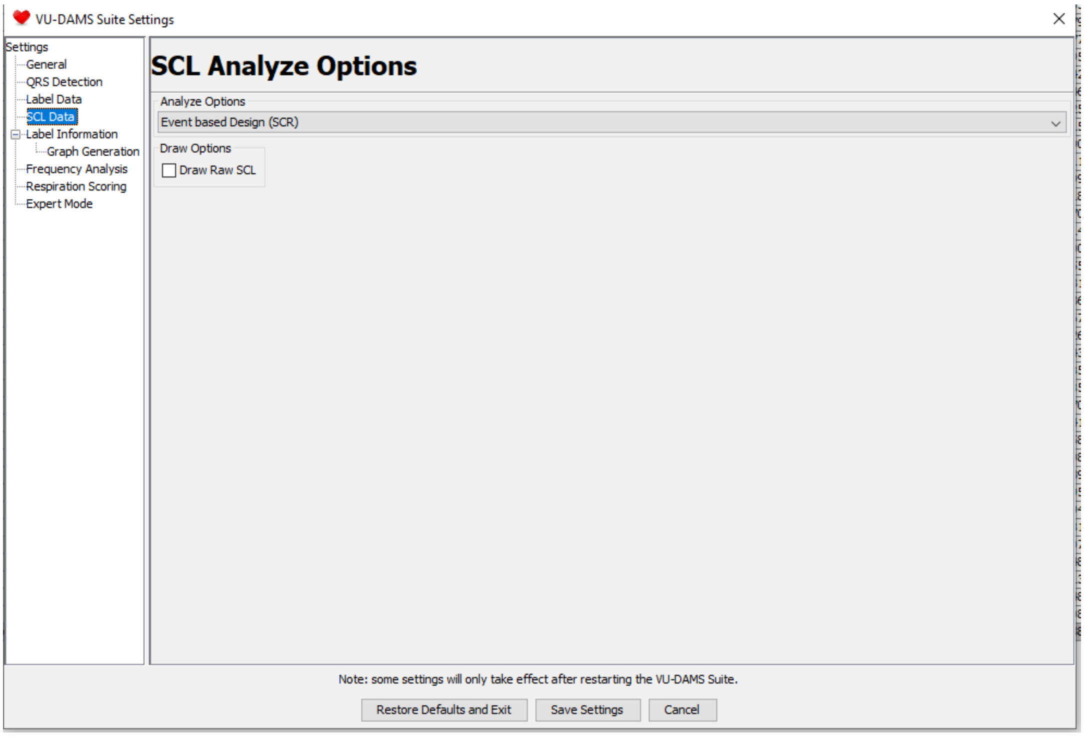
LABEL DATA

In the label data tab settings for the graphs and label categories can be changed.



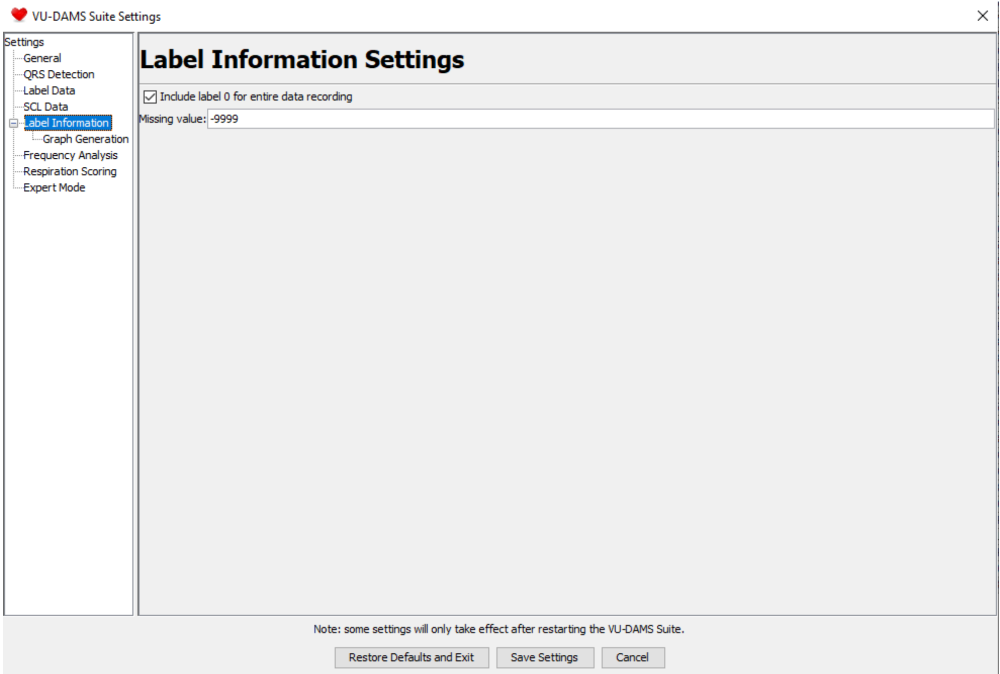
SCL DATA

In this window you can switch between a label based or an event based design.



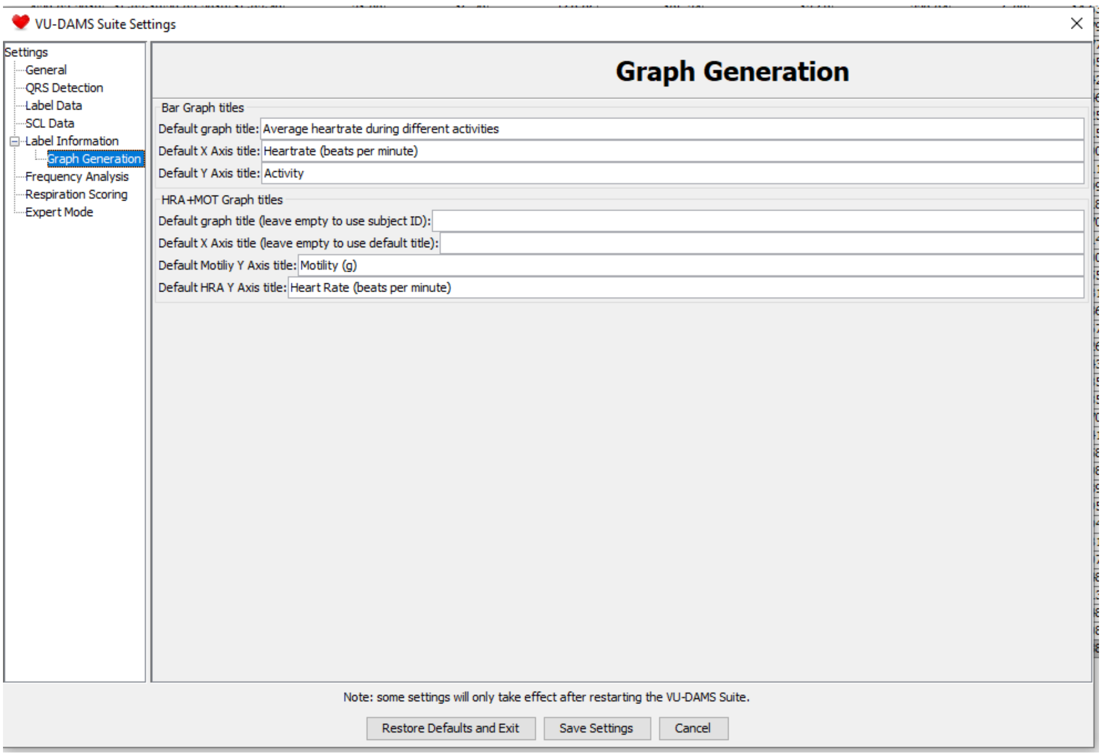
LABEL INFORMATION SETTINGS

In this window you can adjust wether the display the label_ID with number 0, the average of the whole recording. In addition you can adjust the value you want to give to missingness.



GRAPH GENERATOR

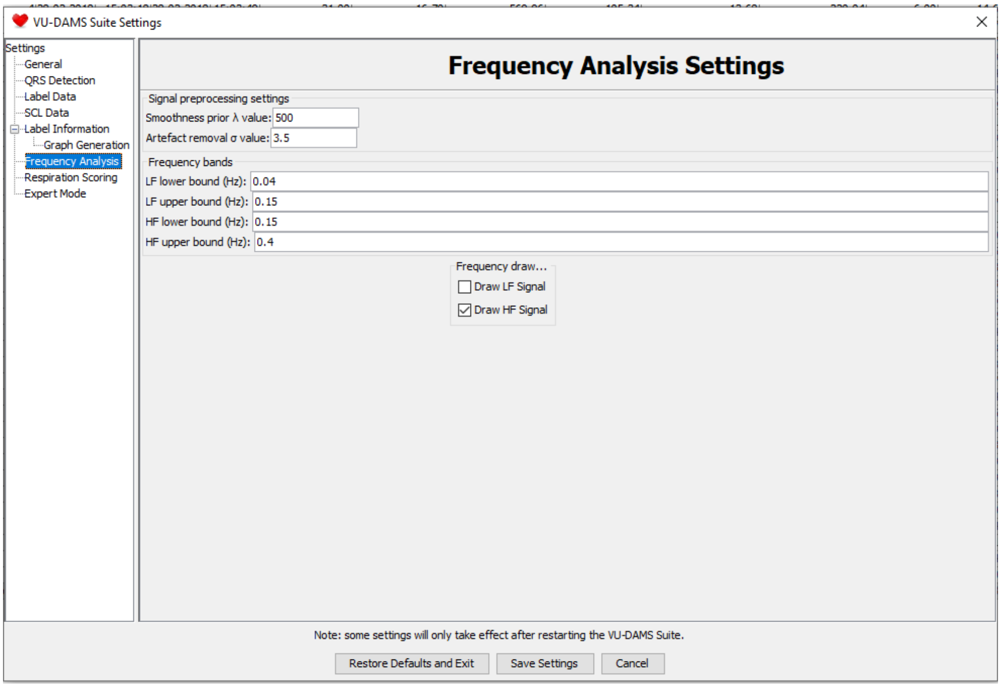
In this window you can change or add default graph titles.



FREQUENCY ANALYSES

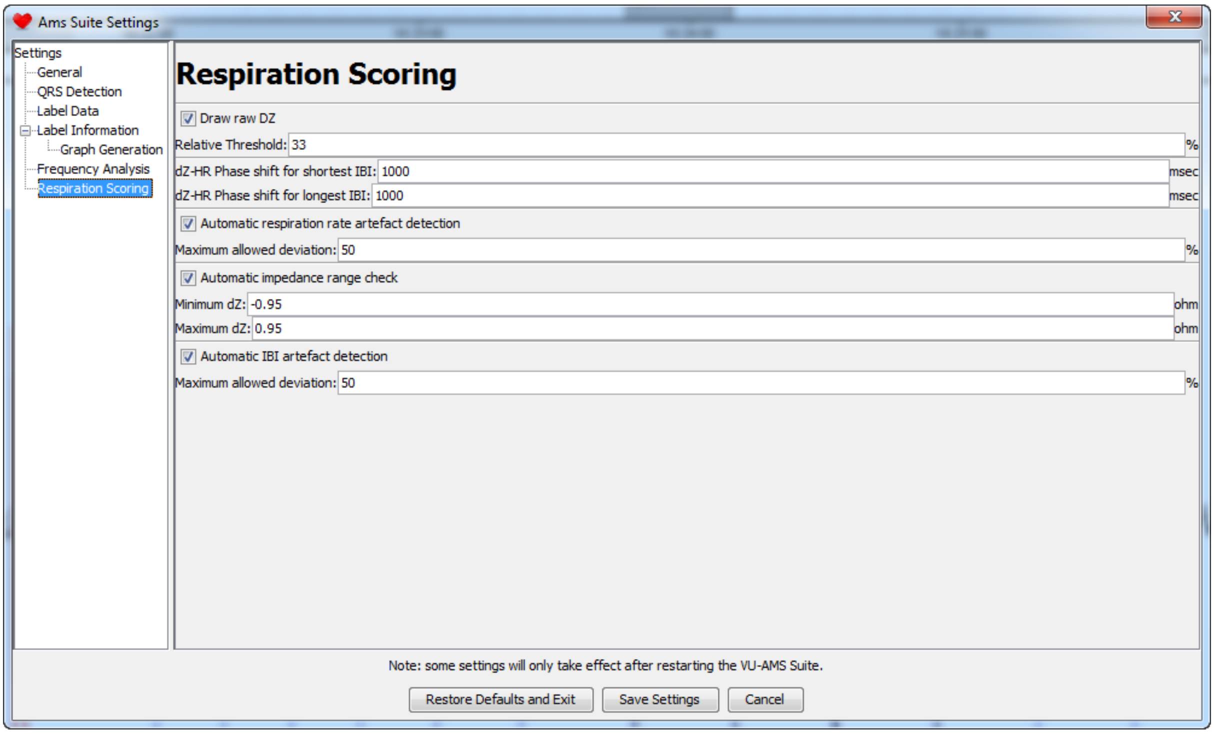
In this window

the frequencies that are used to calculate HF-HRV and LF-HRV can be adjusted.



RESPIRATION SCORING

In this tab you can adjust the respiration scoring parameters.



RELATIVE THRESHOLD [0…1] : 0.33 (DEFAULT)

Purpose: The purpose of this parameter is to alter the sensitivity for the detection of breaths. A breath is defined by two zero crossings (a peak and a valley) in the first derivative of the filtered impedance signal that are separated by a minimum amplitude. The *Respiration Scoring* tab calculates a running average over the 20 seconds preceding the current selected breath cycle of the difference in amplitude at peaks and valleys in the dZ signal. The relative threshold defines the percentage of this average that is used as the minimum amplitude for the 'tidal volume' in a respiratory cycle.

Adjustment: A higher threshold decreases the sensitivity (less of the consecutive peak-valley pairs in the filtered impedance will be counted as true breaths – use when too many small wobbles are counted as breaths), and a lower threshold value increases the sensitivity for amplitude differences (more of the consecutive peak/valley pairs will be counted as true breaths – use when the amplitude of the actual breaths becomes low, for instance in nighttime belly breathing).

DZ-HR PHASE SHIFT (IN MSEC) :1000 (DEFAULT)

Purpose: This defines the delay added to the inspiratory and expiratory phases in which the *Respiration Scoring* tab is allowed to search in the IBI series for a shortest IBI in inspiration or longest IBI in expiration, respectively. Increasing the dZ-HR Phase shift can increase the number of valid RSA values in subjects with low respiration rates whereas at high respiration rates, the default 1000 msec interval may lead to erroneously used IBIs from the next respiratory cycle.

Adjustment: Increasing the phase-shift increases the time-delay.

AUTOMATIC RESPIRATION RATE ARTEFACT DETECTION : IS "ON" WHEN TICKED

Purpose: The purpose of this option is to automatically reject breaths with an unusually small or unusually high respiration rate as compared to the running average of the 20 preceding breaths. For participants with irregular breathing, the maximum allowed deviation might need to be increased in order to prevent false rejects.

Adjustment: The 'maximum allowed deviation' (default 50%) enables the scorer to specify how much the respiration rate of a breath needs to deviate from the running average in order to be excluded by the automatic scoring program. Increasing the percentage means allowing for larger deviations from the running average.

AUTOMATIC IMPEDANCE RANGE CHECK

Purpose: The purpose of this option is to automatically reject clipping (i.e. where the raw respiration signal turns into a flat line at dZ = 1 ohm or dZ = –1 ohm).

Adjustment: Although the default values generally seem to work well, some recordings may require an automatic impedance range check that is a fraction more or a fraction less strict.

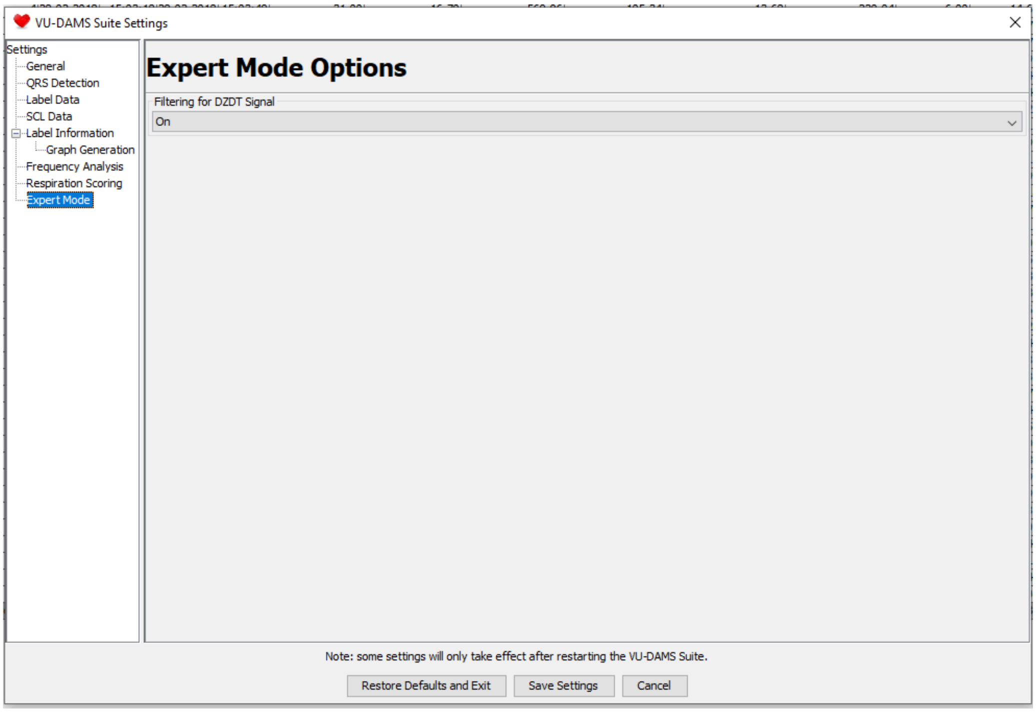
AUTOMATIC IBI ARTEFACT DETECTION

Purpose: The purpose of this option is to automatically reject spikes in the IBI time series, that represent extrasystolic beats or very prolonged beats. Note that these beats do not represent an error in R-peak placement by the DAMS program (which should have been dealt with earlier during manual inspection and correction). They do result in IBIs that are twice the length or half the length of most of the other IBIs. This inflates the RSA value for the breaths in which they occur very strongly. When the irregular IBI check is 'on' these beats are removed from the set of IBIs that are considered when selecting the shortest IBI and longest IBI to compute the RSA.

Adjustment: The allowed magnitude of the difference between consecutive IBIs can be adjusted by changing the 'maximum allowed deviation' (default 50%).

EXPERT MODE

In this window you can decide to turn the filtering of the DZ/DT signal in the ICG scoring tab on or off.


VU-DAMS Suite Settings
Settings
General
QRS Detection
Label Data
SCL Data
Label Information
Graph Generation
Frequency Analysis
Respiration Scoring
Expert Mode
Expert Mode Options
Filtering for DZDT Signal
On
Note: some settings will only take effect after restarting the VU-DAMS Suite.
Restore Defaults and Exit
Save Settings
Cancel

APPENDIX B: VARIABLE OVERVIEW

This section provides an overview of all variables that are provided by DAMS in alphabetical order. For each variable information is given on the full name (if abbreviated), the units of measurement (both the full name and the used abbreviation in the *section 5.7 Results* tab), and in which section of the manual you can find information on this variable. Five different tables are provided, the first table provides information on label information variables, the second table on ECG derived variables, the third on ICG (or ICG and ECG combined) derived variables, the fourth on EDA derived variables and the fifth on motility and external signals.

Table 1. Label information

Variables	**Unit (full name)**	**Unit (abbreviated)**	**Explained in section**
Artefact_Free_Length	seconds	s	
End_Date	Date Time		
End_Time	Date Time		
Experimental_Condition	Character		5.2
Experimental_Condition_Code	Numeric		5.2
Label_Duration_s	seconds	s	
Label_ID	Numeric		
Start_Date	Date Time		
Start_Time	Date Time		
Subject_ID	Numeric		

Table 2. ECG derived variables

Variables	**Full name if abbreviated**	**Unit (full name)**	**Unit (abbreviated)**	**Explained in section**
Average_HR_bpm	Heart rate	beats per minute	bpm	5.3.1
Average_IBI_msec	Inter-beat-interval	milliseconds	msec	5.3.1
Artefact_Free_ECG_percent		percent	%	
Artefact_Free_ECG_s		seconds	s	

HF_ms	High Frequency HRV	milliseconds	ms	5.3.1
LF_ms	Low Frequency HRV	milliseconds	ms	5.3.1
LFHF	Ratio between low frequency and high frequency HRV			
Max_HR_bpm	Heart rate	beats per minute	bpm	5.3.1
Max_IBI_msec	Inter-beat-interval	milliseconds	msec	5.3.1
Min_HR_bpm	Heart rate	beats per minute	bpm	5.3.1
Min_IBI_msec	Inter-beat-interval	milliseconds	msec	5.3.1
No_of_Premature_Atrial_Contractions		count	N	5.3.2.3
No_of_Premature_Ventricular_Contractions		count	N	5.3.2.3
Number_of_IBIs	Inter-beat-interval	count	N	5.3.1
Qonset_point_msec		milliseconds	msec	5.5
QOnsetvalue_mV		milliVolts	mV	5.5
QPoint_msec		milliseconds	msec	5.5
Qvalue_mV		milliseconds	msec	5.5
RMSSD_msec	Root-mean-square of successive differences	milliseconds	msec	5.3.1
RValue_mV		milliVolts	mV	5.5
SDNN_msec	standard deviation of the N-N interval	milliseconds	msec	5.3.1
S_Offset_Val_mV		milliVolts	mV	5.5
SOffset_point_msec		milliseconds	msec	5.5
Spoint_msec		milliseconds	msec	5.5
StdDev_HR_bpm	Standard deviation of the Heart rate	beats per minute	bpm	5.3.1
SValue_mV		milliVolts	mV	5.5
T_Offset_value_mV		milliVolts	mV	5.5
TOffset_point_msec		milliseconds	msec	5.5
TOffset_value_mV		milliVolts	mV	5.5
Tpointmsec		milliseconds	msec	5.5
TValue_mV		milliVolts	mV	5.5
TWave_amplitude_mV		milliVolts	mV	5.5.1

Table 3. ICG derived variables				

Variables	Full name if abbreviated	Unit (full name)	Unit (abbreviated)	Explained in section
Artefact_Free_Respiration_percent		percent	%	
Beats_Discarded_ICG_percent		percent	%	5.5.2.3
BPoint_position_msec		milliseconds	msec	5.5
BPoint_value_Ohmsec		Ohm/millisecond	Ohmsec	5.5
dZdt_min_position_msec		milliseconds	msec	5.5
dZdt_min_value_Ohmsec		Ohm/millisecond	Ohmsec	5.5
Electrode_distance_cm		centimeters	cm	3.2.3
Heather_Index_ohmss		Ohm/millisecond	Ohmsec	5.5.1
LVET_msec	Left-ventricular-ejection time	milliseconds	msec	5.5.1
Minute_Volume_(Kubicek_1966)_lmin		liters per minute	lmin	5.5.1
Minute_Volume_(Nederend_2017)_lmin		liters per minute	lmin	5.5.1
PEP_msec	Pre-ejection period	milliseconds	msec	5.5.1
Respiration_Rate_bpm		breaths per minute	bpm	5.4.1
RSA_msec	Respiratory Sinus Arrythmia	milliseconds	msec	5.3.1
RSA0_msec	Respiratory Sinus Arrythmia	milliseconds	msec	5.3.1
Tidal_Volume_mOhm		Milli Ohm	mOhm	5.5.1
XPoint_position_msec		milliseconds	msec	5.5
XPoint_value_Ohmsec		Ohm/millisecond	Ohmsec	5.5
Z0_Average_Ohm		Ohm	Ohm	5.5

Table 4. EDA derived variables				
Variables	**Full name if abbreviated**	**Unit (full name)**	**Unit (abbreviated)**	**Explained in section**
Artefact_Free_SCL_percent	Skin conductance level	percent	%	
Average_SCL_uS	Skin conductance level	microsiemens	uS	5.6
Latency_sec		Seconds	sec	5.6.2
Max_SCL_uS	Skin conductance level	microsiemens	uS	5.6
Min_SCL_uS	Skin conductance level	microsiemens	uS	5.6
nsSCRs_per_minute_ppm	Non-specific skin conductance responses	peaks per minute	ppm	5.6.1
nsSCR_count	Non-specific skin conductance responses	count	N	5.6.1

Onset_Time		Date/Time		5.6.2
Peak_Time		Date/Time		5.6.2
Rise_Time_uS		microsiemens	uS	5.6.2
Rise_SCL_uS	Skin conductance level	microsiemens	uS	5.6.2
SCL_Onset_uS	Skin conductance level	microsiemens	uS	5.6.2
SCL_Peak_uS	Skin conductance level	microsiemens	uS	5.6.2
SCL_Stimulus	Non-specific skin conductance responses	microsiemens	uS	5.6.2
SCR	Skin conductance response	Logical		5.6.2

Table 5. Motility and external signal				
Variables	**Full name if abbreviated**	**Unit (full name)**	**Unit (abbreviated)**	**Explained in section**
Average_Mot_mg	Motility	milli-g	mg	
Total_Motility_mg		milli-g	mg	
Average_X_Motility_mg		milli-g	mg	
Average_Y_Motility_mg		milli-g	mg	
Average_Z_Motility_mg		milli-g	mg	
External_Signal_Average				
External_Signal_Average_2				
External_Signal_Average_3				